



Hierarchical Forest Management Planning

A Bilevel Wood Supply Modelling Approach

Thèse

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Résumé

Le processus de planification forestière hiérarchique présentement en place sur les terres publiques risque d'échouer à deux niveaux. Au niveau supérieur, le processus en place ne fournit pas de preuve crédible de la durabilité du niveau de récolte actuel. À un niveau inférieur, le processus en place n'appuie pas la réalisation du plein potentiel de création de valeur de la ressource forestière, contraignant parfois inutilement la planification à court terme de la récolte. Ces échecs sont attribuables à certaines hypothèses implicites au modèle d'optimisation de la possibilité forestière, ce qui pourrait expliquer pourquoi ce problème n'est pas bien documenté dans la littérature. Nous utilisons la théorie de l'agence pour modéliser le processus de planification forestière hiérarchique sur les terres publiques. Nous développons un cadre de simulation itératif en deux étapes pour estimer l'effet à long terme de l'interaction entre le gouvernement et le consommateur de fibre, nous permettant ainsi d'établir certaines conditions pouvant mener à des ruptures de stock. Nous proposons ensuite une formulation améliorée du modèle d'optimisation de la possibilité forestière.

La formulation classique du modèle d'optimisation de la possibilité forestière (c.-à-d., maximisation du rendement soutenu en fibre) ne considère pas que le consommateur de fibre industriel souhaite maximiser son profit, mais suppose plutôt la consommation totale de l'offre de fibre à chaque période, peu importe le potentiel de création de valeur de celle-ci. Nous étendons la formulation classique du modèle d'optimisation de la possibilité forestière afin de permettre l'anticipation du comportement du consommateur de fibre, augmentant ainsi la probabilité que l'offre de fibre soit entièrement consommée, rétablissant ainsi la validité de l'hypothèse de consommation totale de l'offre de fibre implicite au modèle d'optimisation. Nous modélisons la relation principal-agent entre le gouvernement et l'industrie à l'aide d'une formulation biniveau du modèle d'optimisation, où le niveau supérieur représente le processus de détermination de la possibilité forestière (responsabilité du gouvernement), et le niveau inférieur représente le processus de consommation de la fibre (responsabilité de l'industrie). Nous montrons que la formulation biniveau peut atténuer le risque de ruptures de stock, améliorant ainsi la crédibilité du processus de planification forestière hiérarchique.

Ensemble, le modèle biniveau d'optimisation de la possibilité forestière et la méthodologie que nous avons développée pour résoudre celui-ci à l'optimalité, représentent une alternative techniquement réalisable à la technologie actuellement utilisée. Notre modèle biniveau et le cadre de simulation itérative représentent un pas vers l'avant en matière de technologie de planification forestière axée sur la création de valeur. L'intégration explicite d'objectifs et de contraintes industrielles au processus de planification forestière, dès la détermination de la possibilité forestière, devrait favoriser une collaboration accrue entre les instances gouvernementales et industrielles, permettant ainsi d'exploiter le plein potentiel de création de valeur de la ressource forestière.

Abstract

The hierarchical forest management (HFM) planning process on public land may currently be failing on two levels. At the top level, HFM may not be providing credible assurance of long-term sustainability of timber supply and forest ecosystem integrity. At a lower level, HFM may be failing to fully realise the value-creation potential from timber-harvesting activities by over-constraining the harvest planning problem. These failures can be traced back to unrealistic assumptions implicitly embedded into long-term wood supply optimisation models, which may explain why this problem has received little attention in the literature. We model the hierarchical forest management planning process as a two-phase rolling-horizon iterative *principal-agent* problem, illustrate failure scenarios of the *status quo* planning process, and propose an improved wood supply model formulation.

The classic wood supply optimisation model formulation (i.e. conventional even-flow wood supply maximisation model) does not explicitly consider the profit-maximising behaviour of the industrial fibre consumer, but instead implicitly assumes the complete consumption of the wood supply in every planning period, regardless of fibre type or value creation potential. We extend the *status quo* wood supply model to explicitly anticipate industrial fibre consumption behaviour, thereby improving the likelihood of the wood supply being entirely consumed in the first planning period, thus restoring the validity of the total-consumption assumption that is embedded in the long-term model formulation. We model the *principal-agent* relationship as a bilevel optimisation problem, where the top level (leader) represents the government wood supply planning process, and the lower level (follower) represents the timber consumption process (i.e. value creation network, or VCN). We show that the bilevel model formulation mitigates the risk of long-term wood supply failure and improves the credibility of the wood supply planning process.

The bilevel wood supply model and solution methodology presented here constitute a technically feasible alternative to the wood supply planning technology currently used. Our bilevel model and iterative simulation framework represent a step forward in terms of value-driven forest management planning. Explicit integration of industrial objectives and constraints early on in the wood supply planning process could facilitate government-industry collaboration to realise the full value-creation potential of the public forest resource.

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*To Jessi, without whom this would
not have been nearly as much fun.*

If we are uncritical we shall
always find what we want: we
shall look for, and find,
confirmations, and we shall look
away from, and not see, whatever
might be dangerous to our pet
theories. In this way it is only too
easy to obtain what appears to be
overwhelming evidence in favor of
a theory which, if approached
critically, would have been
refuted.

Karl Popper

The mere formulation of a problem is far more essential than its solution, which may be merely a matter of mathematical or experimental skill. To raise new questions, new possibilities, to regard old problems from a new angle requires creative imagination and marks real advances in science.

Albert Einstein

It is better to be vaguely right than precisely wrong.

H. Wildon Carr

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Preface

This thesis is submitted in partial fulfillment of the requirements for the degree of *philosophiæ doctor* (PhD) in forest science at Université Laval. The work presented here was directed by Professor Luc LeBel, co-directed by Professor Sophie D'Amours, and realised in collaboration with Dr. Mathieu Bouchard.

This document has been prepared as an article insertion thesis, and includes two journal articles for which I have acted as the principal researcher and principal author. My contributions to these papers include the problem definition, literature review, experimental design, mathematical modelling, experimentation, validation, and the writing of the manuscripts. The co-authors contributed to these papers on several fronts, including problem definition, experimental design, mathematical modelling, and writing of manuscripts. Mathieu Bouchard contributed significantly to the design, implementation, and testing of the mathematical models presented in these papers.

The first article, entitled *On the Risk of Systematic Drift Under Incoherent Hierarchical Forest Management Planning*, was co-authored by Luc LeBel, Sophie D'Amours, and Mathieu Bouchard. This paper has been published in the *Canadian Journal of Forest Research*, 43(5):480–492, 2013. The text inserted into this thesis is mostly identical to the published manuscript, although the presentation of tables and figures has been adapted to best fit the layout of this document, and some references and other passages were edited to improve clarity in the context of this thesis.

The second article, entitled *A Bilevel Model Formulation for the Distributed Wood Supply Planning Problem*, was co-authored by Mathieu Bouchard, Luc LeBel, and Sophie D'Amours. This paper has been published by CIRRELT, as research paper CIRRELT-2015-06. The text inserted into this thesis is mostly identical to the submitted manuscript, although the presentation of tables and figures has been adapted to best fit the layout of this document, and some references and other passages were edited to improve clarity in the context of this thesis.

Introduction

This introductory chapter provides some background information on key topics and describes the motivation for this research project. The final section describes the structure of the remainder of this thesis.

Background

This thesis is about forest management. More specifically, this thesis looks closely at the use of wood supply models by government as a basis for allocating industrial timber licences on public forest land in Canada. We describe the state of the art in wood supply modelling, describe circumstances under which the standard wood supply optimisation model may fail to support sustainable forest management, and propose extensions to the standard model that reduce the risk of wood supply failures.

This section provides some background information, relating key topics in forest management planning, game theory, and operations research. To introduce these topics, and to explain their relationship to each other, let us work backwards from the main contribution of this research project.

Hierarchical Forest Management Planning

The main contribution of this research is a bilevel extension to the classic wood supply optimisation model. The need for such an extension, and the justification for a bilevel modelling approach, stems from the distributed nature of the wood supply planning problem on public land.

In Canada, and in other jurisdictions, government stewards manage public forest on behalf of the people, who ultimately own the land. Civil servants are responsible for a certain upstream portion of the planning process, which they eventually hand off to downstream industrial fibre consumers. The decoupling point between government and industry, although subject to variation among provincial jurisdictions, is closely linked to a timber licence contract binding both parties.

Thus, we can describe wood supply planning on public forest land as a distributed planning problem (Schneeweiß, 2003), with the upper level planning controlled by government and the lower level controlled by industrial fibre consumers. More specifically, it is an instance of the *principal-agent* problem (Laffont and Martimort, 2002), which has its roots in game theory. From an operations research perspective, bilevel programming subsumes the principal-agent problem (Colson et al., 2007), hence the bilevel formulation of our extended wood supply model formulation.

In 1995, the Canadian Council of Forest Ministers (CCFM) adopted a framework of six criteria and indicators to guide the implementation of sustainable forest management (SFM) on public land in

Canada (Canadian Council of Forest Ministers, 1995, 2003). In summary, these indicators relate to (1) biodiversity, (2) ecosystem condition and productivity, (3) soil and water, (4) role in global ecological cycles, (5) economic and social benefits, and (6) social responsibility.

In many jurisdictions, sustainable forest management is implemented using a hierarchical planning framework (Tittler et al., 2001). It has been common practice for several decades that planning tools and forestry decision support systems be designed around the hierarchical planning paradigm (Weintraub and Cholak, 1991; Weintraub et al., 2007). Although it is common in forest management, the notion of hierarchical planning was first formally developed in a manufacturing context (Bitran and Hax, 1977). Hierarchical planning necessarily includes both coherent linkages to lower levels and effective feedback loops to higher levels. In its original manufacturing context, hierarchical planning was typically implemented within a centralised decision-making environment. Under centralised hierarchical planning, there is clear incentive to implement proper linkages and feedback loops. In a distributed decision-making environment, where independent agents with imperfectly aligned goals are responsible for different levels of the planning hierarchy, these incentives may be weakened or absent, as is the case for hierarchical forest management planning on public land.

Despite the widespread and long-standing use of hierarchical planning as a conceptual framework for forest management planning (Barros and Weintraub, 1982; Gunn and Rai, 1987; Weintraub and Davis, 1996), the linkages between long- and short-term forest management planning remain weak, and adaptive feedback loops are essentially non-existent. These deficiencies may compromise the effectiveness of the planning process in achieving its stated goals.

Weak linkages are related to the distributed nature of the decision-making environment. Furthermore, the government’s effective planning horizon is likely much shorter than its nominal planning horizon, which may further compromise the rational deployment of the hierarchical planning paradigm in the context of sustainable forest management on public land. We can gain further insight into these issues by analysing the relationships that bind these actors from a game-theoretic perspective.

Agency Theory

Thus far, we have presented three actors involved in SFM, which we can simplistically, for the sake of discussion, refer to as *public* (i.e. owners of the public forest land), *government* (i.e. stewards with legal mandate to sustainably manage the forest resource on behalf of the public), and *industry* (i.e. consumers of fibre from public forest). We can use principal-agent theory to describe the relationship between these three actors, and help predict future behaviour under a range of hypothetical scenarios.

Agency theory is a branch of game theory that studies the relationship between two parties, where one party (the *principal*) contracts a second party (the *agent*) to carry out some task on his behalf (Laffont and Martimort, 2002). The *principal-agent problem* occurs in the presence of *imperfectly aligned interests* and *information asymmetry*. Imperfectly aligned interests are ubiquitous—interests are almost never perfectly aligned. Information asymmetry refers to the impossibility of the principal completely observing the agent’s actions. The agent, whom we assume is self-interested and rational, will always exploit the information asymmetry to further his interests.

The principal-agent problem aptly describes the relationship between the public (principal) and the government (agent) in a forest management context. The public has delegated the responsibility of

implementing sustainable forest management to the government. However, the technical complexity of forest management and the limited transparency of government activities make it impossible for the principal to fully observe the agent. Furthermore, the agent (i.e. government) is at least partly motivated by political factors, which would provide him with the incentive to compromise long-term sustainability in favour of short-term objectives. The public has no obvious means of detecting when and to what extent the government may compromise its sustainable forest management duties for political reasons. This principal-agent relationship, between the public and government stewards to whom the responsibility to sustainably manage public forests has been delegated, has not been documented in the literature. The very nature of this relationship suggests that the government is unlikely to ever fully realise its mandate as steward of our public forests, which may compromise or impede our ability to achieve sustainable forest management objectives. At the very least, the public should be made aware of the extent of the problem, if any, and given the opportunity to implement appropriate mitigatory measures to limit the effects of this potential prejudice. This problem could potentially be modelled as an optimal contract design problem, based on the idea that government behaviour can, to a certain extent, be controlled by judicious use of penalties and incentives. Although we will not further expound these ideas in the context of this thesis, this remains an important topic that deserves further study.

The principal-agent problem also aptly describes the relationship between government and industrial fibre consumers. Note that in this case, we assume that government is acting as a perfect agent on behalf of the people (i.e. we are ignoring the political component described earlier, and assuming that the government is purely motivated by its duty to sustainably manage the public forest on behalf of the public). Hereinafter, the reader may assume this second principal-agent interpretation (i.e. the government as principal, and industrial fibre consumer as agent), unless otherwise specified. Bogle and van Kooten (2012) use the principal-agent paradigm to model the relationship binding government and industrial fibre consumers, although their analysis is confined to modelling government incentive policy in the context of liquidating dead wood affected by the recent pine beetle crisis in British Columbia.

The principal is responsible for implementing sustainable forest management, which he does by setting policies. However, the principal cannot (due to lack of infrastructure) and will not (due to lack of motivation) go so far as to harvest trees and transform the fibre into valuable forest products. This is why he contracts the agent. The contract binding the agent to the principal essentially sets an upper bound on the volume of fibre the agent may extract from the forest (i.e. annual allowable cut, or AAC), along with policy constraining the execution of harvesting activities.

Annual allowable cut (AAC) is determined by the principal using a complex model (hereinafter referred to as the *wood supply model*), that simulates an alternating sequence of harvesting activity and forest growth. Time is typically discretised into planning periods, which typically correspond to the established rolling-horizon replanning cycle length (e.g. 5-year planning periods are common). Wood supply models therefore output a vector of predicted future forest states.

The wood supply model, which is typically deterministic, can *only* be relied upon to output an accurate vector of future forest states if both future harvesting activity and future forest growth are known with certainty. This condition is never respected in practice.

Although it is impractical for the principal to anticipate wood supply model response to all possible

combinations of input data error, it is possible to empirically test model sensitivity to a limited number of input data error scenarios. However, this can quickly become a complex and time-consuming endeavour. There is a growing body of research examining the impact of various sources of forest growth model uncertainty (including natural disturbances) on deterministic wood supply model output (Ouhimmou et al., 2010; Dhital et al., 2013; Boychuk and Martell, 1996; Yousefpour et al., 2012), however very little attention has been paid to the way we model the sequence of future harvesting activity.

By definition, deterministic wood supply models implicitly assume that the location, timing, and nature of all future harvesting activity are known with certainty. Harvesting affects the state of the residual forest, which in turn affects forest growth for upcoming periods, which affects the future availability of stands for harvesting. Thus harvest and growth components of the model are closely linked. A small change to the simulated vector of harvest activity can potentially affect predictions of both growth and harvesting in subsequent periods. Harvesting treatment location and times are typically modelled as decision variables in the wood supply model, which implies that the vector of harvest activity output from the wood supply model is a good approximation of future harvest activity. This implied statement is not supported by historical data.

If one compares past AAC and harvest levels in Canada, it is easy to verify that wood supply models used in practice are systematically over-estimating the level of harvest activity (Canadian Council of Forest Ministers, 2005). Furthermore, the bias is skewed in favour of softwood fibre. For example, only 80% of softwood AAC and 45% of hardwood AAC were consumed¹ in Canada from 1990 to 2012 (National Forestry Database, 2014).

This species-skewed negative bias, despite being common knowledge, has been largely ignored by both government and industry planners for several decades. Furthermore, the classic wood supply model fails to account for the fact that it is embedded in a rolling-horizon replanning process (i.e. only a species-skewed subset of the first period of the optimal solution is ever implemented, with no explicit anticipation of upcoming planning cycles). We cannot properly test the sustainability of the wood supply planning process without accounting for both the species-skewed consumption bias and the rolling-horizon replanning context.

One of the contributions of this research project is the development of a modelling framework that can simulate multiple two-stage interactions between principal and agent (i.e. in the first stage the principal allocates the timber licence, in the second stage the agent consumes a profit-maximising subset of fibre allocation). After simulating two-stage principal-agent interaction, our framework simulates rolling the planning horizon forward one period of forest growth, using the yield curves from the wood supply model. This principal-agent-growth sequence can be repeated indefinitely, potentially providing heretofore unavailable insight into the long-term impact of various forest management policy options.

To our knowledge, no two-stage rolling-horizon wood-supply simulation has been published to date. A one-stage rolling-horizon wood supply simulation is described in Daugherty (1991), however he assumes perfect implementation of the first period solution to the wood supply optimisation problem.

We use our framework to simulate that the aforementioned consumption bias can, in fact, induce future wood supply failures after several rolling-horizon replanning cycles. These wood supply failures are

¹See Table 1.1 and Figure 2.1 for a more detailed representation of this data.

undetectable using the standard wood supply planning process, because it does not account for the industrial fibre consumption bias. Later, we also show that we can mitigate the risk of wood supply failures by more accurately predicting harvest levels in the wood supply model. To achieve this, we must first examine the root cause the problematic consumption bias.

By definition, AAC only specifies an upper bound on harvest volume. The agent is free to consume any profit-maximising subset of allocated fibre. The agent maximises his profit, subject to resource and market constraints. Thus, the difference between AAC and actual fibre consumption represents fibre that is either impossible or unprofitable to consume. Ignoring this bias compromises the rational nature of the modelling exercise, and, by extension, compromises the credibility of the sustainable forest management process. In other words, the principal is not currently using all the information at his disposal to anticipate and avoid potential wood supply failures, and in this respect he is failing in his duty as steward of the public forest resource.

The obvious solution to this problem would be to eliminate the consumption bias in the wood supply model. To achieve this, we must somehow *filter out* the undesirable fibre (i.e. the subset of AAC that the agent will not willingly consume). This implies extending the wood supply model to anticipate industrial fibre-consumption behaviour.

Bilevel Optimisation

From an operations research perspective, our principal-agent problem can be formulated as an optimal contract design problem. The optimal contract (from the perspective of the principal) would maximise (species-wise) fibre allocation that the agent will consume entirely. Anticipating profit-maximising agent behaviour implies a second optimisation model, which, when embedded within the existing wood supply model as a constraint, induces a bilevel structure (Colson et al., 2007). Even the simplest bilevel problems are known to be $\mathcal{NP}$ -hard, and our problem is rather complex. Furthermore, many bilevel problem instances (including our bilevel wood supply problem) feature non-convex solution spaces², which makes them even more difficult to solve to global optimality.

Examples of bilevel programming in the forestry literature are rare. Bogle and van Kooten (2012) recently used a bilevel programming approach to model the principal-agent relationship between government and timber licencees in areas affected by mountain pine beetle outbreaks in British Columbia. Their analysis is limited to this specific context, and their mathematical model and test dataset³ are simpler than the more general case presented here. They assume fixed exogenous unit values of timber, whereas we dynamically generate timber unit value in our agent optimisation sub-model⁴. They implement a grid search algorithm (GSA) first described in Bard et al. (2000) to solve the bilevel problem. The GSA does not guarantee convergence on global optimal solutions, and the computational effort required to solve the problem grows geometrically with the number of upper-level decision variables⁵.

²See §2.2.1.4 for a more elaborate discussion of solution space convexity in the context of our bilevel wood supply model formulation.

³Bogle and van Kooten (2012) test their bilevel model on a small synthetic dataset, made up of 50 even-sized stands in two stand types.

⁴We assume exogenous unit values for a range of value-added products that are generated from our simulated value-creation network (e.g. lumber, paper, chips, etc.).

⁵Bard et al. (2000) reports test results for an instance with only two upper-level variables.

Motivation

Personal motivation for this research project stems from first-hand professional experience. After working as a consulting forester and expert wood supply analyst, for both industrial and government clients, it became clear to this researcher that the forest management planning process on Canadian public land was failing to provide credible evidence of sustainability. Conversations with operations foresters provided anecdotal confirmation that mixed-wood stands were systematically under-represented in annual harvest plans, as these tended to yield more low-quality hardwood fibre than the local market could absorb. Although such forms of high-grading are technically permitted, to a certain extent, by current regulations in most jurisdictions, they clearly undermine the intent of sustainable forest management policy. Although the situation is common knowledge amongst both government and company foresters, this topic is rarely discussed. The government of Quebec recently published a report confirming the perpetuation of this wood supply high-grading trend for the 2008–2013 period ([Bureau du forestier en chef, 2014](#)).

Inasmuch as the current regulatory framework provides valuable short-term flexibility for both government and industry planners, there is limited motivation for wood supply planning practitioners to hasten the advent of a new generation of wood supply planning models, particularly if the primary beneficiaries of the *status quo* system suspect—and they would likely be correct—that further constraining the planning process in the name of sustainability may well reduce current fibre allocation levels and short-term revenue-generating opportunities.

The FORAC research consortium provided much of the funding and resources for this project. The mission of the consortium is to promote the design and management of innovative value creation networks for a sustainable forest products industry. Ensuring a stable long-term supply of potentially valuable fibre is fundamental to any sustainable forest sector development strategy. Furthermore, FORAC has always promoted a culture of collaboration between industry and government stakeholders. By adopting a game-theoretic approach to wood supply modelling, we hope this research can help pave the way for a new era of value-driven collaborative fibre supply planning. Both government and industry stakeholders can potentially benefit from the work presented here.

As the steward of the public forest, government has the responsibility to use the best available data and technology in support of the wood supply planning process. The bilevel wood supply model and solution methodology presented here constitute a technically feasible and conceptually superior alternative to the wood supply planning technology currently used by government.

Together, our bilevel model and iterative simulation framework represent a step forward in terms of value-driven forest management planning technology. Explicit integration of industrial value-creation network objectives and constraints early on in the wood supply planning process will facilitate government-industry collaboration to realise full value-creation potential from the public forest resource.

Problem Statement and Objectives

For decades, the distributed wood supply planning problem has been solved using an essentially unchanged optimisation model formulation (i.e. the *classic* model), which systematically over-estimates industrial fibre consumption. Although this bias is well documented, the potential negative effects of

incoherence between planned and executed harvesting activities on the credibility of the wood supply planning process has not been tested.

The objective of this thesis is twofold: *(a)* to confirm that the fibre consumption bias embedded in the classic model jeopardises the credibility of the distributed wood supply planning process, and *(b)* to propose an extended model formulation that can restore credibility to the distributed wood supply planning process.

Thesis Structure

The next three chapters are a collection of articles, which either have been, or will be, submitted for publication in peer-reviewed scientific journals.

Chapter 1 introduces the distributed wood supply planning problem, including the mathematical formulations for the classic wood supply optimisation model and our agent-anticipation optimisation model. We describe a two-stage rolling-horizon replanning simulation framework, which we use to simulate the effect of the species-skewed fibre consumption bias on wood supply sustainability. We conclude this chapter by conjecturing that the classic wood supply model could be improved if it were extended to explicitly anticipate industrial fibre consumption behaviour. The main contributions of this chapter are the mathematical formulation of the distributed wood supply planning problem in terms of agency theory, the implementation of a wood supply replanning simulation framework, and the presentation of a realistic case study showing that the current wood supply planning process may be failing to predict certain types of wood supply failures in the presence of a species-skewed fibre consumption bias.

Chapter 2 further explores the source of antagonism between the principal and the agent, and presents a mathematical formulation and solution methodology for the bilevel distributed wood supply planning problem. We compare performance of classic and bilevel models using our two-stage rolling-horizon replanning simulation framework. We conclude that the bilevel model successfully mitigates the risk of wood supply failure, although there is some residual instability in long-term wood supply. The main contributions of this chapter are the mathematical formulation of the bilevel wood supply problem, the development of a solution methodology to solve the bilevel problem to optimality, and the presentation of a realistic case study showing that our bilevel model formulation can mitigate the risk of wood supply failure identified in Chapter 1.

Chapter 3 explores a number of potential wood supply policy options using a combination of our replanning simulation framework and our bilevel wood supply model. We test the effect of proportion of AAC attributed, planning horizon length, tightness of even-flow constraints, stochastic variation of fibre consumption volume, and collaborative agent behaviour on stability and magnitude of the wood supply. The main contribution of this chapter is to demonstrate that the replanning simulation framework and bilevel wood supply model developed in previous chapters can be used in tandem to provide heretofore unavailable insight into the distributed wood supply planning problem policy space.

In the final chapter, we present general conclusions from this research project and describe several promising directions for further research.

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Chapter 1

On Risk of Systematic Drift Under Incoherent Hierarchical Forest Management Planning

This chapter presents an article entitled *On the Risk of Systematic Drift Under Incoherent Hierarchical Forest Management Planning*, published in the *Canadian Journal of Forest Research*, 43(5):480–492, 2013. The authors are Gregory Paradis, Luc LeBel, Sophie D’Amours, and Mathieu Bouchard.

Résumé

En théorie, les systèmes de planification hiérarchiques intègrent des mécanismes de liaison efficaces, assurant ainsi une désagrégation cohérente de l'attribution de volumes aux usines lors de la planification détaillée des opérations de récolte. En pratique, les mécanismes de liaison entre la planification à long- et à court-terme peuvent être inefficaces, menant donc à des plans incohérents sur le plan du volume récolté, de la représentation des essences, et du potentiel de création de valeur des billes livrées aux usines. Cette incohérence entre la planification et l'exécution de la récolte peut induire une dérive systématique de l'état du système forestier (c.-à-d. divergence entre les trajectoires projetées et réalisées), compromettant donc la crédibilité et la performance du processus de planification de l'aménagement forestier. Nous décrivons le processus de planification forestière en matière de la théorie du jeu, et nous simulons l'interaction entre la planification de l'approvisionnement et la planification de la récolte à l'aide d'un modèle itératif à deux étapes. Nous présentons une étude de cas, et nous montrons l'existence d'un effet de dérive systématique, que nous attribuons à l'inefficacité des mécanismes reliant la planification à long- et à court-terme. Nous montrons qu'il est possible d'améliorer la cohérence des plans en manipulant les mécanismes de liaison, et proposons des avenues de recherche futures pouvant améliorer la performance du processus de planification hiérarchique à l'aide de nouvelles formulations de modèles basés sur la théorie du jeu.

Abstract

In theory, hierarchical forest management planning should be implemented with effective linkages between planning levels, to ensure coherent disaggregation of long-term wood supply allocation as input for short-term demand-driven harvest planning. In practice, these linkages may be ineffective, and solutions produced may be incoherent in terms of both the volume and the value creation potential of harvested timber. The systematic incoherence between planned and implemented forest management activities may induce drift of the forest system state (i.e. divergence of planned and actual system state trajectories), thus compromising the credibility and the performance of the sustainable forest management planning process. We describe hierarchical forest management from a game-theoretic perspective, and present an iterative two-stage model simulating interaction between long- and short-term planning processes. Using an illustrative case study, we confirm the existence of a systematic drift effect, which we attribute to the ineffective linkages between long- and short-term planning. In several simulated scenarios, the planning process fails to ensure long-term wood supply sustainability, it fails to reliably meet industrial fibre demand over time, and it exacerbates incoherence between wood supply and fibre demand after repeated rolling-horizon planning cycles. We show that manipulating linkages between long- and short-term planning processes can reduce incoherence. We also describe future work on game-theoretic wood supply model formulations that may improve hierarchical planning process performance.

1.1 Introduction

Hierarchical forest management (HFM) on public land may currently be failing on two levels. At the top level, HFM may not be providing credible assurance of long-term sustainability of timber supply and forest ecosystem integrity. At a lower level, HFM may be failing to fully realise value-creation potential from timber-harvesting activities through over-constraining the harvest planning problem. These failures can be traced back to unrealistic assumptions regarding long-term timber consumption behaviour. These problematic assumptions are implicitly embedded into the optimisation models used to determine annual allowable cut (AAC) levels, which may explain why this problem has received little attention in the literature.

Operations research (OR) has been used extensively in forest resource planning in the past several decades (Weintraub and Romero, 2006). Many forestry applications of OR are developed in a hierarchical planning framework (Weintraub and Cholak, 1991; Weintraub et al., 2007). Justification for hierarchical decomposition of planning problems is based on problem size, the correspondence of solution decision levels to organisational structure, and the simplicity of the planning approach in a realistic forest planning context (Weintraub and Cholak, 1991). However, decomposing a planning problem into a hierarchy of planning levels introduces risk of distorting global solution space. Thus, hierarchical planning systems may produce globally infeasible or sub-optimal solutions.

Forestry applications of OR in the literature have tended to focus on either forest management issues *viz.* sustainability of timber and non-timber forest values, land use allocation, silvicultural strategies, etc. (Bettinger and Chung, 2004; Gunn, 2007), or on forest products industry supply-chain planning (D’Amours et al., 2008; Carlsson et al., 2009). Nevertheless, obvious interdependencies exist between long-term forest management planning and short-term forest industry supply chain planning. In particular, the coherence of long- and short-term planning is highly dependent on assumptions regarding the relationship between supply and demand, and the assumptions regarding the spatial distribution of harvesting activities. Relatively little research has been published on the integration of forest management planning and forest products industry supply chain planning, although several interesting research topics in this area are described in Nelson (2003) and Weintraub and Murray (2006). Recent publications (Troncoso et al., 2011; Gunn, 2009; Martin, 2013; Bogle, 2012) indicate some recent and ongoing work integrating strategic wood supply analysis and supply chain planning.

We believe improved integration of long-term wood supply planning and short-term fibre demand planning is key to improving the performance and credibility of the hierarchical forest management planning process. Throughout this paper, we use the terms *performance* and *credibility* in reference to the hierarchical forest management planning process. We use the term *performance* to describe the realisation of value-creation potential, and the term *credibility* to describe the extent to which assurance of long-term sustainability is supported by currently available information (i.e. believable). Usage of the term *credibility* in this context is coherent with its usage in Daugherty (1991) and Davis et al. (2001a).

A key component of the hierarchical planning paradigm is the existence of effective linkages between planning levels (Bitran and Hax, 1977). These linkages should ensure the coherence between hierarchical planning levels, and limit the risk of inducing distortions in solution space. In practice, the execution of a hierarchical forest planning process tends to be distributed across teams of planners with stable,

but divergent, planning objectives. Planning levels are typically linked by several constraints, including allowable cut levels and operational harvesting regulations. Furthermore, strategic plans are reviewed every few years. Nonetheless, existing linkages combined with a regular replanning cycle may not fully correct biases induced by systematic differences between planned and implemented activities. More specifically, improperly designed linkages between planning levels may be distorting solutions, producing long-term projections that are systematically biased relative to as-implemented course of action. The long-term impact, on forest system state, of systematic bias between planned and implemented forest harvesting activities is largely unknown, and has received relatively little attention in the literature (Bettinger and Chung, 2004; Weintraub and Romero, 2006; Pittman et al., 2007).

Daugherty (1991) explored *dynamic inconsistency* in forest planning using an iterative linear programming simulation of sequential replanning, assuming exact implementation of first-period optimal solutions at each iteration, discounted profit-maximisation objective function, and even-flow harvest volume constraints. He argued that maximising discounted profit in conjunction with even-flow constraints creates dynamically inconsistent long-term model formulations, which are not a credible basis for near-term policy decisions, and notes that dynamic inconsistency in long-term models may be avoidable using alternative objective function formulations or different harvest flow control approaches. McQuillan (1986) makes similar observations, also based on long-term model formulations featuring discounted profit-maximisation objective functions and even-flow harvest volume constraints.

Strategic forest planning on public forest tends to focus on establishing timber resource allocation such that long-term sustainability constraints are respected. An important component of strategic forest planning in several jurisdictions is the annual allowable cut level calculation, often based on complex simulations of forest growth and projected management activity levels. The current purpose of AAC determination within the forest management planning process is twofold: to ensure long-term sustainability of the timber supply, and to demonstrate long-term integrity of the forest ecosystem.

It is common practice to maximise the AAC through optimal scheduling of forest management activities, using an aggregate representation of the forest landscape. In this context, the validity of AAC projections depends on the validity of model assumptions regarding current and future harvest levels, spatial distribution of harvest areas within the forest landscape, and silviculture activity levels. These long-term projections are only valid under specific conditions, *viz.* the model is dynamically consistent, the first period of the optimal AAC solution is implemented exactly and deterministic assumptions regarding forest inventory, forest growth, and silviculture treatment effects hold true. Thus, the credibility of long-term AAC solutions as a basis for short-term policy decisions depends on the extent to which these conditions are respected. If executed forest management activities differ significantly from simulated activities, then the AAC modelling process is unlikely to provide credible information regarding the sustainability of wood supply or the integrity of forest ecosystems. Regardless of credibility level, AAC solutions always constrain short-term harvest planning solution space, thereby potentially limiting short-term value-creation opportunities induced by timber harvesting activities. Constraining short-term value-creation opportunities on the basis of low-credibility long-term projections is of questionable merit, and further compromises the rationality of the forest management policy decision-making process.

At the short-term harvest planning level, AAC is typically interpreted as an upper bound on the allowable harvest level, *viz.* short-term harvest plans may include any subset of AAC. This interpretation of AAC does not respect one of the solution validity conditions described earlier, to wit, that the optimal

AAC solution is implemented exactly. In practice, implemented harvest level may be significantly lower than AAC if aggregate timber demand is lower than AAC, composition of harvested stands in AAC model is incoherent with timber demand (e.g. no local market for certain species-product combinations), or portions of projected AAC harvest volume are economically or operationally unattractive. Hence, long-term planning models tend to systematically over-estimate harvest activity levels, and short-term harvesting plans tend to distort long-term planning assumptions in favour of demand satisfaction and cost minimisation (or profit maximisation).

If left unresolved, this biased incoherence between planned and implemented harvest and silviculture activity levels may induce systematic drift of the forest system state (i.e. long term trajectory of system state systematically differs from projected system state) after several iterations of implementing incoherent strategic and operational plans, thus reducing the credibility of the long-term planning effort and potentially over-constraining short-term value-creation opportunity. This *systematic drift effect* (SDE), induced by the incoherence between hierarchical forest planning levels, is not discussed in the literature. The objectives of this paper are: to examine the impact of *status quo* hierarchical decomposition of forest management planning on the performance and credibility of the overall planning process using the SDE concept as a basis for evaluation, to identify weaknesses in the linkages between planning levels, to simulate SDE using an illustrative case study, and to propose potential improvements to the planning process.

1.2 Problem Definition

In many jurisdictions, responsibility for long-term and short-term planning on public forest is distributed across government and industrial agents with divergent planning objectives. Long-term wood supply planning is typically implemented first by government planners, followed by short-term demand-driven harvest planning by industry planners. We model this distributed hierarchical planning process, with the purpose of approximating the net effect of a two-stage planning hierarchy in a rolling-horizon replanning context. More specifically, we examine the credibility of long-term planning projections under varying levels of incoherence between timber supply and demand.

We present several assumptions about the hierarchical forest management planning process, which reflect our understanding of the current situation in jurisdictions where public forests are managed by government stewards for production of timber resources:

- long-term planning is the responsibility of government planners (stewards of public land);
- short-term harvest planning is the responsibility of forest industry planners;
- the objective of long-term planning is to ensure the long-term sustainability of timber supply and to demonstrate the long-term integrity of forest ecosystems;
- output from the long-term planning process includes species-wise AAC (i.e. upper bounds on species-wise harvest levels), and treatment-wise minimum area thresholds on annual silviculture treatments;
- the long-term AAC model is formulated as an aspatial (i.e. stratum-based) species-wise even-flow harvest maximisation problem;
- the objective of short-term harvest planning is to satisfy timber demand at maximum profit.

We present an illustrative case study showing that incoherence between long- and short-term forest

management planning can negatively impact credibility and performance of the planning process. In a worst-case scenario, incoherent planning may induce sudden (unforeseen) wood supply failure. Alternatively, incoherent hierarchical planning may over-constrain the short-term planning process, impeding the realisation of full value-creation potential. Realisation of full sustainable value-creation potential may be key to navigating periods of economic crisis in the forest industry. We also show that extending the *status quo* long-term planning model to explicitly anticipate certain aspects of the short-term planning process (i.e. fibre demand) can improve credibility and performance of the hierarchical planning process.

The remaining subsections of §1.2 describe *status quo* long- and short-term planning processes, and present the distributed hierarchical forest management problem from a game-theoretic perspective.

1.2.1 Long-term Planning Process

Strategic planning on public forest land traditionally focuses on establishing forest resource allocation (with emphasis on the timber resource) such that long-term sustainability objectives are respected (Gunn, 2007; Davis et al., 2001b). An important component of strategic forest planning is the annual allowable cut level calculation, based on complex simulations of forest growth and projected management activities, over a long planning horizon (e.g. 150 years). AAC is often defined as maximum even-flow timber harvest level on a given land base. Despite its widespread use, this model formulation may induce incoherence between stated management goals and planning process outcome, especially when long-term planning is integrated with supply chain planning (Gunn, 2009). The purpose of AAC calculation in the planning process is to ensure long-term sustainability of the timber resource, however sustainability is not necessarily ensured by a non-declining yield AAC model formulation (Gunn, 2009). Actual harvest levels may fluctuate significantly from one planning period to another, and tend to be significantly lower than harvest levels simulated by AAC models, therefore a non-declining yield AAC model is not likely to provide a credible projection of either future harvest levels or future forest system states. Furthermore, maximising even-flow AAC level through optimal allocation of harvesting activities over time does not guarantee efficient tradeoff between availability of timber and other values (i.e. economic, social, ecological, etc.).

In fact, AAC maximisation tends to rely on an *allowable cut effect* (ACE) to justify potentially unsustainable AAC simulations (Luckert and Haley, 1995). ACE can be defined as the allocation of anticipated future timber yields to current harvest levels, and is used to justify short-term increases in AAC. Depending on initial age-class structure of the forest under management, ACE may also induce AAC models to prescribe large short-term investments to establish high-yield plantations to obtain marginal long-term increases in (simulated) available timber at a critical period. These silvicultural investments may be difficult to justify economically, especially if the anticipated harvest level is significantly lower than the AAC.

The validity of AAC projections depends on the validity of underlying assumptions regarding harvest levels, distribution of harvest areas within the forest landscape, and other silviculture activity levels (e.g. establishment of high-yield plantations, stand improvement treatments, etc.). AAC scenarios accurately predict future forest conditions only if projected harvest levels represent an *unbiased* approximation of future management activity levels.

In practice, AAC constitutes a starting point for a political negotiation process between stewards of public forest and industrial consumers of timber from public forest. Aggregation of forest inventory data into stratum-based representations (i.e. failure to account for spatially-explicit operational constraints associated with harvest block layout and landscape planning), lack of integration with industrial fibre demand planning, and use of an ACE as a basis for determining short-term silviculture and harvesting policy tends to produce AAC estimations that are systematically higher than observed harvest levels (Canadian Council of Forest Ministers, 2005). Although spatial disaggregation and additional planning constraints at lower planning levels can restore feasibility, harvest levels tend to drop on the order of 20%, relative to AAC, following spatially explicit feasible allocation of harvest operations on the landscape (Walters and Cox, 2001). It is not clear how this feasibility-restoration process impacts global optimality, nor is it clear that strategic planning objectives (i.e. sustainability) are respected following spatially explicit allocation of harvest operations. Species-wise AAC levels are linked (due to harvesting of multiple species in mixed stands), and sustainability is not guaranteed for all possible combinations of species-wise harvest levels, even if AAC upper-bound harvest constraints are respected. Upper-bound harvest level constraints (as defined by AAC) may be neither sufficient nor necessary to ensure long-term sustainability of timber harvest or ecological integrity.

Table 1.1 compares recent AAC and harvest volume, by province, in Canada. Actual harvest levels are in many cases significantly lower than projected harvest levels, where actual timber demand differs significantly from projected timber demand. Also note that the proportion of hardwood AAC harvested is systematically lower than the proportion of softwood AAC harvested, indicating that the AAC determination process simulates harvest intensity and distribution that is unrepresentative of actual harvesting activities. Lower historical hardwood harvesting levels are likely linked to bounded economic viability of hardwood fibre utilisation opportunities. Factors contributing to limit hardwood harvesting value-creation opportunities include differences between supply and demand (species distribution, quality distribution), high unit cost of partial cut harvesting operations, and increased dispersion and fragmentation of harvest blocks due to spatial layout of partial cutting blocks (proportionally higher setup costs, road construction cost, transportation cost). In some areas, only the most valuable hardwood logs (e.g. veneer grade) are economically viable given current local market conditions, while other hardwood co-products are harvested at a net loss.

Table 1.1 – Comparison of recent AAC and harvested volume, by species (softwood, hardwood) and Canadian province (Canadian Council of Forest Ministers, 2005)

	<i>Units</i>	<i>NL</i>	<i>PE</i>	<i>NS</i>	<i>NB</i>	<i>QC</i>	<i>ON</i>	<i>MB</i>	<i>SK</i>	<i>AB</i>	<i>BC</i>
Softwood											
Harvest	thousand m ³	1958	431	562	3712	24702	16568	1563	2259	12090	64941
AAC	thousand m ³	2573	300	865	3686	31602	22887	5639	3864	13670	66653
Deviation	thousand m ³	-615	+131	-303	+26	-6900	-6319	-4076	-1605	-1580	-1713
Ratio	%	76	144	65	101	78	72	28	58	88	97
Hardwood											
Harvest	thousand m ³	62	151	99	1457	4132	4406	638	1313	6100	1199
AAC	thousand m ³	N/A	169	455	1633	12084	13058	3261	3244	10210	2843
Deviation	thousand m ³	N/A	-9	-356	-176	-7952	-8652	-2623	-1931	-4110	-1645
Ratio	%	N/A	94	22	89	34	34	20	40	60	42

Furthermore, a significant portion of AAC may be operationally unavailable (i.e. small isolated stands, low volume density, distance from existing road network, timber value, wrong species mix, etc.). Including operationally unavailable timber in the AAC is a source of incoherence between strategic and operational plans. The process of selecting a spatially explicit, feasible, demand-coherent subset of AAC for the first-period harvest schedule may produce a sample of stands that is unrepresentative of assumptions made during AAC simulation (i.e. biased in favour of values that were omitted from AAC simulation, *viz.* proximity of harvest blocks to roads and mills, timber value, specific timber product demand, etc.).

Rolling-horizon replanning of AAC (e.g. typically 5-year planning period) may help counteract some effects of stochastic variations (e.g. market prices) and data error (e.g. growth and yield model error), but will not necessarily correct the effect of systematic incoherence between long-term projections and actual harvest activities. The hierarchical planning process has not been adjusted to account for incoherence between planned and actual activities, despite widespread knowledge of discrepancies between AAC and industrial fibre consumption. Typically, AAC is simply re-calculated at the start of every planning cycle using updated inventory information without any significant changes to model structure or assumptions (i.e. maximise even-flow wood supply, ignoring current and historical fibre consumption).

1.2.2 Short-term Planning Process

The short-term harvest planning process involves decisions regarding the selection and spatial layout of harvest blocks, the allocation of harvesting machinery to harvest blocks, road construction and upgrading, and transportation of logs from harvest blocks to mills. The objective of short-term harvest planning is typically to satisfy mill demand for logs at minimum cost, subject to AAC constraints and various operational constraints from government regulations, *viz.* size and layout of harvesting blocks, and requirement to conduct certain post-harvest silviculture treatments (Epstein et al., 2007).

Complex and irregular solution spaces in large combinatorial optimisation problems (such as the operational forest planning problem) have multiple local optima (Richards and Gunn, 2003; Paradis, 2005), and optimisation model solutions in operational forest management planning (i.e. spatially and temporally explicit harvest and delivery schedules) can be sensitive to small changes in problem formulation and to constraints inherited from upper planning levels. Model response may be difficult to predict if tradeoff relationships between competing objectives (e.g. maximise AAC, minimise harvest cost) are unknown and the hierarchical planning process is implemented through sequential optimisation of nested planning levels, particularly in the absence of thorough sensitivity analysis and adaptive feed-back or feed-forward loops (Gunn, 1991).

In this context, realisation of short-term value-creation potential can be improved pro-actively if the long-term (upstream) planning process anticipates short-term (downstream) planning objectives and constraints (Beaudoin et al., 2008). Using a planning approach designed specifically to coordinate long- and short-term planning, taking into account the distributed nature of the hierarchical forest management planning process, it may be possible to simultaneously improve credibility of long-term plans and short-term realisation of value-creation potential.

1.2.3 The Principal-Agent Problem

Within the context of hierarchical forest management planning, the relationship between long-term planners (i.e. government) and short-term planners (i.e. industry) can be described as an instance of the *principal-agent problem* (Laffont and Martimort, 2002). The *principal* (i.e. government steward of publicly-owned forested land) owns the timber resource, and is responsible for long-term planning. The *agent* (i.e. industrial consumer of timber) pays a stumpage fee to gain access to the timber resource, and is responsible for planning and executing harvesting activities. A contract (i.e. government-regulated hierarchical planning process, culminating in emission of a timber licence) binds the two parties. From the perspective of the principal, the objective of the contract is to control the behaviour of the agent, such that short-term harvesting activities are aligned with long-term planning goals. Under certain supply and demand conditions, *projected* and *actual* agent behaviour may differ significantly, thus the principal may be failing to meet long-term objectives (i.e. credibly demonstrate sustainability of long-term planning).

We model the hierarchical forest management planning process as a principal-agent problem. The model presented in the next section formulates this principal-agent problem as a two-stage rolling-horizon simulation process. We present an illustrative case study which simulates some long-term consequences of the interaction between the principal and the agent, in a *status quo* planning process context.

1.3 Model Formulation

We describe a two-stage rolling-horizon replanning simulation framework. Principal and agent behaviour are modelled as optimisation problems. Objective functions and constraint structures are intentionally designed to be as simple as possible, while capturing the essence of the *status quo* forest management planning process.

We use agent-based representations (Frayret et al., 2007; Forget et al., 2009; Ouhimmou et al., 2009) of long- and short-term planning processes to encapsulate modelling of decision-making behaviour within the hierarchical forest management process. Agent-based design decouples *optimisation model formulation* from *interaction behaviour* of long- and short-term planners, thereby allowing us to test the impact of different long- and short-term planning model formulations on the overall performance of the hierarchical planning process (i.e. iterative simulation of a two-stage planning process). Performance of the hierarchical planning process is evaluated using several scenarios, simulating interaction of various combinations of long- and short-term model formulations to confirm the existence of a systematic drift effect and to determine its potential impact on the outcome of the planning process.

1.3.1 Long-term Planning Model

Long-term planning model formulations can be classified into three groups (Gunn, 2009; Johnson and Scheurman, 1977; Garcia, 1990) according to the nature of the decision variables:

- *Model I*: variables represent a sequence of actions on a given forest unit for the entire planning period.
- *Model II*: variables represent a sequence of actions on an even-aged forest unit from its beginning to the moment when it is cut, or to the moment it dies.
- *Model III*: variables represent individual actions (or groups of few actions) on a given forest unit.

The long-term model formulation is similar to the *Model I* LP formulation. The territory is divided into a set Z of spatial zones. For every zone $i \in Z$ we have a set of prescriptions P_i , where each prescription is a sequence of forest operations to be applied to the stands in the zone for the entire long-term planning horizon.

We formulate the model as follows.

Maximise

$$\sum_{i \in Z} \sum_{k \in P_i} c_{ik} x_{ik} \quad (1.1)$$

subject to

$$\sum_{k \in P_i} x_{ik} = 1, \quad \forall i \in Z \quad (1.2)$$

$$y_o \leq \sum_{i \in Z} \sum_{k \in P_i} \alpha_{ikot} x_{ik} \leq (1 - \varepsilon_o) y_o, \quad \forall o \in O', t \in T \quad (1.3)$$

$$l_{ot} \leq \sum_{i \in Z} \sum_{k \in P_i} \alpha_{ikot} x_{ik} \leq u_{ot}, \quad \forall o \in O, t \in T \quad (1.4)$$

where

Z := set of spatial zones

P_i := set of available prescriptions for zone $i \in Z$

O := set of forest outputs

$O' \subset O$:= set of sustainable forest outputs

T := set of time periods in the planning horizon

ε_o := admissible level of variation on yield of output $o \in O$

α_{ikot} := quantity of output $o \in O$ produced in period $t \in T$ by prescription $k \in P_i$ in zone $i \in Z$

l_{ot} := lower bound on yield of output $o \in O$ in period $t \in T$

u_{ot} := upper bound on yield of output $o \in O$ in period $t \in T$

c_{ik} := global value of including cost and benefits of prescription $k \in P_i$ in zone $i \in Z$

x_{ik} := fraction of zone $i \in Z$ on which prescription $k \in P_i$ is applied

$y_o := \sum_{i \in Z} \sum_{k \in P_i} \alpha_{iko1} x_{ik}$, which corresponds to first-period harvest volume for output $o \in O'$

The set P_i of prescriptions for zone i is not a static set, but rather is generated optimally using a *column generation*¹ scheme (Desaulniers et al., 2005; Dantzig and Wolfe, 1960). Prescriptions therefore are obtained by solving a subproblem for an optimal prescription for a given pricing of the outputs. The subproblem is in essence a *Model III* LP formulation stripped of bounds and even-flow type constraints (equivalent to finding a shortest hyper-path in a hyper-graph), and can be solved by dynamic programming. The set $O' \subset O$ represents the species groups for which we determine AAC and enforce even-flow constraints.

¹Column generation is an approach used to solve large optimisation problems, involving iteratively solving a subproblem (using a subset of the decision variables, which correspond to columns of the LP matrix) and adding new columns based on analysis of subproblem solutions.

Constraint (1.2) requires prescriptions applied to a zone to cover the entire territory. Note that doing nothing for the entire planning horizon is considered a prescription that could generate some outputs (e.g. carbon flows). Constraint (1.3) enforces an even-flow policy on targeted species groups. Constraint (1.4) bounds minimal and maximal periodic yield on the general outputs.

In the above formulation, periodic harvest volume is stabilised using *even-flow* constraints, which is common in practice. The objective of the optimisation problem is to maximise species-wise AAC over a planning horizon made up of 30 five-year planning periods. This corresponds to the *status quo* strategic planning model in many jurisdictions.

1.3.2 Short-term Planning Model

We use two distinct formulations of the short-term planning model, depending on the scenario. The planning horizon is limited to a single 5-year period in both cases. The first model formulation is used to simulate all non-integrated scenarios (i.e. 1.1 through 2.2), and is implemented as a constrained version of the long-term model to simulate consumption of a specified subset of AAC at each planning iteration. The second model formulation is used to simulate the two integrated scenarios (i.e. 3.1 and 3.2), and estimates VCN profit at each planning iteration using a much more complex simulation of fibre consumption behaviour. Both model formulations are described in detail in the following sections.

1.3.2.1 Formulation S1

The first short-term model formulation (henceforth referred to as *formulation S1*) is a modified version of the long-term LP planning model. Equations (1.1) to (1.4) therefore are sufficient to describe short-term model formulation S1.

The planning horizon is reduced to a single period, and two sets of constraints are activated: the first set of constraints forces species-wise harvest level to correspond to industry fibre demand, and the second set of constraints forces silviculture treatment area in the short-term solution to be proportional to area prescribed in the first period of the long-term solution (relative to the proportion of AAC harvested in the short-term plan). Both of these additional constraints are special cases of constraint (1.4).

Note that the solution is not constrained to schedule harvest and silviculture activities from the same forest units that were scheduled in the first period of the long-term solution, however harvest and silviculture decisions are subject to the same treatment operability limits as the long-term model (i.e. same model, with shortened planning horizon and added demand-satisfaction and silviculture treatment area constraints). The short-term problem may occasionally be infeasible for some iterations of certain scenarios. In the event of infeasibility, our column-generation-based solver engine returns a best attempt at restoring feasibility. This best-attempt solution is then implemented by the short-term planning agent. This produces a reasonable solution in the event of infeasibility, allowing a simulation to proceed with remaining planning cycle iterations.

1.3.2.2 Formulation S2

The second short-term model formulation (henceforth referred to as *formulation S2*) is an abstract supply chain problem, which we formulate as follows.

Maximise

$$\sum_{i \in W} c_i x_i \tag{1.5}$$

subject to

$$\sum_{i \in W_i} \alpha_{ij} x_i = s_j, \quad \forall j \in P \tag{1.6}$$

$$l_i \leq x_i \leq u_i, \quad \forall i \in W \tag{1.7}$$

where

c_i := net contribution to profit for one unit of abstract process i

x_i := quantity of abstract process i used

W := set of abstract processes (consume and produce abstract products)

α_{ij} := quantity of abstract product j used (−) or produced (+) by abstract process i

s_j := quantity of abstract product j introduced to (−) or removed from (+) the system

P := set of abstract products (product, period, location)

u_i := lower bound on the quantity of abstract process i

l_i := upper bound on the quantity of abstract process i

The abstract processes represent the transformative capacity of the supply chain. The abstract products can be, at a given location and time, either a physical product or a resource. A physical product can represent any of the product states from the raw log to the final forest product. A resource can represent machines or employees. A process could be, for example, a sawing recipe at a given mill at a certain period requiring a specific kind of log input, some sawing machine utilisation and two hours of manpower to produce a certain type of lumber.

This model is used in our integrated scenarios (3.1 and 3.2). In integrated mode, our solver engine iteratively finds the maximum long-term species-wise even-flow AAC that maximises first-period profit of the value creation network.

1.3.3 Iterative Rolling-horizon Replanning Simulation

We developed an agent-based iterative rolling-horizon replanning simulation platform so that we could simulate the long-term effect of applying the *status quo* hierarchical planning process on public land. The simulation platform uses agent-based representations of long-term planners (i.e. government) and short-term planners (i.e. industry) to simulate the behaviour of major stakeholders in the hierarchical forest management planning process. The platform is implemented as a code module that iteratively communicates with *SilviLab Solver Engine* (SSE) (Simard et al., 2012) and *LogiLab Solver Engine* (LSE) (Jerbi et al., 2012) to generate optimal wood supply solutions (i.e. determine AAC), simulate implementation of short-term planning solutions, age the forest, and roll the planning horizon forward.

The simulation algorithm can be summarised as follows:

1. Solve a basic AAC model (even-flow harvest volume maximisation).
2. Extract first period decisions from the optimal solution, and calculate species-wise AAC and prescribed silviculture treatment levels.

3. Solve the short-term planning model (satisfy industrial fibre demand, subject to species-wise AAC as upper-bound harvest constraints and silviculture treatment area constraints).
4. Simulate implementation of the short-term solution (update post-treatment forest states using transition logic from the long-term wood supply model, simulate species-wise harvested volume output).
5. Simulate rolling horizon forward one period (simulate evolution of forest state using growth and yield curves from long-term wood supply model).

One simulation iteration is equivalent to a single 5-year two-stage planning cycle. During the first stage of each planning cycle, government planners determine an AAC solution, which is used to constrain harvest and silviculture activities for the upcoming 5-year period (steps 1 and 2 of the algorithm). During the second stage of each planning cycle, industry planners harvest and process a subset of the AAC (steps 3 and 4 of the algorithm). Step 5 of the algorithm is necessary to accurately simulate the passage of time between planning cycle iterations.

The long-term planning agent behaviour is modelled using the optimisation model formulation described in §1.3.1. Short-term planning agent behaviour is modelled using one of the short-term optimisation model formulations described in §1.3.2 (S1 or S2, depending on scenario). Simulation of the implementation of short-term solution uses treatment response and forest growth curves from the long-term wood supply model. The model is deterministic, *viz.* it is possible to simulate perfect implementation of a long-term plan, which we present in scenario 1.1.

We repeat steps 1–5 for as many iterations as there are planning periods in the long-term planning horizon (30 iterations, for our test model). The iterative simulation runs locally on an Apple MacBook Pro² notebook computer, and communicates with a remote SSE server to generate and download optimal solutions. The locally-executed simulation platform is coded in the Python programming language (version 2.7). SSE is coded in C++, and calls ILOG CPLEX (version 12.3) to solve LP sub-problems.

1.4 Dataset Description

This section describes test datasets used for the two-stage iterative simulation scenarios. Simulation of long-term planning is based on existing forest data compiled for the purpose of wood supply analysis. Simulation of the value creation network is based on data from a variety of sources including government databases and industry records. Cost coefficients and unit conversions were subject to expert professional review, and are generally representative of current values in Quebec, Canada.

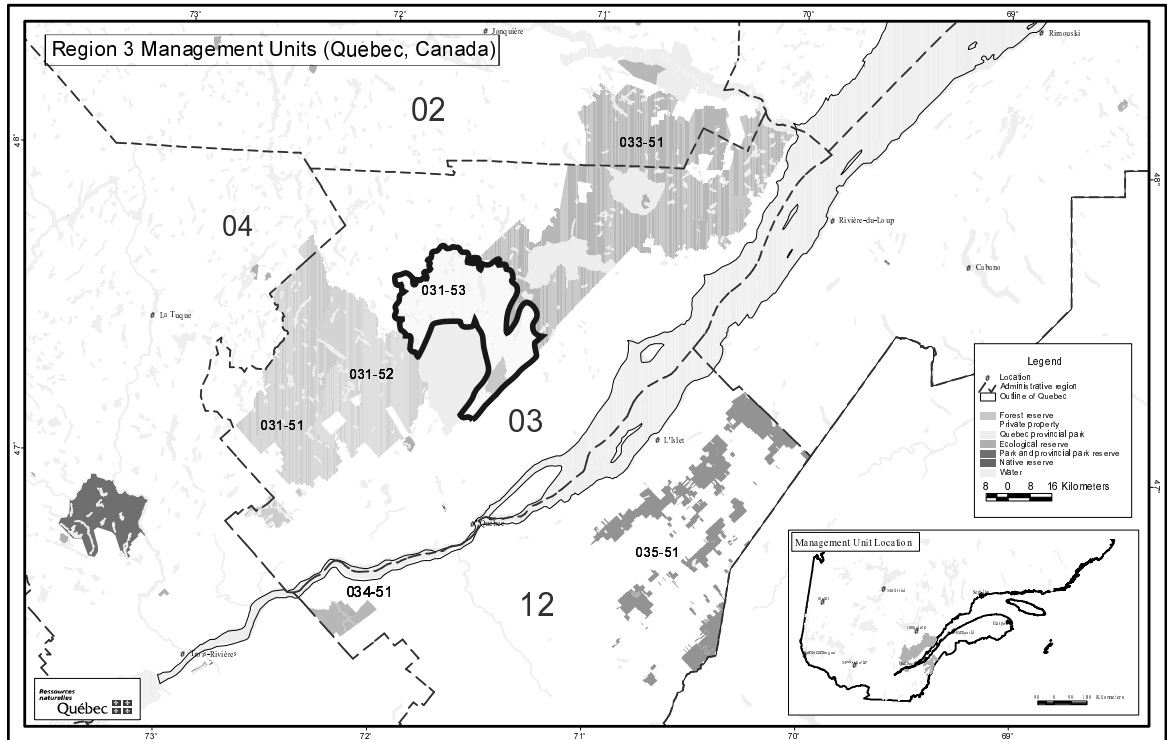
1.4.1 Forest Dataset

For the long-term model, we used a dataset adapted from a wood supply model developed by government planners, for forest management unit 031–53 in Quebec, Canada. Figure 1.1 shows the location of this management unit.

Initial forest inventory, silviculture treatment eligibility and operability, yield curves, and state transition matrix were all compiled by government wood supply analysts for the 2013–2018 AAC planning period.

²2.66 GHz Intel Core i7 processor, 8 GB 1067 MHz DDR3 RAM.

Figure 1.1 – Map showing location of forest management unit 031–53 in Quebec, Canada



The test area is in the boreal forest region. The majority (88%) of initial growing stock is softwood³, with presence (12%) of hardwood species⁴. Some pure softwood stands occur naturally, and plantations are generally pure spruce. A significant proportion of the forest cover is made up of mixed-wood stands containing different proportions of hardwood mixed in with the softwood. Total productive area in this management unit is 102 thousand hectares. The most recently published official AAC (determined by government planners) is 101 thousand m³ for softwood and 10 thousand m³ for hardwood.

We simulate one harvesting treatment (clearcut) and two silviculture treatments (planting, pre-commercial thinning). No species-wise selective cutting is possible (i.e. hardwood must be harvested if present) in mixed-wood stands.

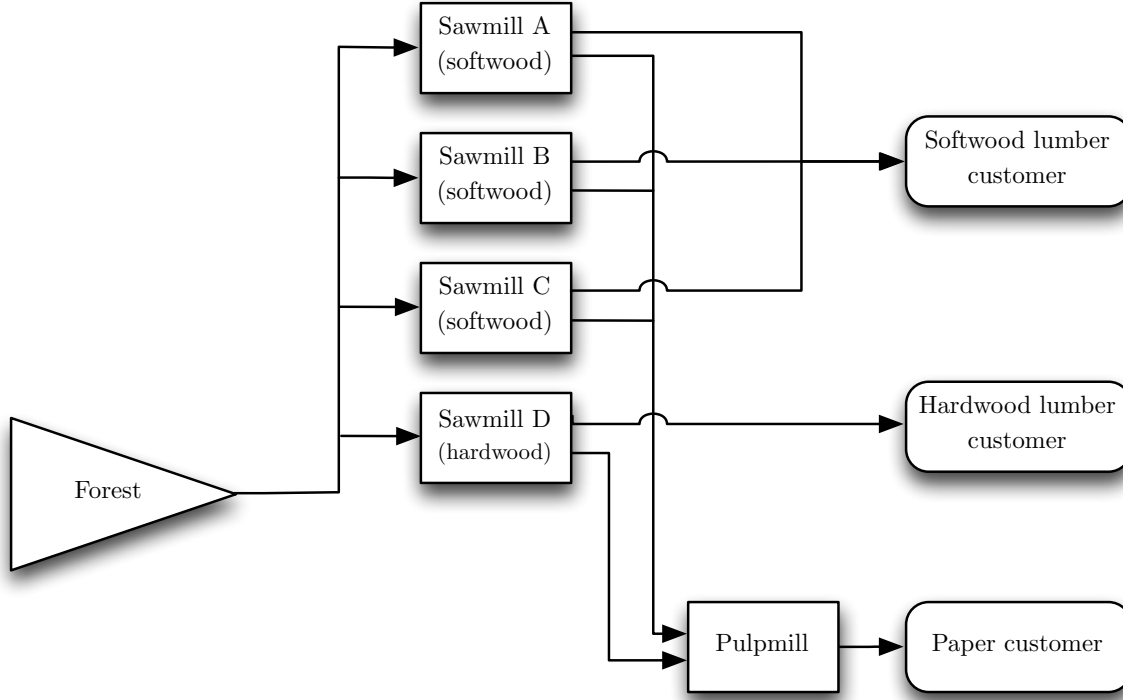
1.4.2 Value Creation Network Dataset

For the short-term planning model, we compiled an illustrative case study using realistic parameters (production capacities, unit cost of raw logs, transformation process input/output maps, unit prices of finished products). Parameter values used were originally collected from industry partners in the course of previous research projects. Figure 1.2 provides a schematic representation of our value creation network test dataset, illustrating potential product flow paths through the network. Note that the entire value creation network is encapsulated within the short-term planning agent.

³Mostly black spruce (*Picea mariana*) and balsam fir (*Abies balsamea*).

⁴Mostly white birch (*Betula papyrifera*) and poplar (*Populus tremuloides*).

Figure 1.2 – Schematic representation of test value creation network dataset



1.5 Experimental Design

Using our illustrative case study dataset, we tested for existence of a SDE by simulating the *status quo* hierarchical planning process. We compared the output from our optimisation model to the original government wood supply model, and confirmed that our model perfectly replicates source model structure (i.e. initial inventory, forest growth curves, treatment eligibility, state transitions following treatment). Setting aside our simplification of objective function and constraint structure, our model perfectly replicates a *status quo* long-term wood supply model, and thus provides a reasonable basis for our illustrative case study.

We defined six simulation scenarios, which vary in terms of demand types, demand levels, and integration of long- and short-term planning models. Table 2.2 summarises scenario parameters used in the experiment. Each simulation is actually made up of 30 two-stage (long-term, short-term) rolling-horizon replanning iterations. All scenarios maximise the sum of even-flow softwood and even-flow hardwood volume. The six scenarios can be grouped into three scenario-pairs, which we describe below.

Scenarios 1.1 and 1.2 maximise AAC independently of demand (i.e. long- and short-term plans not integrated, short-term model formulation S1). Demand for short-term planning is assumed to be infinite (i.e. 100% of AAC is harvested at each iteration). Scenario 1.1 simulates perfect implementation of period-1 solution at each iteration (decision variable values preserved). In contrast, scenario 1.2 simulates *imperfect implementation* of period-1 solution at each iteration—first-period decision variables are re-optimised using model formulation S1, subject to constraints forcing species-wise harvest volume and silviculture treatment area to match the first-period solution of the long-term model solution. As noted previously in §1.3.2.1, the short-term model will not necessarily schedule the same first-period

forest units (i.e. strata) as the long-term model. Thus implicit spatial distributions of long- and short-term harvesting solutions may differ, both at local (i.e. harvest block) and landscape-level scales (considering ecologically-driven spatial clustering of forest unit types on the landscape).

Similarly to the previously described scenario-pair, scenarios 2.1 and 2.2 maximise the AAC independently of demand (i.e. long- and short-term plans not integrated, short-term model formulation S1). Demand for short-term model is intentionally set to an unbalanced subset of the AAC (80% of softwood AAC, 40% of hardwood AAC), which is similar to historical harvest-to-AAC ratio observed in Quebec and other Canadian provinces (see Table 1.1). Scenarios 2.1 and 2.2 differ only in demand type. Scenario 2.1 assumes constant demand (i.e. demand level fixed for all rolling-horizon replanning iterations), whereas scenario 2.2 assumes proportional demand (i.e. demand level updated at each rolling-horizon replanning iteration to equal 80% of softwood AAC and 40% of hardwood AAC).

Scenarios 3.1 and 3.2 use the integrated model formulation (short-term model formulation S2), which finds the long-term sustainable wood supply solution that maximises first-period profit of the value creation network. As model formulation S2 maximises profit over a single planning period, we do not discount profit values as this would have no impact on optimisation model decisions. Scenario 3.1 is the base scenario for this series. Scenario 3.2 adds an upper-bound constraint on hardwood output from the long-term model. The value of this constraint parametre is selected to match the hardwood processing capacity in the value creation network (i.e. a simple anticipation mechanism to demonstrate the potential benefits of improved linkages on hierarchical planning process performance).

1.6 Results and Discussion

Figures 1.3 to 1.8 present simulation results for six scenarios. The disposition of figures is identical for all scenarios. The first subfigure (*a*) for each scenario shows the initial (i.e. iteration-0) AAC solution. The second subfigure (*b*) for each scenario shows the first period of the optimal wood supply solution (i.e. AAC) for all 30 planning iterations. The third subfigure (*c*) for each scenario shows the implemented harvest level for all 30 planning iterations. Scenarios 3.1 and 3.2 also show profit in this subfigure on a secondary axis. The fourth subfigure (*d*) for each scenario shows the difference between initial and replanned AAC. The fifth subfigure (*e*) for each scenario shows the difference between replanned AAC and harvest. The sixth subfigure (*f*) for each scenario shows the difference between initial AAC and harvest. Softwood volume is shown with white bars, hardwood volume with black bars, and total volume with small circles. Profit (where applicable) is shown with the $\times$ symbol.

Table 1.2 – Scenario descriptions

Scenario	Demand Type	Demand Level
1.1	Infinite	100% of AAC (exact implementation of first-period solution)
1.2	Infinite	100% of AAC (re-optimised with short-term model)
2.1	Constant	80% of initial softwood AAC, 40% of initial hardwood AAC
2.2	Proportional	80% of initial softwood AAC, 40% of initial hardwood AAC
3.1	Constant	Integrated model (basic)
3.2	Constant	Integrated model (with long-term hardwood supply constraint)

Table 1.3 compares harvested volume and profit for the first seven planning iterations, for scenarios 3.1 and 3.2. We do not include the first two scenario-pairs here, as only the third scenario-pair uses integrated short-term model formulation S1, which allows us to compare profit and harvest volume.

Note that profit values presented here are not discounted, as short-term planning model formulation S2 maximises profit over a single 5-year planning period. Although it would be possible to apply an arbitrary discount rate *a posteriori* to profit values in Table 1.3, this would have no impact on iterative simulation solutions. Our long-term model formulation does not include any profit performance indicators.

Although we simulated a large number of scenarios, we retained only three scenario-pairs for illustrative purposes. The first scenario-pair (1.1 and 1.2) is presented as a base case, to show the relative stability of a long-term solution when the actual harvest at each planning cycle closely (or perfectly) matches the AAC solution. The second scenario-pair shows the relative instability of long-term wood supply when timber demand is a skewed subset of wood supply, and simulates the possibility of wood supply failures under these conditions. The third scenario-pair shows the potential benefit (on stability of wood supply and value-creation potential) of integrating a simple demand-anticipation mechanism into the long-term planning process.

As can be expected for scenario 1.1 (perfect implementation of period-1 solution at each iteration), replanned AAC solutions track perfectly along the initial optimal solution until a critical period is reached (i.e. period 7), after which the AAC increases sharply and remains relatively stable despite replanning at each period, indicating negligible influence of end-of-horizon effects on simulation results and low systematic drift effect (SDE). The situation is noticeably different for scenario 1.2 (imperfect implementation of first-period optimal solution): the AAC steadily declines until the critical period, then increases sharply after the critical period as new growing stock becomes available, and resumes steady decline (see Figure 1.4b). This *declining non-declining yield* phenomenon is discussed in the literature (McQuillan, 1986; Daugherty, 1991; Gunn, 2007), however little has been done to address this issue in practice. Note that documented cases of *declining non-declining yield* in the literature are all based on models with profit maximisation objectives (with even-flow constraints), whereas our model uses a volume maximisation objective (with even-flow constraints).

Table 1.3 – Comparison of harvested volume and profit for periods 1–7 (scenarios 3.1 and 3.2)

Iteration	Scenario 3.1		Scenario 3.2		Difference	
	Volume (thousand m ³)	Profit (M\$)	Volume (thousand m ³)	Profit (M\$)	Volume (%)	Profit (%)
1	501	19.7	533	17.7	6	-10
2	507	19.9	530	17.6	5	-11
3	505	19.3	535	17.7	6	-8
4	496	18.5	525	17.5	6	-5
5	507	17.1	518	17.1	2	1
6	527	15.1	516	16.7	-2	11
7	494	15.1	507	16.1	3	6
Mean	505	17.8	523	17.4	4	-3
CV	2	11	2	3	0	-7

Scenarios 1.1 and 1.2 demonstrate that *status quo* implementation of the AAC model, if used within a *status quo* rolling-horizon hierarchical planning context, does not guarantee non-declining future harvest levels *even if the first period of optimal solutions are implemented almost perfectly* and deterministic model assumptions are assumed to hold. Undesirable behaviour of the AAC model under these optimistic base-case conditions (i.e. high level of coherence between long- and short-term plans) shows that the *status quo* planning process is sensitive to even the smallest deviations from the long-term optimal solution during its implementation. The infinite demand assumption behind these first two scenarios is unrepresentative of the *status quo* planning context in many jurisdictions—harvest volumes are generally well below the AAC, and the choice of harvest blocks may include significant biases relative to optimal AAC solution (i.e. closest to access roads, least dispersed, etc.).

Scenarios 2.1 and 2.2 simulate impact of incoherence between supply and demand assumptions in planning models. For both scenarios, species-wise demand distribution is a skewed subset of supply. We project current demand forward in two different ways. Scenario 2.1 assumes a constant demand pattern (i.e. species-wise demand is not updated at each rolling-horizon replanning iteration). Scenario 2.2 assumes a proportional demand pattern (i.e. species-wise demand is updated at each rolling-horizon replanning iteration, always using 80% and 40% of most recently calculated softwood and hardwood AAC). Scenario 2.1 simulates an unsustainable trajectory—Figure 1.5c shows a wood supply failure in period 6, followed by regular wood supply failures later in the planning horizon. These wood supply failures are induced by the systematic omission of hardwood-rich forest units from short-term harvest plans, and are not predicted by the long-term wood supply model. In fact, the long-term wood supply model had been predicting *increasing* AAC levels⁵ for several rolling-horizon planning iterations when the first wood supply failure occurs (see Figure 1.5b). Scenario 2.2 simulates less severe wood supply failures, in part because we allow species-wise demand to scale up and down (as a proportion of current AAC) at each rolling-horizon replanning iteration, thereby ensuring a certain adaptive coherence between supply and demand (despite a systematic species-skewed gap between planned and implemented harvest levels).

Scenarios 3.1 and 3.2 demonstrate some potential advantages of integrating long- and short-term planning. In our integrated model, the long-term wood supply plan is driven by short-term value creation, *viz.* the wood supply solution is iteratively adjusted until it converges on an even-flow wood supply plan that maximises first-period profit. Note that process parameters have been set in our model such that softwood processing is generally more profitable than hardwood processing (consistent with current situation in Quebec according to industry data). Scenario 3.1 shows a significant decline in profits between periods 1 and 7, which we explain by low hardwood demand (relative to projected hardwood harvest levels) in the long-term wood supply plan. At every iteration, the short-term planning agent systematically avoids harvesting (less profitable) hardwood-rich stands. Lower-value hardwood growing stock accumulates, causing a systematic decline of value-creation potential in the medium term.

We added a simple constraint to the wood supply model in scenario 3.2, which sets an upper bound on long-term hardwood harvest projection to (approximately) match hardwood demand in the value creation network. The impact of improved coherence between long- and short-term planning processes

⁵Increasing AAC can be explained by the unanticipated accumulation of growing stock in the forest, due to harvest levels systematically being below AAC.

is obvious: optimising long-term hardwood supply planning to more closely match long-term demand eliminates much of the softwood high-grading behaviour observed in scenario 3.1, resulting in stabilised profit and volume flow, and increased mean AAC for the first seven planning iterations (see Figures 1.7c and 1.8c, and Table 1.3).

The last five scenarios all exhibit some form of *systematic drift effect* (SDE) that is not predicted by the long-term wood supply optimisation model. Evidence of SDE in our illustrative case study includes a monotonic increase or decrease in the AAC over several iterations, a monotonic change in the AAC composition (i.e. ratio of hardwood to softwood), increasing incoherence between supply and demand, and increasing infeasibility of the demand-satisfaction problem (i.e. instability of simulated harvest levels). Undetected forms of drift are likely to exist within these scenarios for performance indicators not presented here (e.g. standing timber inventory, age class distribution, landscape patch distribution, value-creation potential, harvest block size distributions and locations, wildlife habitat suitability, etc.).

1.7 Conclusion

The illustrative case study presented here shows that the *status quo* hierarchical forest planning process (i.e. two-stage rolling-horizon iterative replanning) is incoherent and dysfunctional: it fails to demonstrate the long-term sustainability of government-endorsed short-term harvest levels, fails to reliably meet industrial fibre demand over time, and exacerbates incoherence between wood supply and fibre demand over several planning iterations. We identify systematic differences between planned and actual harvesting activities as an important factor contributing to this dysfunction, which manifests itself as either instability in long-term wood supply or lost short-term value creation opportunity.

We show that coherence between long- and short-term planning solutions can be improved by manipulating linkages between planning levels (e.g. adding species-wise demand constraints to the model). Although simplistic in its implementation, our integrated model illustrates the potential value (in terms of both credibility of the long term planning process and performance of the short-term planning process) of anticipating short-term planning objectives (e.g. demand satisfaction) within a long-term planning model. Based on our preliminary simulation results, we feel there is much potential for further refinement of the linkages between long- and short-term planning processes.

Much of the difference between planned and actual harvesting activities can be explained by the principal-agent relationship between government and industry planners. This principal-agent relationship could potentially be modelled as a bilevel optimisation problem, and the long-term planning process extended to include an anticipative-iterative dimension. This would allow the principal to develop long-term forest policy that is more likely to be compatible with anticipated agent behaviour, thereby inducing a reduction in SDE, and improving hierarchical planning process credibility and value-creation potential.

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Figure 1.3 – Scenario 1.1: simulated wood supply for 30 sequential cycles

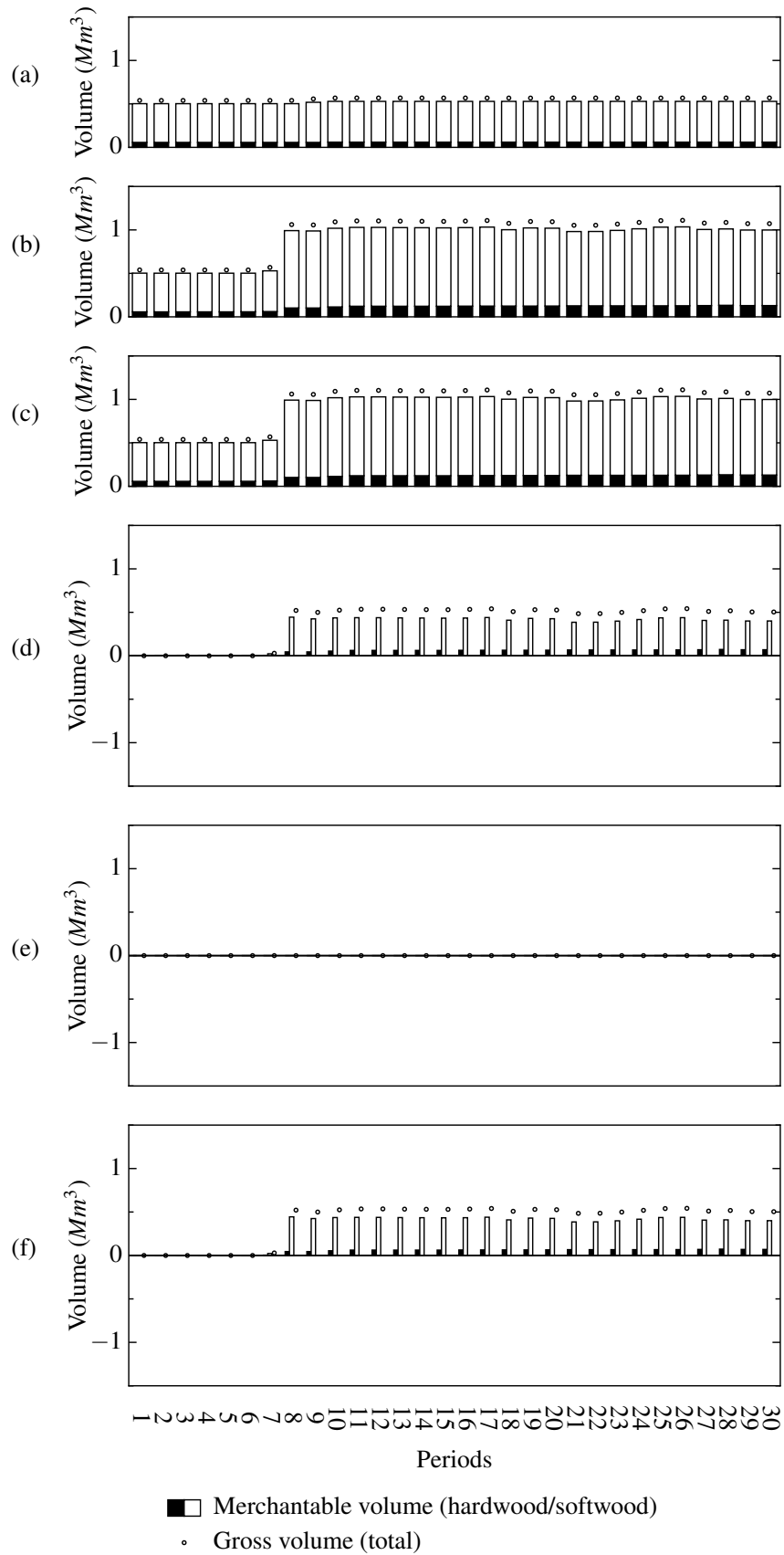


Figure 1.4 – Scenario 1.2: simulated wood supply for 30 sequential planning cycles

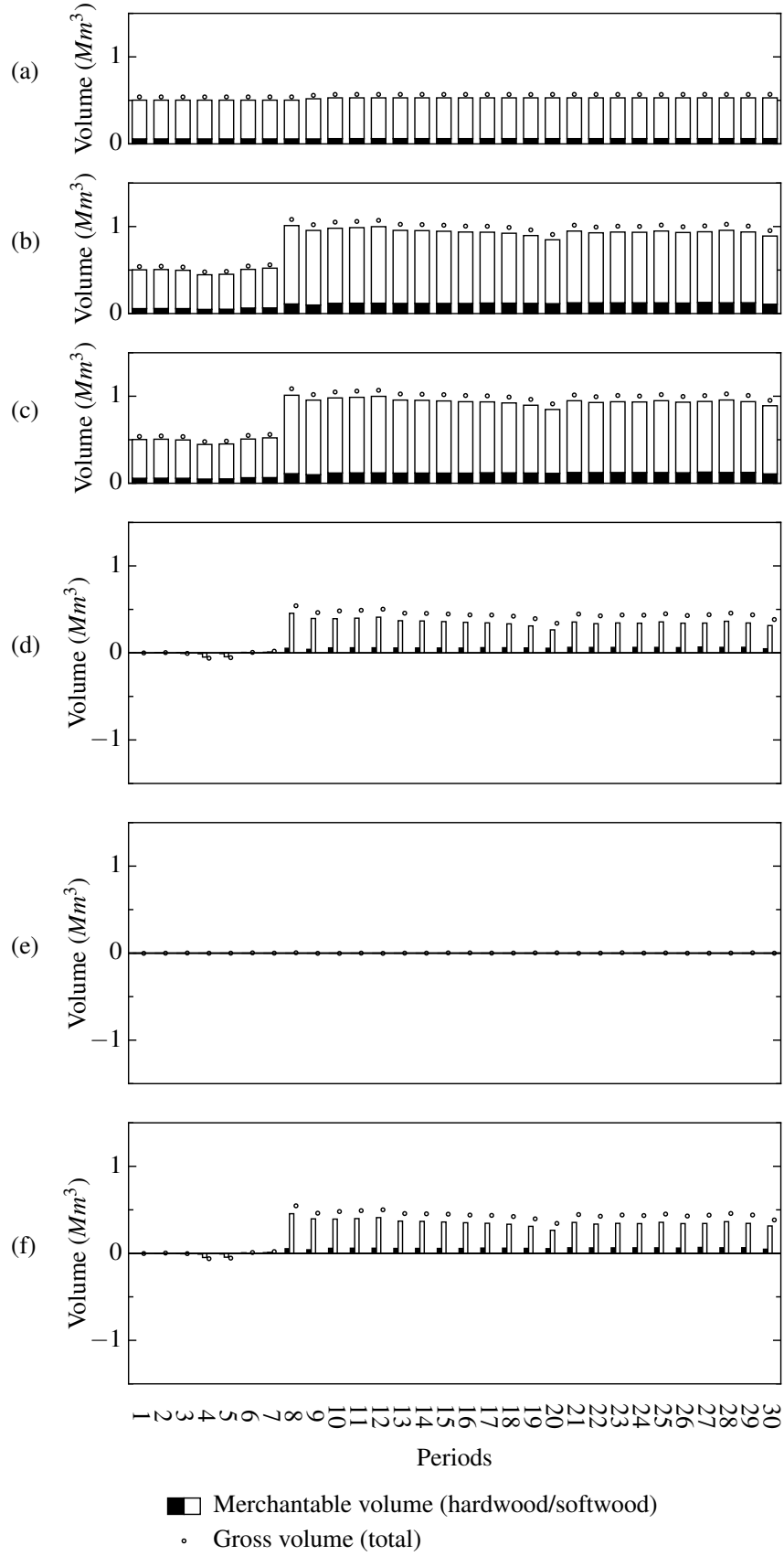


Figure 1.5 – Scenario 2.1: simulated wood supply for 30 sequential planning cycles

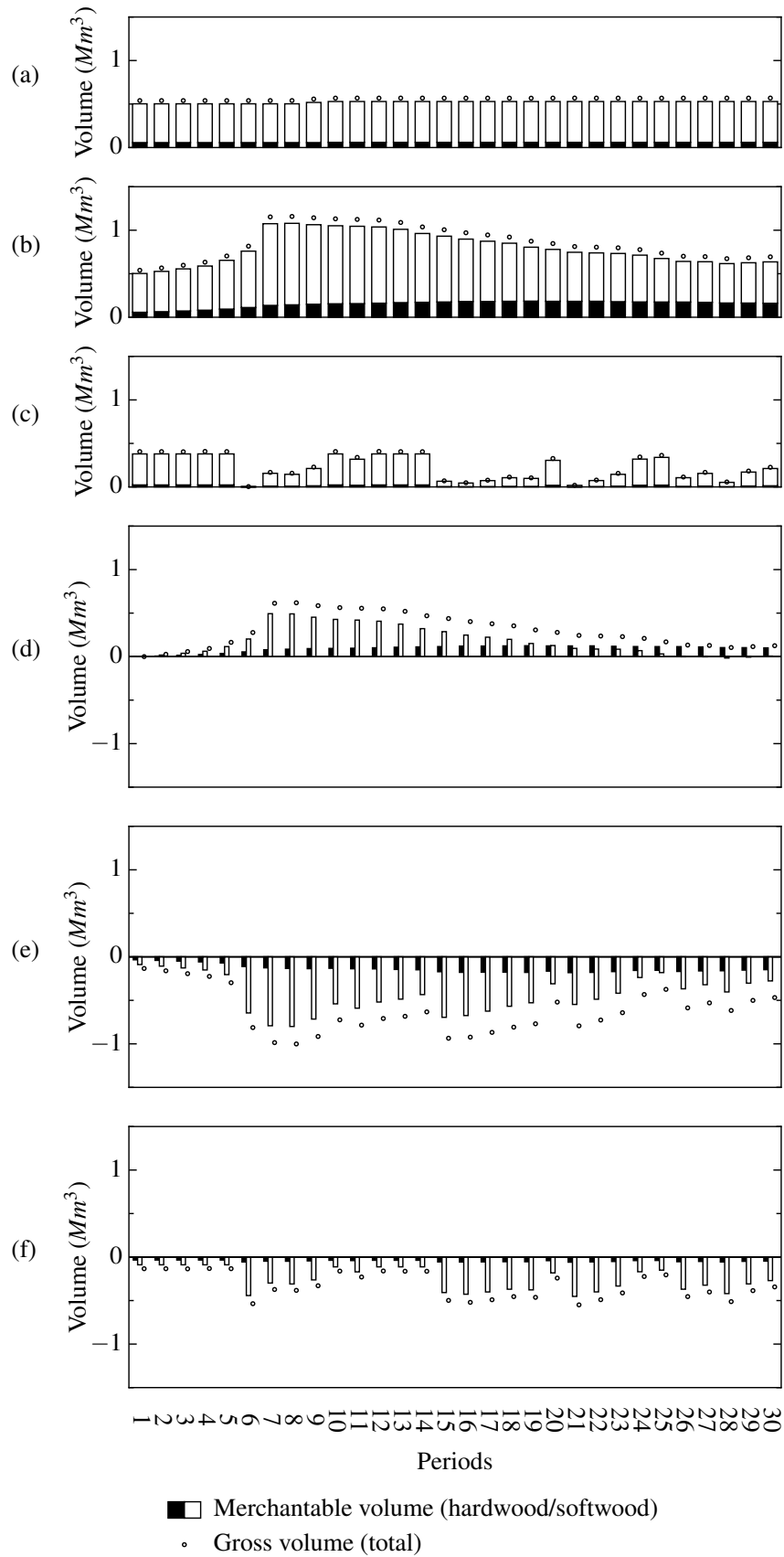


Figure 1.6 – Scenario 2.2: simulated wood supply for 30 sequential planning cycles

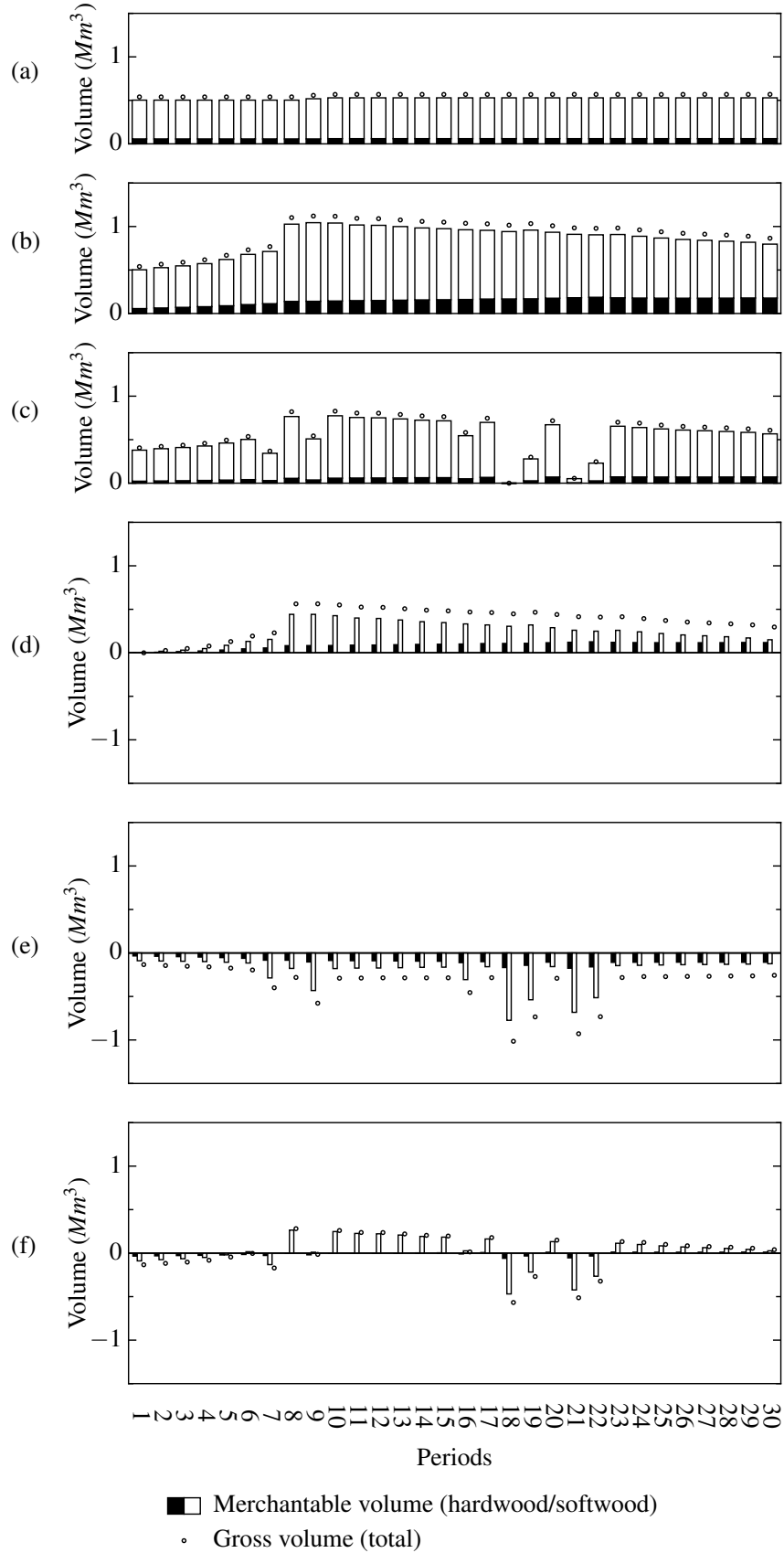


Figure 1.7 – Scenario 3.1: simulated wood supply for 30 sequential planning cycles

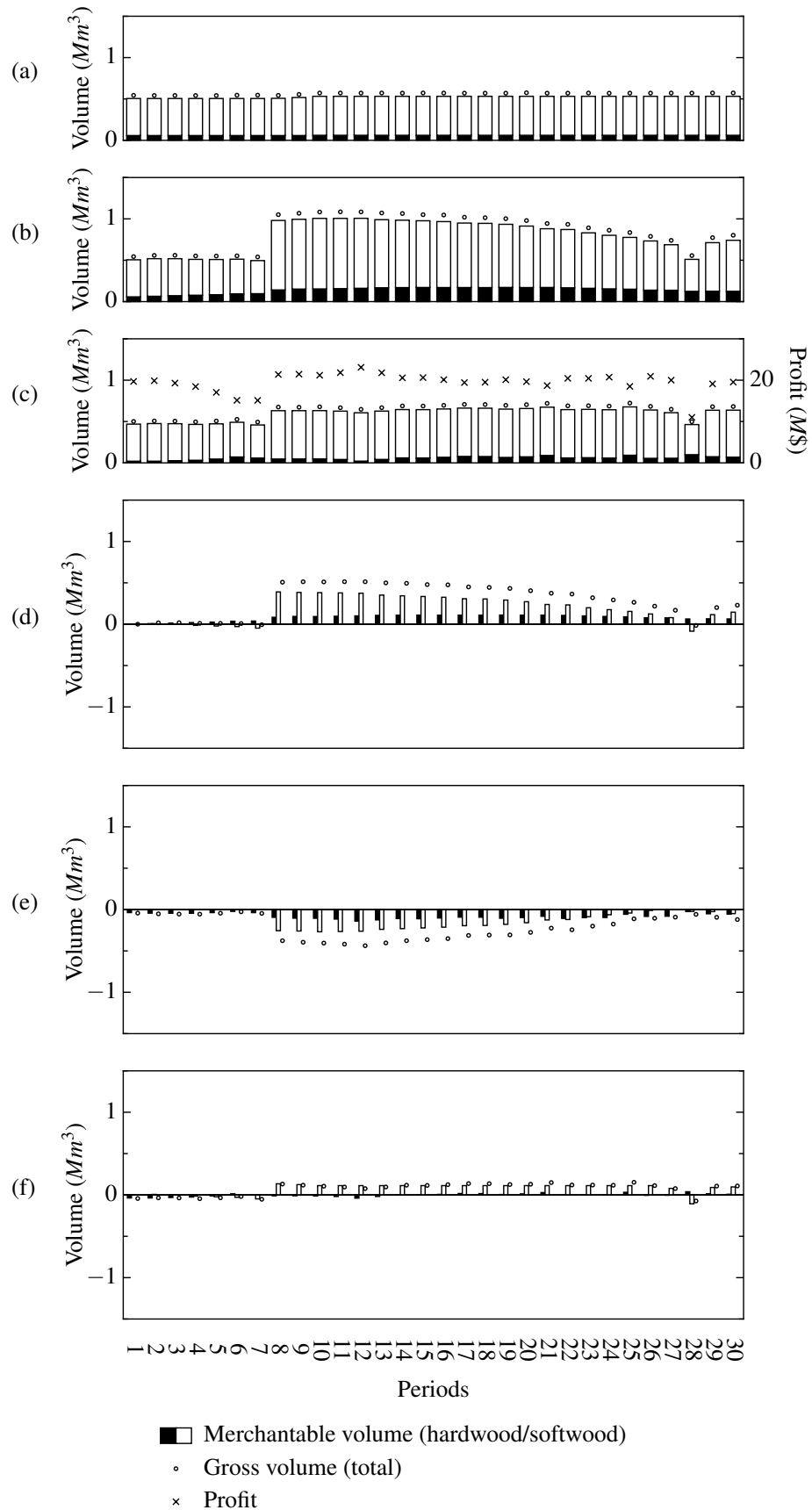
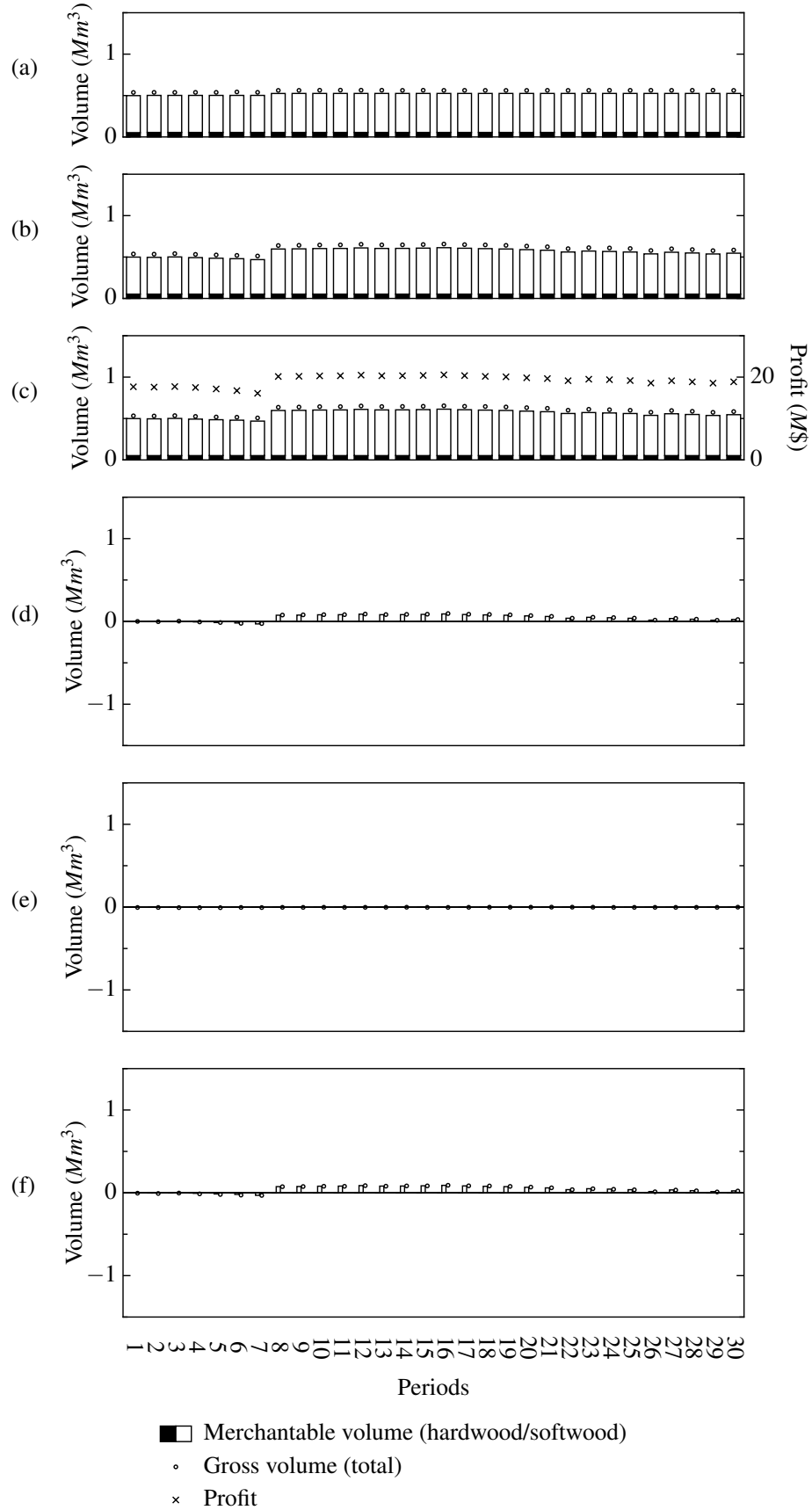


Figure 1.8 – Scenario 3.2: simulated wood supply for 30 sequential planning cycles



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Chapter 2

A Bilevel Model Formulation for the Distributed Wood Supply Planning Problem

This chapter presents an article entitled *A Bilevel Model Formulation for the Distributed Wood Supply Problem*, which has been published by CIRRELT as research paper CIRRELT-2015-06. The authors are Gregory Paradis, Mathieu Bouchard, Luc LeBel, and Sophie D'Amours.

Résumé

La formulation classique du modèle d'optimisation de la possibilité forestière maximise le rendement soutenu, et pose l'hypothèse implicite que la demande de fibre est infinie. Dans plusieurs juridictions, cette hypothèse représente mal la réelle consommation de fibre, qui est typiquement un sous-ensemble de l'attribution totale favorisant certains groupes d'essences. L'incapacité de prévoir l'effet de ce biais entre la possibilité forestière et la consommation réelle est liée à une hausse du risque de ruptures de stock. En particulier, nous examinons le *problème de planification distribuée de l'approvisionnement en bois*, qui est une variante du problème général où des agents indépendants jouent les rôles de propriétaire de la ressource forestière et de consommateur de la fibre (p. ex., la planification de l'approvisionnement en bois sur les terres publiques au Canada, où les délégués gouvernementaux contrôlent l'approvisionnement en bois et les entreprises du secteur forestier consomment la fibre). Nous utilisons la théorie de l'agence pour décrire la source d'antagonisme entre le propriétaire de la ressource (le *principal*) et le consommateur industriel (l'*agent*). Nous démontrons que le problème de planification distribuée de l'approvisionnement en bois peut être modélisé plus justement avec une formulation biniveau, et nous présentons une extension du modèle d'optimisation classique pouvant anticiper le comportement du consommateur de fibre industriel. Le cas général du problème d'optimisation biniveau est $\mathcal{NP}$ -difficile, non-convexe, et non-linéaire—il est difficile à résoudre à l'optimalité globale. En imposant certaines restrictions sur la topologie du réseau de l'agent, nous démontrons que le problème biniveau général peut être décomposé en sous-problèmes convexes. Nous présentons une méthodologie pour résoudre le cas spécial convexe à l'optimalité globale, et nous comparons les performances expérimentales des formulations classiques et biniveaux à l'aide d'un jeu de données réaliste. Les résultats expérimentaux montrent que le temps de solution du modèle biniveau est comparable au temps de solution du modèle classique, et que le modèle biniveau peut atténuer le risque de ruptures de stock.

Abstract

The classic wood supply optimisation model maximises even-flow harvest levels, and implicitly assumes infinite fibre demand. In many jurisdictions, this modelling assumption is a poor fit for actual fibre consumption, which is often a species-unbalanced subset of total fibre allocation. Failure to anticipate this bias in volume and species mix of industrial wood fibre consumption has been linked to increased risk of wood supply failure. In particular, we examine the *distributed wood supply planning problem*, which is a variant of the general wood supply planning problem where the roles of forest owner and fibre consumer are played by independent agents (e.g. wood supply planning on public forest land in Canada, where government stewards control wood supply and forest products industry firms consume the fibre). We use agency theory to describe the source of antagonism between public forest land owners (the *principal*) and industrial fibre consumers (the *agent*). We show that the distributed wood supply planning problem can be modelled more accurately using a bilevel formulation, and present an extension of the classic wood supply optimisation model which explicitly anticipates industrial fibre consumption behaviour. The general case of the bilevel wood supply optimisation problem is $\mathcal{NP}$ -hard, non-linear, and non-convex—it is difficult to solve to global optimality. By imposing certain restrictions on agent network topology, we show that the general case can be decomposed into convex sub-problems. We present a solution methodology that can solve this special case to global optimality, and compare output and solution times of classic and bilevel model formulations using a computational experiment on a realistic dataset. Experimental results show that solution time for the bilevel problem is comparable to solution time for the classic single-level problem, and that the bilevel formulation can mitigate risk of wood supply failure.

2.1 Introduction

In Canada, provincial government authorities allocate timber licences (TL) to industrial fibre consumers. These TLs, which grant rights to harvest fibre on public forest land, set species-wise upper bounds on periodic harvest volume, however there is typically no policy requirement to set matching lower bounds. Timber licences are typically valid for a pre-determined period (e.g. 5 years), after which point licences may be renewed, subject to re-evaluation of available wood supply. Maximal sustainable TL volume is commonly referred to as *annual allowable cut*, or AAC.

The term *wood supply planning* describes the process by which AAC is determined. In practice, this often amounts to solving a linear programming (LP) optimisation model, which finds the maximum even-flow harvest levels (Gunn, 2007). Periodic fluctuation of harvest levels in the LP solution is controlled using *even-flow constraints*, which specify an upper bound on the difference between highest and lowest periodic harvest volumes. Even-flow constraints are conceptually associated with the *sustainability* of the forest management process, although the scientific basis for this association is questionable.

The concept of sustainable forest management (SFM) has long been a central theme of public forest policy in Canada (Canadian Council of Forest Ministers, 2008). In concrete terms, sustainable forest management policy is implemented via silviculture treatments, notably harvesting treatments. According to the *Criteria and Indicators of Sustainable Forest Management in Canada* (CCFM C&I) (Canadian Council of Forest Ministers, 2003), harvesting is generally deemed to be sustainable if it is below AAC¹. This notion, that any harvest level below AAC is sustainable, is also implicitly used at the forest management unit (FMU) scale when government uses species-wise even-flow AAC as contractual upper-bounds in TLs.

The classic model simulates a finite alternating sequence of harvesting and growth, and implicitly assumes that all available fibre will be consumed in every planning period (regardless of quantity, quality, cost, or value creation potential). In practice this assumption is rarely respected. Figure 2.1 shows the species-wise proportion of Canadian AAC consumed from 1990 to 2012. On average, 80% of softwood AAC and 45% of hardwood AAC were consumed, indicating a clear industrial preference for softwood during this period. The data show a species-skewed negative consumption bias, relative to AAC. This bias is related to the infinite-demand assumption implicitly embedded in the classic wood supply optimisation model. The bias can, to a certain extent, be attributed to a poor alignment between industrial fibre demand and wood supply planning. Local industrial processing capacity may be insufficient to consume some parts of AAC. Other parts of AAC may be economically unattractive (i.e. have a negative net value) or be operationally inaccessible (due to fragmented forest landscape, prohibitive access cost, or constrained by regulations that govern harvesting). Thus, the optimal solutions will likely never be executed, and the long-term state of the forest will be systematically different from that predicted by the wood supply model.

Mathey et al. (2009) use a spatially-explicit harvest scheduling model to estimate financial outputs for various timber supply levels, using a case study dataset from northern Ontario. Their simulation results confirm that a subset of AAC may be uneconomic.

¹In particular, see criterion 5.3.1 of the CCFM C&I, which compares aggregated national AAC and harvest volumes as an indication of sustainability of forest management practices.

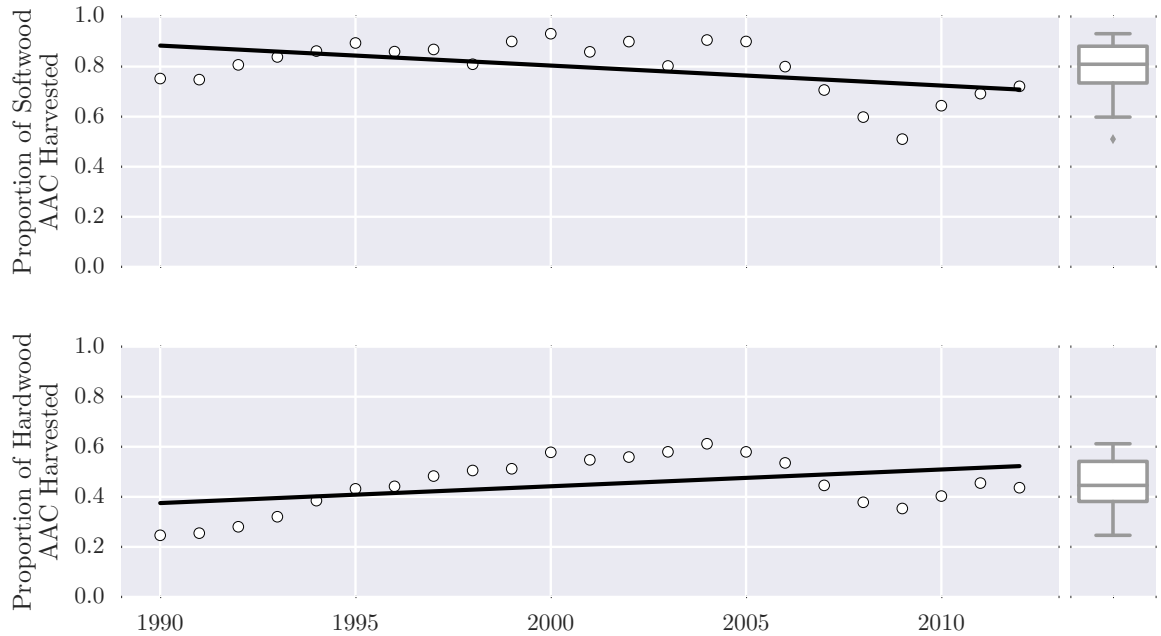


Figure 2.1 – Proportion of species-wise AAC consumed in Canada for period 1990 to 2012 [source: National Forestry Database (2014)]

In Chapter 1, we simulated repeated wood supply planning cycles, using a principal-agent paradigm to model the interaction between government wood supply planners and industrial fibre consumers. We showed that the classic wood supply model may fail due to the aforementioned species-skewed negative fibre consumption bias, and concluded that the wood supply planning process currently in place on public land in Canada does not provide credible assurance of the long-term sustainability of the wood supply. Given the pervasiveness of this bias in practice, the classic wood supply optimisation model formulation does not constitute a rational basis for the implementation of sustainable forest management.

For management and planning purposes, the forest landscape is subdivided into stands. The stand is the basic silviculture decision unit, and can be described as a contiguous forested area with uniform vegetation and growth characteristics. Projection of future forest condition and fibre availability is based on aggregated result of stand-level growth and yield simulation. This requires four distinct types of information: (1) detailed starting inventory of forest condition (i.e. stand ages and types), (2) hypothetical projection of stand condition over time for each type², (3) hypothetical state transitions induced by stand events (i.e. shift in stand age and stand type), and (4) hypothetical intensity, location and timing of planned future stand events (e.g. clear-cut harvesting of stand i in period j).

Wood supply optimisation models typically use the first three information types (i.e. starting inventory, yield curves, and state transitions) as input, leaving the fourth information type (i.e. location and timing of future stand events) as variables in the objective function. Assembling this information into a coherent model, and subsequent analysis of model output, is referred to as the *wood supply planning problem*. We focus on a particular problem variant, which we call the *distributed wood supply planning*

²Foresters typically refer to these as *yield curves*.

problem (DWSPP), where the roles of forest land owner and industrial fibre consumer are played by independent agents. The DWSPP is common in Canada and other jurisdictions, where public forest land is managed by government stewards on behalf of the general population.

We can describe the DWSPP, from a game-theoretic perspective, as an instance of the principal-agent problem (Laffont and Martimort, 2002; Schneeweiß, 2003). The role of principal is played by the forest owner (or government steward), and the role of agent is played by the industrial fibre consumer. The principal has the long-term responsibility to ensure a sustainable wood supply (hence the even-flow constraints in the wood supply model), but aims to maximise economic activity by exploiting the forest resource (hence the wood-supply-maximisation objective function). The agent aims to maximise short-term profit by transforming a subset of wood supply into forest products. Antagonism between the principal and the agent stems from either (a) binding agent capacity constraints³ or (b) the presence of negatively-valued subsets of the wood supply⁴. Either of these factors may deter the agent from consuming the entire wood supply, which in turn induces the problematic negative fibre consumption bias.

There remains a gap in the literature with respect to the incidence of the principal-agent problem on the DWSPP, although a number of recent papers use a bilevel approach to model forest-sector decision problems. Bogle (2012) recently modelled optimal government policy response to a mountain pine beetle epidemic in British Columbia, Canada, using a principal-agent framework. Emphasis is placed on determining optimal government policy to incite fast liquidation of rapidly deteriorating beetle wood⁵. Their approximate solution methodology, which is tested only on a very small synthetic dataset, does not guarantee convergence on optimal solutions and is may be intractable for typical (i.e. large) datasets. Zhai et al. (2014) use a bilevel approach to model hierarchical planning in the case of fast-growing plantation management. Their explicit treatment of multiple lower level decision makers is interesting, however they do not explicitly address the case where only a subset of the harvest quota is economically attractive. Yue and You (2014) present a bilevel optimisation model to analyse the impact of adding new biorefinery capacity to an existing timber supply chain.

In Chapter 1, we also use a principal-agent framework to model failure of the classic wood supply model, and conjecture that extending the model to explicitly anticipate the principal-agent relationship should improve the coherence of the distributed wood supply planning process. Beaudoin et al. (2010) successfully used an agent-based modelling approach to anticipate interaction between participants in a distributed fibre procurement process at the tactical-operational planning level. They leverage this anticipation to better coordinate the planning process between horizontally-linked agents (i.e. firms sourcing fibre from the same procurement area in the public forest), resulting in local and global

³We model agent behaviour using a network flow model, which we describe in more detail in §2.2. Each business unit in the agent network encapsulates one or more processes. Product flows between business units are defined by directional links. Both processes and links have capacity constraints, which can become saturated. When saturated capacity constraints in the agent network limits further improvement in the (profit-maximising) agent objective function, we can describe them as *binding agent capacity constraints*.

⁴If total cost of pushing a unit of fibre through the agent network exceeds potential revenue from sale of products to end-clients, then this unit of fibre has a negative net unit value, and will not be willingly consumed by a profit-maximising agent.

⁵The mountain pine beetle acts as a vector for a fungus, which spreads through the sapwood and typically kills the affected trees (Byrne et al., 2006). Although these trees can still be harvested and transformed into valuable forest products, the blue fungus that kills the trees stains the wood, making it less attractive than a similar volume of healthy clear wood. Furthermore, the physical properties of the dead wood can negatively affect the quality of lumber products manufactured from this material, with degradation worsening overtime such that beetle wood is typically considered to have a 5 to 10 year shelf life (Trent et al., 2006).

profitability gains relative to the *ad hoc* procurement planning process currently in place in many jurisdictions. We conjecture that an agent-based anticipation approach could help coordinate the vertical integration between government (principal) and industry (agent) planners, at a strategic-tactical planning level. The principal-agent paradigm has not, to the best of our knowledge, been used to model the gap between planned and executed harvest volume in the context of the DWSPP.

We endeavour to close this gap by extending the classic wood supply optimisation model to explicitly anticipate industrial fibre consumption behaviour. Our anticipation function is based on a network flow model called *LogiLab*, developed by the FORAC Research Consortium (Jerbi et al., 2012). Thus, we model agent behaviour as material and financial flows through a forest product value-creation network. The agent seeks to maximise profit, subject to wood supply, capacity, and demand constraints. The model assumes centralised network planning, and exogenous end-product prices.

From an operations research perspective, bilevel programming subsumes the principal-agent problem (Colson et al., 2007). Hence, any principal-agent problem can be formulated as a bilevel optimisation problem. Although a bilevel modelling approach could potentially address the principal-agent aspect of the DWSPP, solving a bilevel problem to global optimality is typically not trivial. Even the simplest bilevel optimisation problems are known to be $\mathcal{NP}$ -hard, and non-convex, non-linear solution spaces are common (Dempe, 2003; Colson et al., 2007). This chapter addresses the shortcomings of the classic formulation identified in Chapter 1, namely the species-skewed fibre consumption bias that is implicitly embedded in the classic wood supply model. We developed both a bilevel model formulation for the DWSPP, and a novel methodology to solve the bilevel problem to global optimality.

We hypothesised that the bilevel model formulation would improve stability of the wood supply. To verify this hypothesis, we designed a computational experiment comparing output from classic and bilevel model formulations after 30 sequential rolling-horizon replanning cycles.

The remainder of this chapter is organised as follows. The mathematical formulation of the bilevel problem, solution methodology, and experimental methods are presented in §2.2. Results from the computational experiment are presented in §2.3, followed by discussion in §2.4. Concluding remarks are presented in §2.5.

2.2 Methods

We present a mathematical formulation of the bilevel wood supply optimisation model in §2.2.1, followed by a solution methodology we developed to solve the bilevel problem in §2.2.2. We also present the methodology for a computational experiment, in which we compare the performance of classic and bilevel model formulations, in §2.2.3.

2.2.1 Bilevel Wood Supply Problem Formulation

The bilevel wood supply problem extends the DWSPP to anticipate industrial fibre consumption. We present mathematical formulations for both upper level (principal) and lower level (agent) problems before defining the bilevel optimisation problem solution space.

2.2.1.1 Upper-Level Problem Formulation

The following formulation of the upper level model is a simplified representation of the *de-facto* standard (i.e. *classic*) wood supply planning model implemented in many jurisdictions, including public forest land in Canada⁶. This is equivalent to a *Model I* formulation, using the modelling terminology in Davis et al. (2001).

Maximise

$$\sum_{i \in Z} \sum_{k \in P_i} c_{ik} x_{ik} \quad (2.1)$$

subject to

$$\sum_{k \in P_i} x_{ik} = 1, \quad \forall i \in Z \quad (2.2)$$

$$y_o \leq \sum_{i \in Z} \sum_{k \in P_i} \alpha_{ikot} x_{ik} \leq (1 - \varepsilon_o) y_o, \quad \forall o \in O', t \in T \quad (2.3)$$

$$l_{ot} \leq \sum_{i \in Z} \sum_{k \in P_i} \alpha_{ikot} x_{ik} \leq u_{ot}, \quad \forall o \in O, t \in T \quad (2.4)$$

where

Z := set of spatial zones

P_i := set of available prescriptions for zone $i \in Z$

O := set of forest outputs

$O' \subset O$:= set of sustainable forest outputs

T := set of time periods in the planning horizon

ε_o := admissible level of variation on yield of output $o \in O$

α_{ikot} := quantity of output $o \in O$ produced in period $t \in T$ by prescription $k \in P_i$ in zone $i \in Z$

l_{ot} := lower bound on yield of output $o \in O$ in period $t \in T$

u_{ot} := upper bound on yield of output $o \in O$ in period $t \in T$

c_{ik} := global value of including cost and benefits of prescription $k \in P_i$ in zone $i \in Z$

x_{ik} := fraction of zone $i \in Z$ on which prescription $k \in P_i$ is applied

$y_o := \sum_{i \in Z} \sum_{k \in P_i} \alpha_{iko1} x_{ik}$, which corresponds to first-period harvest volume for output $o \in O'$

The objective function (2.1) maximises harvest volume over the planning horizon. Constraint (2.2) is an accounting constraint, and simply ensures that the entire forest area is assigned to a prescription (including the null prescription, i.e. do nothing). Constraint (2.3) models an *even-flow* policy, which constrains simulated harvest level to be more or less level throughout the planning horizon (i.e. stabilise periodic flow of timber from the forest). We have expressed the even-flow constraint in terms of y_o , which represents first-period harvest volume for output $o \in O'$ (i.e. species-wise periodic allowable cut⁷). Constraint (2.4) bounds minimal and maximal periodic yield on the general outputs (e.g. area converted to plantation in a given period).

⁶Adapted from Chapter 1.

⁷Annual allowable cut, or AAC, is essentially the same as periodic allowable cut, but expressed on an annal basis.

Thus, y_o corresponds to the wood supply, for a given species group, that the principal might offer the agent in a given planning cycle— y_o constitutes the primary linkage interface between upper and lower level models, as we will see in the next section. This is analogous to the actual wood supply planning process, where AAC constitutes the primary policy interface between the principal and the agent.

2.2.1.2 Lower-Level Problem Formulation

The following formulation of the lower-level (agent) problem is adapted from Jerbi et al. (2012).

Maximise

$$\sum_{u \in U} \left(\sum_{p \in P | d_{up} > 0} \rho_{up} D_{up} - \sum_{w \in W_u} c_w Y_{uw} \right) - \sum_{e \in E} \sum_{p \in P} c_{ep}^f F_{ep} \quad (2.5)$$

subject to

$$\beta_{uo} + \sum_{w \in W_u} (\gamma_{pw} - \alpha_{pw}) Y_{uw} + \sum_{e \in \delta_u^+} F_{ep} - \sum_{e \in \delta_u^-} F_{ep} - D_{up} = 0, \quad u \in U, p \in P \quad (2.6)$$

$$\sum_{w \in W_u} \gamma_{kuw} Y_{uw} \leq q_{ku}, \quad u \in U, k \in K \quad (2.7)$$

$$D_{up} \leq d_{up}, \quad \forall u \in U, p \in P \quad (2.8)$$

$$\sum_{p \in P} F_{ep} \leq f_e^u, \quad \forall e \in E \quad (2.9)$$

$$f_{ep}^l \leq F_{ep} \leq f_{ep}^u, \quad \forall e \in E, p \in P \quad (2.10)$$

$$\sum_{u \in U} \beta_{uo} \leq y_o, \quad \forall o \in O' \quad (2.11)$$

where

U := set of business units

K := set of resource capacity types (machine capacities, stock limits)

W := set of processes (machines, inventories)

$W_u \subset W$:= set of processes available at business unit $u \in U$

P := set of products

$O' \subset P$:= set of sustainable forest outputs, from the upper-level model

E := set of links between business units

$\delta_u^+ \subset E$:= set of inbound links for business unit $u \in U$

$\delta_u^- \subset E$:= set of outbound links for business unit $u \in U$

q_{ku} := capacity of type $k \in K$ at business unit $u \in U$

f_{ep}^l := lower bound on flow of product $p \in P$ through link $e \in E$

f_{ep}^u := upper bound on flow of product $p \in P$ through link $e \in E$

f_e^l := upper bound on flow of all products through link $e \in E$

c_w := unit cost of process $w \in W$

c_{ep}^f := unit cost of transporting product $p \in P$ on link $e \in E$

$\alpha_{pw} :=$ quantity of product $p \in P$ required for one unit of process $w \in W$
 $\gamma_{pw} :=$ quantity of product $p \in P$ produced for one unit of process $w \in W$
 $\lambda_{kuw} :=$ capacity of type $k \in K$ utilised by process $w \in W$ at business unit $u \in U$
 $d_{up} :=$ demand for product $p \in P$ at business unit $u \in U$
 $\rho_{up} :=$ price of product $p \in P$ at business unit $u \in U$
 $\beta_{uo} :=$ external supply of sustainable forest output $o \in O'$ at business unit $u \in U$
 $y_o :=$ maximum external supply of sustainable forest output $o \in O'$ (from upper-level model)

and

$Y_{uw} :=$ quantity of process $w \in W$ performed at business unit $u \in U$
 $D_{up} :=$ quantity of product $p \in P$ sold by business unit $u \in U$
 $F_{ep} :=$ flow of product $p \in P$ on link $e \in E$

The objective function (2.5) maximises network profit (i.e. revenue from sale of products, net of production and transportation cost). Constraint (2.6) ensures flow conservation in the network. Constraint (2.7) limits process utilisation to production capacity upper bounds. Constraint (2.8) limits product sales to demand upper bounds. Constraint (2.9) limits total flow to link capacity upper bounds. Constraint (2.10) ensures that product-wise flow upper and lower bounds are respected for each link. Constraint (2.11) limits consumption of fibre by agent network to the species-wise wood supply determined in the upper-level model.

Note that storage of unsold products and end-product consumption by external clients are modelled as special cases of processes. Fibre procurement from the forest is also modelled as a process, although we present it here using a separate parametre β_{uo} , as this facilitates description of the linkage with the upper-level model, via parametre y_o .

We have chosen to model industrial fibre consumption in the lower level for the first planning period only. We could have chosen to extend anticipation of agent behaviour to an arbitrary number of periods without loss of generality.

2.2.1.3 Bilevel Solution Space

The bilevel feasible region is the subset of the upper-level feasible region for which the lower-level model consumes the entire wood supply. In other words, a bilevel-feasible wood supply solution must be upper level feasible *and* be entirely consumed by the lower-level model (i.e. $\sum_{u \in U} \beta_{uo} = y_o, \forall o \in O'$).

The classic definition of AAC fails to account for the species-skewed negative fibre consumption bias (i.e. fails to account for realistic industrial fibre consumption behaviour). We propose an extended definition of AAC, which is *the maximum periodic harvest level that respects species-wise even-flow constraints, and will be entirely consumed by profit-maximising industrial consumers*. Given this extended definition of AAC, even-flow wood supply solutions that will not be entirely consumed by the agent are no longer feasible. If deterministic assumptions hold true⁸, this extended definition will completely eliminate the problematic species-skewed negative fibre consumption bias.

⁸The assumption of determinism is implicit to linear programming, which we use to model both upper and lower levels of the problem.

Although defining bilevel AAC is relatively straightforward, the resulting bilevel solution space is non-convex, which makes it difficult to solve the bilevel wood supply problem to optimality.

2.2.1.4 Bilevel Solution Space Convexity Issues

The objective of this research is to formulate and solve a bilevel optimisation model for the DWSPP. Early model development efforts focused on what we will hereafter refer to as the *general bilevel case*, which we hoped to solve to global optimality using an iterative solution methodology, based on the pioneering work of Fortuny-Amat and McCarl (1981) and Bard and Falk (1982). After developing the mathematical formulation for the general bilevel case, we recognised the potential for non-convexity of the bilevel solution space. This limited the worst-case performance of our initial solution methodology approach to local optimal solutions. Given the strategic importance of wood supply planning, we considered local optimal solutions to be of limited interest, particularly in the absence of reliable bounds on the optimality gap. Thus, we decided to focus subsequent efforts on identifying special cases of the general bilevel problem that we could solve to global optimality.

Non-convex cases can occur when the agent value-creation network features one or more convergent processes with saturated capacity constraints⁹. For example, this could occur if all chips in a network must flow through a single pulpmill (i.e. the convergent process), with a single digester that runs non-stop (i.e. shared resource with saturated capacity constraints). Appendix 2.B presents a counter-example proving non-convexity of the general bilevel solution space for the DWSPP.

By restricting the problem domain to non-problematic (i.e. convex) instances, we can define a special case of the bilevel problem that can be solved more easily. The special case, which is defined by excluding problematic instances from the general problem domain, occurs in two types of instances. Special case 1 corresponds to strictly divergent networks (i.e. no convergence of product flows at any process), and can be easily identified by analysing network topology of the agent model. Special case 2 corresponds to partially convergent networks with sufficient capacity at convergent processes so these will not be saturated at the bilevel optimal solution. Special case 2 may only be observable through empirical analysis of the agent model, and may not be detectable *a priori*, as checking for potential capacity constraint saturation requires us to first solve the lower level model.

Both types of species cases are separable in O' (i.e. the sustainable forest outputs used to segment AAC in the upper level model). Thus, we can develop a solution methodology for the special case that decomposes the lower level problem into convex species-wise subproblems, which can be solved to determine optimal species-wise upper bounds on harvest levels in the upper level problem. These species-wise upper bounds essentially represent valid cuts for the upper level solution space. These cuts form the basis for our solution methodology to solve the special cases. We cannot generate similar valid cuts for the general case, using this methodology, due to non-convexity of the general bilevel solution space.

The species-wise upper-bound volume for output $o \in O'$ (i.e. solution to lower level subproblem for output o) is equal to $\arg\max_{v_o} p(v_o)$, where $p(v_o)$ is a concave piece-wise linear function describing profit. The piecewise profit function is concave for each separable output o , for any non-negative consumption volume v_o (see Figure 2.2). At the leftmost end of each curve (i.e. the lower extremity

⁹A convergent process accepts input from two or more sustainable forest outputs $o \in O'$ (e.g. a mix of softwood and hardwood).

of the piecewise linear function domain), the agent utilises his limited resources to produce his most profitable product mix until one of his resource constraints is saturated. The agent then switches production to the next most profitable product mix. The slope of each line segment in the piecewise function represents unit profit from a given product mix. By definition, the slopes of these line segments will be monotonically decreasing, hence the concave shape of the piece-wise linear profit function.

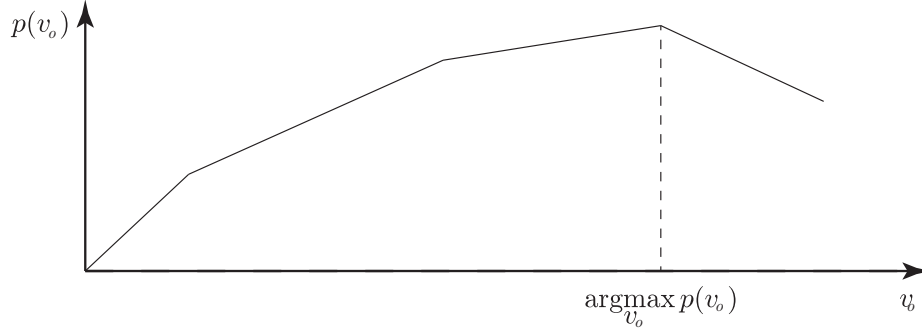


Figure 2.2 – Illustration of concave profit function of a hypothetical product o

2.2.2 Solution Methodology

We developed a solution methodology to solve the bilevel problem presented in §2.2.1. Our methodology is implemented on top of an iterative simulation framework, which we describe in Chapter 1. This framework, which was developed to simulate principal-agent interaction after several sequential two-stage rolling-horizon distributed wood supply planning cycles, links SilviLab and LogiLab software platforms¹⁰ to model principal and agent behaviour. See Appendix 2.C for more detailed framework implementation notes.

We originally set out to design an iterative solution methodology for the general case, however convexity issues¹¹ led us to focus algorithmic development efforts on the (convex) special cases, which we can solve to global optimality. By limiting the problem domain to the special cases, we eliminate the possibility of interaction between outputs $o \in O'$ (i.e. the bilevel problem becomes separable in O'). This property allows us to compute output-wise upper bounds on agent consumption behaviour, which can be used to optimally constrain the upper-level problem to produce the global optimal bilevel wood supply solution. This is the basis of our solution methodology for the special case. Algorithm 1 describes this solution methodology using pseudo-code.

Algorithm 1 essentially augments the upper-level model formulation presented in §2.2.1.1 with output-wise upper bounds (i.e. $y_o \leq \mu_o, \forall o \in O'$) on harvesting. Values for μ_o are derived from the optimal solutions to *lower-level subproblems* (referenced in line 2 of Algorithm 1).

Each subproblem is similar to the lower-level model formulation presented in §2.2.1.2, but with constraint (2.11) relaxed¹² and non-targeted outputs¹³ disabled. Disabling non-targeted outputs

¹⁰SilviLab and LogiLab software platforms are developed by the FORAC Research Consortium to model wood supply problems and forest products industry value creation network problems, respectively. See Chapter 1 for a more detailed description of these software platforms.

¹¹See §2.2.1.4 and Appendix 2.B.

¹²Equivalent to infinite external supply of output o .

¹³Non-targeted outputs correspond to $O' \setminus o$, that is the set O' excluding targeted output o .

ensures that only the targeted output o affects the subproblem solution (see Appendix 2.C for more detailed notes on lower-level model implementation, and how we disable the non-targeted outputs). The optimal solution to each lower-level subproblem yields the maximum volume μ_o of output o that would be voluntarily consumed by the agent if an infinite supply of output o was available. Conceptually, μ_o corresponds to $\operatorname{argmax}_{v_o} p(v_o)$ (see Figure 2.2).

For the special case, each μ_o constitutes a valid cut for the bilevel solution space. Applying similar cuts to the general case would possibly exclude the global optimal solution, due to non-convexity of the solution space. Algorithm 2 describes a methodology to solve the general case to global optimality by simple enumeration (i.e. iterate over the set of all possible convex subproblems, and return the best solution). We use the term *super-saturated process* to describe a process $p \in P_S$ (where $P_S \subset P$), whose capacity constraints are exceeded when subproblem solutions are summed in line 5 of Algorithm 2. It is possible that some process capacity constraints are violated when subproblem solutions are summed, due to non-convexity of general case solution space (i.e. the general case is not strictly separable in o , hence the need for enumeration to find feasible solutions). Subproblem definition is the same as for line 5 of Algorithm 1.

Algorithm 2 is of limited practical interest, as computational effort required to solve realistically-sized instances by enumeration would be prohibitive. However, efficiency of this algorithm could potentially be improved by replacing subproblem enumeration with a custom branching algorithm, thereby taking advantage of problem structure. Although we have not tested this approach, further development of computationally tractable methodologies to solve the general case represents an interesting direction for further research.

In practice, most lower-level datasets will correspond to one or the other of the special cases, which can be solved with relative ease using Algorithm 1. The test dataset used in our computational experiment, which we describe in the next section, is an example of special case 2.

2.2.3 Computational Experiment Methodology

This section describes the computational experiment we conducted to compare the performance of classic and bilevel wood supply models.

2.2.3.1 Test Dataset

We tested our bilevel solution methodology on a realistic synthetic dataset from Quebec, Canada. The study area is a forest management unit (FMU 031–53) located in the boreal forest region. It covers an area of approximately 102 thousand hectares. Approximately 88% of initial growing stock is from

Algorithm 1: Bilevel model solution algorithm (special case)

Output: Global optimal wood supply solution x^*

- 1 **foreach** output $o \in O'$ **do**
 - 2 Determine upper bound μ_o on agent consumption of output o (i.e. solve lower-level sub-problem).
 - 3 Set upper bound constraint $y_o \leq \mu_o$ on quantity of output o produced by the upper level problem.
 - 4 **end**
 - 5 Solve constrained upper-level problem.
-

Algorithm 2: Bilevel model solution algorithm (general case)

Output : Global optimal wood supply solution x^*

```
1 foreach output  $o \in O'$  do
2   | Generate subproblem (isolate output  $o$  in lower-level problem).
3   | Solve subproblem.
4 end
5 Sum subproblem solutions.
6 Build super-saturated process set  $P_s$  (i.e. find violated capacity constraints).
7 foreach combination of super-saturated process  $p \in P_s$  and output  $o \in O'$  do
8   | Generate subproblem (restrict certain combinations of output  $o$  and process  $p$ ).
9   | Solve subproblem.
10  | if subproblem feasible (i.e. no violated capacity constraints) then
11    | Add solution  $x$  to feasible solution set  $X$ .
12  | end
13 end
14 return Global optimal solution  $x^*$  (i.e. best solution in feasible set  $X$ ).
```

softwood species, with the remaining 12% of initial growing stock in hardwood species. Although some pure softwood stands are present, forest cover is primarily composed of softwood-rich mixed-wood stands.

Output from the upper-level (forest) model is aggregated into two outputs: *softwood* and *hardwood*. The lower-level (industrial) model has limited capacity for transforming hardwood (approximately 1/3 of potential sustainable wood supply). The classic wood supply model therefore systematically over-estimates short-term hardwood fibre consumption.

We use the same test dataset as in Chapter 1, which is an instance of special case 2. Although chip flows from both hardwood and softwood sawmills converge at the pulp mill, we have determined empirically that its capacity is sufficient to avoid saturation problems. We can therefore use Algorithm 1 to solve the bilevel problem to optimality for our test instance.

2.2.3.2 Iterative Simulation Framework

We use the two-stage rolling-horizon simulation framework presented in Chapter 1 as a testbed in which to compare the performance of classic and bilevel wood supply model formulations. At each simulated planning cycle, the principal and the agent make their moves sequentially, in a two-stage game. For our computational experiments, we chose to simulate 30 (5-year) replanning cycles, as this corresponds to the length of our wood supply model planning horizon. The framework simulates forest growth between each 5-year rolling-horizon replanning cycle. The principal has the advantage of the first move, which means he can set AAC to any level of his choosing.

The simulation algorithm can be summarised as follows:

1. *First stage: the principal determines his wood supply offer.* We simulate the wood supply planning process by solving a wood supply optimisation model. Which model we solve at this stage—either the classic (single-level) model or the extended (bilevel) model—depends on the scenario. The wood supply offer is communicated to the agent in terms of species-wise upper bounds on volume that can be harvested in the second stage.

2. *Second stage: the agent consumes a subset of the wood supply.* We simulate fibre consumption by solving a network flow model, to determine the profit-maximising subset of wood supply that the agent would willingly consume. Species-wise upper bounds from the first stage (i.e. AAC) are applied at this stage. We also implement *line-wise profitability constraints* (see §2.C), which simulate existence of multiple profit centres in the agent network.
3. Simulate rolling horizon forward one period (simulate evolution of forest state using growth and yield curves from long-term wood supply model).

2.2.3.3 Experimental Methodology

We present five scenarios. Within each scenario, simulation parameters¹⁴ for the industrial fibre consumption network are held constant for all 30 planning cycles. Table 2.1 summarises simulation parameters for each scenario.

Scenario 1 simulates *status quo* behaviour for both principal and agent, and acts as a control scenario. At each planning cycle, the principal maximises even-flow AAC (30-period horizon) using the classic wood supply model, then the agent maximises first-period profits (1-period horizon) by consuming an optimal subset of the wood supply offered by the principal. The principal does not consider the agent's fibre consumption capacity when determining AAC.

Scenario 2 presents a perfect-implementation bilevel scenario—rather than being allowed to replan harvesting on a one-period horizon (as is the case for the other scenarios), the agent is forced to exactly implement the first period of the principal's wood supply solution. This scenario shows the best-case performance of the bilevel model solution.

Scenario 3 is the basic bilevel scenario. The principal uses the bilevel model to determine AAC, and the agent is allowed to replan harvesting on a one-period horizon, choosing the profit-maximising subset of available wood supply. Due to the optimal formulation of the bilevel model and perfect anticipation of volume consumption, the agent always chooses to harvest the entire wood supply. However, the agent may select to harvest this volume from a different combination of forest types than that which was prescribed in the first period of the principal's optimal solution. This reflects the distributed nature of forest management planning on public forest land in many jurisdictions.

Scenarios 4 and 5 are variants of scenario 3, simulating reduction of softwood supply allocated to the agent to 80% and 60% of bilevel AAC. Adjusting AAC allocation indirectly creates a buffer stock to protect against the effects of agent harvest replanning (i.e. compensation for the principal's incomplete control of the fibre procurement process).

2.3 Results

We present experimental results in two stages. First, we present detailed results comparing output from the first planning cycle of scenarios 1 and 3. Next, we show results of simulating 30 sequential rolling-horizon planning cycles for scenarios 1 through 5.

For scenario 1, the control scenario, potential hardwood fibre supply is 64 583 m³, whereas actual consumption is only 20 800 m³. The difference between planned and executed hardwood fibre con-

¹⁴Mill capacities, costs, prices, client demand, etc.

Table 2.1 – Summary of scenario parametres

Scenario	Principal Model	Agent Model	Description
1	Classic	Basic	Control scenario (<i>status quo</i>)
2	Bilevel	Slave	Best-case bilevel scenario
3	Bilevel	Basic	Base bilevel scenario
4	Bilevel (80% attribution [†])	Basic	Reduced attribution bilevel scenario
5	Bilevel (60% attribution [†])	Basic	Reduced attribution bilevel scenario (stable)

[†]Of bilevel softwood AAC.

sumption volumes is due to the limited processing capacity at the (single) hardwood sawmill in the agent processor network. The entire softwood fibre supply is consumed by the agent, as we would expect, as both end-product demand and processing capacity for the softwood line are high enough to accommodate more softwood fibre than the forest can supply. This phenomenon (of consuming certain components of the wood supply entirely while other components are only partially consumed) can be observed to varying extents in practice, as processing capacity and market demand are often misaligned with the proposed wood supply. To consume the full softwood supply, while only harvesting a third of the hardwood supply, the agent must favour harvesting stands that have a higher proportion of softwood and lower proportion of hardwood.

For scenario 3 (i.e. the basic bilevel scenario), results show that our bilevel anticipation mechanism completely eliminates the over-estimation of hardwood fibre consumption volume. Fibre consumption by the agent is exactly equal to wood supply volumes offered by the principal (20 800 m³ for hardwood, and 323 759 m³ for softwood). This is the desired outcome from the bilevel model, and represents a global optimal solution for this instance. Note that the agent plans his own harvesting in the second stage of the simulation (using his single-period profit-maximising model), so harvest areas will not typically match the first period of the principal’s plan, although harvest volume is exactly equal in scenario 3.

Table 2.2 presents intermediate results from each step of the bilevel solution method¹⁵, for scenario 3. We present this data as an example of how we derive the optimal upper bounds to the wood supply problem from solutions to output-wise sub-problems. Hardwood consumption capacity (20 800 m³) is the binding constraint in this case. Our anticipation mechanism shows that the softwood line would have willingly consumed up to 584 861 m³ of softwood in the first planning period, however the species-wise even-flow constraints on the upper-level wood supply model limit long-term softwood harvest level to 323 759 m³. As expected, fibre volume consumed by the agent in the second phase of the simulation is exactly equal to the wood supply. The bilevel model eliminates the gap between planned and executed fibre consumption levels, thereby fulfilling its intended purpose. Figure 2.3 presents detailed results from the first planning cycle of scenarios 1 (classic model) and 3 (basic bilevel model).

We solve the bilevel model in less than twice the time required to solve the classic model. The classic model can be solved in a single step, which corresponds to approximately 13 seconds of CPU time for our test setup. The bilevel model requires $|O'| + 1$ steps to solve, which corresponds to approximately $(4 + 6) + 10$ seconds of CPU time using our test setup. We ran our tests on an Intel[®] Xeon[®] E5-2670 processor (20 MB cache, 2.60 GHz).

¹⁵See Algorithm 1.

Figures 2.5 and 2.6 show sequential replanning simulation results for five scenarios described in §2.2.3.3. For each scenario, Figure 2.5 plots species-wise AAC and fibre consumption for each of the 30 rolling-horizon replanning cycles. The same data are shown in Figure 2.6 using box-plots to illustrate the variability of periodic AAC and fibre consumption data across scenarios. The boxes encompass the inter-quartile range (IQR) with the median marked. The whiskers extend to 1.5 IQR past the nearest quartile. Observations outside this range are marked as outliers using a dot symbol.

Figure 2.4 shows planned and executed softwood harvest volumes for bilevel scenarios 3, 4, and 5. We include this figure to show that withholding a portion of bilevel AAC tends increase mean AAC, decrease mean harvested volume, and improve wood supply stability throughout the horizon (as show by increased tightness of the boxplots).

Table 2.2 – Bilevel solution method (intermediate results)

Stage	Description	Volume (m ³)	
		Hardwood	Softwood
1 (principal)	Upper bound on <i>hardwood</i> consumption	20 800	–
1 (principal)	Upper bound on <i>softwood</i> consumption	–	584 861
1 (principal)	Maximum even-flow wood supply levels	20 800	323 759
2 (agent)	Agent fibre consumption	20 800	323 759

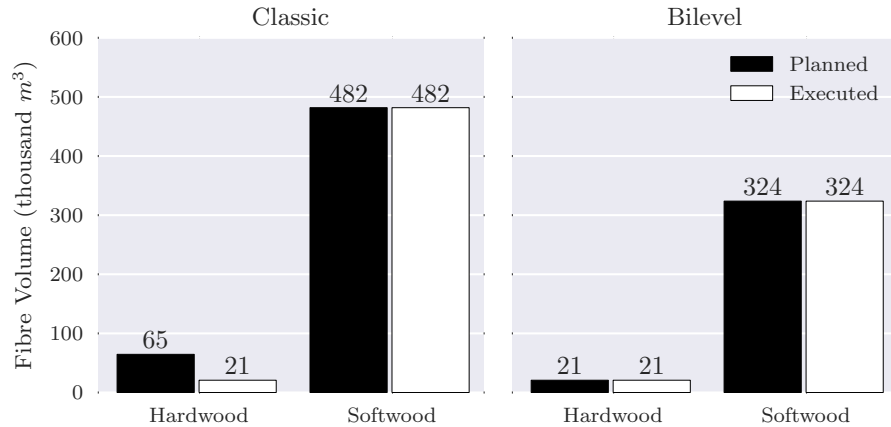


Figure 2.3 – Comparison of planned and executed volumes

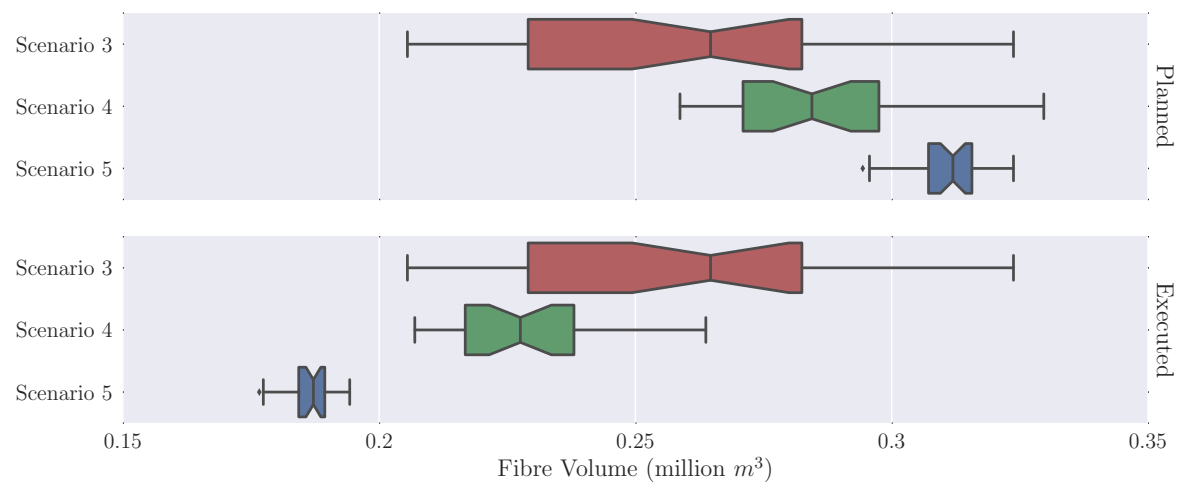


Figure 2.4 – Planned and executed softwood harvest volume for bilevel scenarios 3, 4 and 5 (correspond to 100%, 80%, and 60% softwood AAC attribution policies)

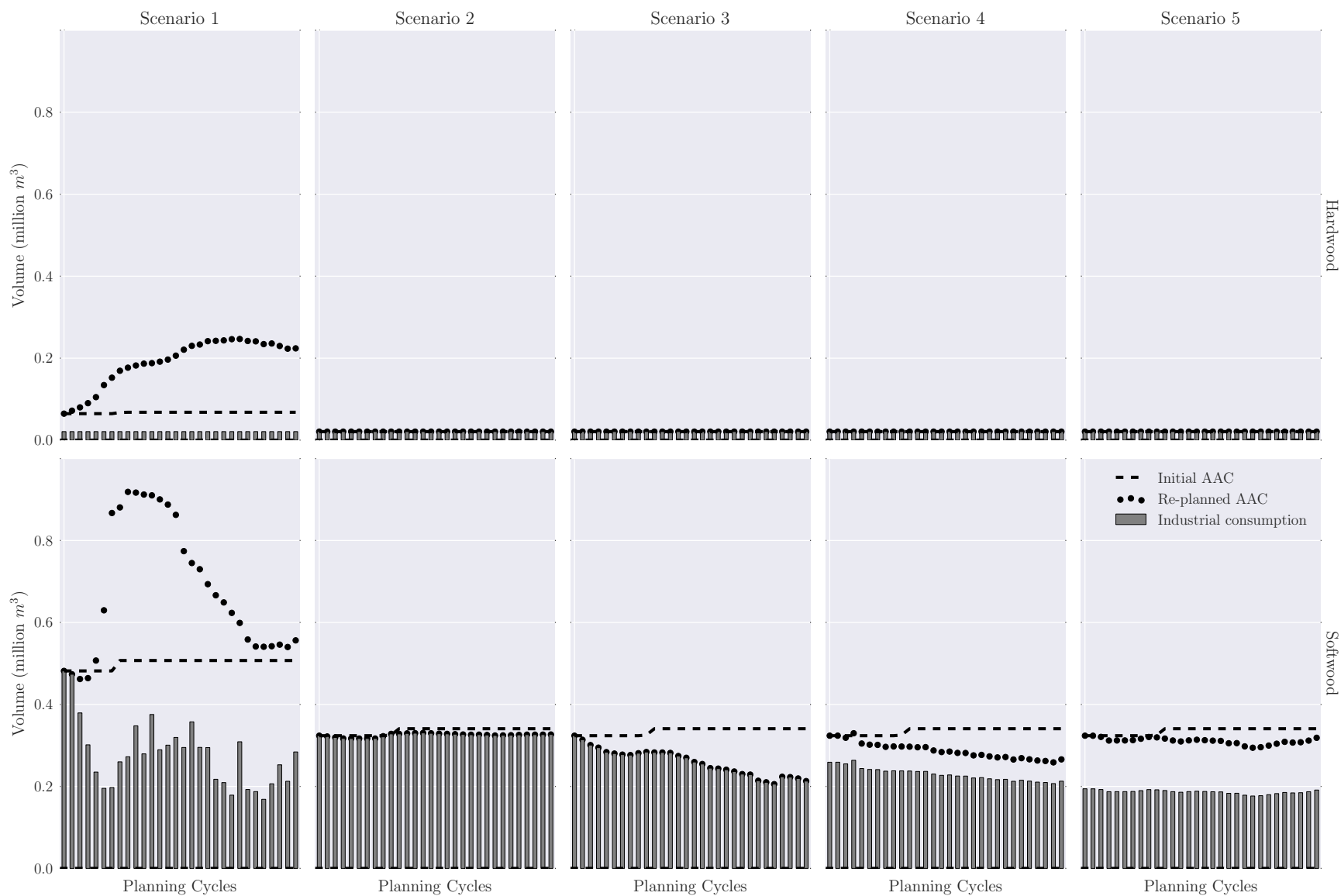


Figure 2.5 – Species-wise AAC and fibre consumption for scenarios 1 to 5 (time series)

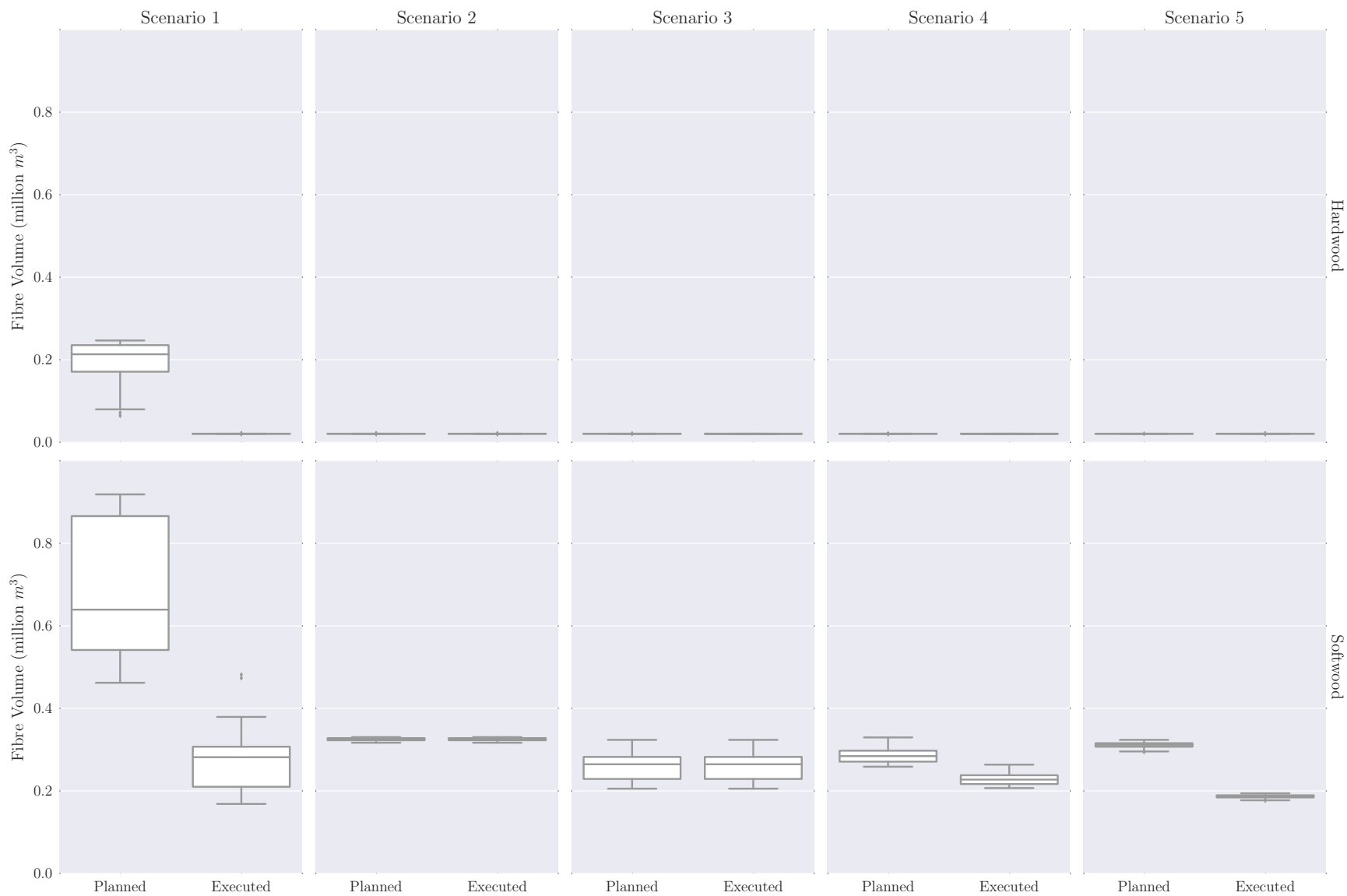


Figure 2.6 – Species-wise AAC and fibre consumption for scenarios 1 to 5 (boxplots)

2.4 Discussion

Scenario 1 shows the relative instability of the classic wood supply model. This is attributable to the species-skewed gap between AAC and fibre consumption volumes. Despite the harvest levels being systematically lower than AAC, the agent’s preference for harvesting high-softwood-content stands gradually shifts the composition of the residual forest cover towards a higher hardwood content. This undesirable shift in forest composition is not predicted by the classic wood supply model. In Chapter 1, we use the term *systematic drift effect* to describe this phenomenon.

The principal uses the bilevel model to determine AAC for scenarios 2 through 5. By virtue of its formulation, the bilevel model completely eliminates the volume gap between AAC and fibre consumption. Note that we simulate perfect anticipation of agent fibre consumption *volume*. For scenarios 3 through 5, we allow the agent to plan his own harvesting in the second phase of each planning cycle simulation—this explains the residual instability in long-term wood supply.

Scenario 2 forces the agent to harvest the exact forest units that form the basis of the first period of the principal’s optimal bilevel solution. The purpose of this scenario is to show that wood supply tracks almost perfectly along the initial bilevel AAC solution (even after 30 rolling-horizon replanning cycles) under best-case conditions (i.e. when the principal controls wood procurement planning and execution all the way to the mill gate). In practice, the decoupling point between the principal and the agent is not typically located this far downstream. Scenarios 3 through 6 simulate a more conventional decoupling point.

Scenario 3 shows vastly improved stability softwood supply levels, relative to the control scenario. Note the gradual downward trend of the softwood fibre supply for scenario 3, which contrasts with the even supply profile simulated in scenario 2. The contrast between scenarios 2 and 3 shows that the bilevel model is sensitive to deviations from the optimal wood supply model solution. Sensitivity to deviations from the optimal solution is typical of deterministic optimisation models, as optimal solutions are invariably located along the boundary of the feasible region—even the slightest deviations from the optimal solution (or error in constraint right-hand-side values) can induce problem infeasibility.

Scenarios 4 and 5 show the effect of reducing the proportion of AAC that is allocated to the agent, in an attempt to compensate for the residual drift seen in scenario 3. Reducing allocation is an indirect way for the principal to induce a buffer stock in the standing timber inventory. This tends to move the executed (second-stage) solution away from the feasible boundary of the planned (first-stage) solution space, thereby improving the robustness of the distributed wood supply planning process. Scenario 4 shows a marked reduction in drift compared with scenario 3. Residual drift is virtually eliminated in scenario 5. Intuitively, withholding 40% of AAC seems like a high penalty to eliminate the residual drift in the bilevel model. We conjecture that, using a more direct management approach to maintaining a buffer stock in standing timber inventory, as described in [Raulier et al. \(2014\)](#), it may be possible to achieve higher stable bilevel AAC levels. This represents a promising direction for further research.

By stabilising the long-term wood supply, scenario 5 succeeds in restoring credibility to the wood supply planning process, albeit at a relatively high cost in terms of withheld AAC. Furthermore, scenario 5 makes less optimistic assumptions regarding agent behaviour than scenarios 2 (which simulates a perfectly compliant agent). Scenario 5 respects the even-flow pattern prescribed by the wood supply model constraints. Assuming that the even-flow constraints are valid and necessary (although not

sufficient) conditions for sustainability of the forest management plan¹⁶, and that the principal’s responsibility to ensure sustainability must absolutely supersede any desire to maximise short-term TL volume allocations, scenario 5 represents the only example of a principal-feasible policy in this study.

We simulated the distributed wood supply planning process as a two-stage sequential game, where the principal proposes his wood supply in the first phase and the agent consumes a profit-maximising subset of the wood supply in the second phase. At this point, we can conjecture that stable increases in AAC may be achievable if the principal and the agent were allowed to iteratively adjust their respective supply and demand offers within a given planning cycle. This represents a promising direction for further wood supply policy research. From a game-theoretic perspective, extending the two-stage game simulated in this study to include an iterative negotiation dimension corresponds to a *repeated game* or *supergame* in game theory. Under certain conditions *supergames* are known to converge on *socially optimum* equilibrium solutions (i.e. collaborative solutions) that are globally superior to the (optimal) selfish behaviour in the context of non-repeated (i.e. one-shot) games (Fudenberg and Tirole, 1991). The concept of supergames could also be used on a larger scale, to model principal and agent anticipation of upcoming planning cycles (and, potentially, memory of past planning cycles). Ultimately, both scales could be nested (i.e. iterative negotiation within each planning cycle, combined with anticipation of upcoming planning cycles). Although technically challenging, these hypothetical nested-supergame models might be harnessed for practical application using a *metagaming* approach (Howard, 1971), potentially providing a wealth of valuable insight to guide high-level government policy-makers.

2.5 Conclusion

In Chapter 1, we conjectured that extending the classic wood supply model formulation to anticipate industrial fibre consumption would improve wood supply stability (i.e. mitigate risk of wood supply failure). The current chapter tests this conjecture.

We framed this problem using agency theory, and proposed mathematical formulations to describe the optimisation problems of the principal and the agent. We then combined principal and agent problems into a bilevel optimisation model.

Using a counter-example, we showed that the general case of the bilevel problem is non-convex. We presented a solution algorithm to solve the general case to global optimality, through enumeration of feasible solutions. However, an enumeration-based strategy is computationally intractable for realistically-sized instances. By imposing a restrictive condition on the topology of the agent’s problem, we isolated a special case of the bilevel problem, which can be decomposed into output-wise convex subproblems. We presented an algorithm that solves the special case to global optimality.

We tested our solution methodology on a synthetic dataset of realistic size and complexity, and compared results to output from the classic (single-level) wood supply optimisation model. Using a

¹⁶There has been considerable debate over the validity and necessity of including even-flow or non-declining yield constraints in wood supply optimisation models (Gunn, 2009). Nonetheless, one or the other of these constraint formulations has traditionally been included in almost all wood supply models in Canada since the advent of the use of linear programming to optimise wood supply planning (with the notable exception of the province of Ontario). We have included even-flow constraints in both the classic and bilevel optimisation model formulations used in this study, as this allows us to measure the impact of extending the *status quo* wood supply model formulation to include explicit anticipation of industrial fibre consumption behaviour. For more information on the effects of even-flow constraints and alternative model formulations, we invite the reader to consult Luckert and Williamson (2005).

series of five scenarios, we showed that the bilevel model improves long-term wood supply stability, although instability is not completely eliminated by the bilevel model. We showed that the principal can compensate for this residual instability by withholding (i.e. not attributing) a large fraction of bilevel AAC. We conjecture that a similar stabilising effect could be achieved more efficiently (i.e. at a lower cost in terms of withheld bilevel AAC) using a more direct buffer stock modelling approach.

The bilevel solution algorithm for the special case converges on a global optimal solution in less than twice the time required to solve the classic (single-level) model formulation. Considering that these wood supply models are solved infrequently (i.e. once per planning cycle), this increase in solution time is not obviously problematic. The bilevel model has the same output data format as the classic model and can be solved using comparable computational effort. As such, the bilevel model formulation constitutes a technically compatible and conceptually superior alternative to the classic model.

The current study examines the performance of a bilevel model formulation in the context of a two-stage principal-agent game. We recommend that research effort on bilevel wood supply model formulations be extended to *supergame* contexts, both in terms of intra-cycle principal-agent negotiations, and inter-cycle anticipation of future wood supply planning games. To cope with the complexity of using these hypothetical nested-supergame models in a practical government-policy-setting environment, we suggest the adoption of a metagaming approach to wood supply planning as an appropriate starting point for further study.

2.6 Acknowledgements

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Appendices

2.A Sources of Principal-Agent Antagonism

The principal has the long-term responsibility to ensure a sustained wood supply (hence the even-flow constraints in the wood supply model), but aims to maximise economic activity from the exploitation of the forest resource (hence the wood-supply-maximisation objective function). The agent aims to maximise short-term profit by transforming the wood supply into forest products (e.g. lumber, paper, etc.). The antagonism between the principal and agent is linked to either (a) binding agent capacity constraints or (b) the presence of negatively-valued subsets of the wood supply. Either of these factors will prevent the profit-maximising agent from consuming the entire wood supply, which in turn induces the problematic negative consumption bias described in Paradis et al. (2013).

The test dataset used in the computational experiments features binding agent capacity constraints. Our test dataset has two lines (which we will refer to as *hardwood* and *softwood*, based on an aggregation of tree species that grow in our test forest). All the hardwood harvested from the forest must pass through a single hardwood sawmill. The hardwood sawmill capacity is approximately one third of the

maximum sustainable hardwood supply level determined by the principal using the classic wood supply optimisation model. The softwood line is profitable and has sufficient capacity to process the entire softwood supply offered by the principal. The agent therefore has an incentive to utilise his entire softwood allocation, but limit his hardwood consumption to the capacity of the hardwood sawmill. We have only permitted clear-cut harvesting in our test model, which means the agent only has take-all and leave-all options for each harvestable forest unit. In order to achieve the correct hardwood/softwood mix in his harvesting plan, the agent may select harvest units that have a lower proportion of hardwood than the harvest units appearing in the first period of the principal's optimal wood supply solution. This species-biased deviation from the principal's wood supply plan increases risk of future wood supply shortages.

In the case of wood supply offers with negatively-valued subsets, the only way the principal has to motivate the agent to act is by allowing him to harvest part of the forest (i.e. the agent can choose to consume any subset of wood supply offered by the principal). In practice, it is difficult (impossible) for the principal to force the agent to consume timber at a net loss, so this is a real problem. The principal is incited to propose plans where part of the output is not interesting for the agent, as this allows him to increase his wood supply offer (this is desirable, given his objective function). Including this negatively-valued part in the short-term wood supply allows the principal to increase simulated long-term wood supply offer¹⁷. However, the agent only plans his consumption on a short-term basis, and has no incentive to consume the negatively-valued subset of wood supply. It may be impossible for the principal to offer the globally optimal plan *and* force the agent to use all of the wood offered. By failing to consume the uninteresting part of the supply, the agent may compromise feasibility of the principal's wood supply plan.

We illustrate this second source of antagonism with an example. Suppose the principal P can offer H_1 to the agent A , which has a value $v(H_1) = 10$. To this offer, the principal can add H_2 , which has a value $v(H_2) = 2$. However, H_1 and H_2 can only be consumed sustainably if they are bundled with H_3 , which has a value $v(H_3) = -1$. The best long-term solution for both parties is for the agent to consume H_1 , H_2 and H_3 for $v(H_1 \cup H_2 \cup H_3) = 11$. However, the principal knows that if he offers all three lots, the profit-maximising agent will only take H_1 and H_2 for $v(H_1 \cup H_2) = 12$. The principal knows that this is unsustainable, so he only offers H_1 which is sustainable but has a lower value of 10.

Note that if the principal could bundle the uninteresting part with a more interesting surplus, then the antagonism would disappear. However, this bundling would require a more highly-constrained contract binding the agent to the principal. This bundling option is not typically available to the principal in practice, leaving him with no rational choice but to lower the wood supply offer until the agent willingly consumes it all. Determining the maximum species-wise even-flow wood supply offer that will be totally consumed by the agent is not a trivial problem. The antagonism between the two levels can induce non-convexity and non-linearity in the solution space, when we constrain the principal's problem such that a wood supply contract is principal-feasible only if the agent consumes it entirely.

¹⁷For example, simulating harvesting of relatively unproductive or over-mature parts of the forest and regenerating them into higher-productivity stands in a wood supply model may increase the simulated availability of fibre in a future time period. This *allowable cut effect* is a well documented, but potentially problematic, forest policy instrument. For more information, see Luckert and Haley (1995).

2.B Proof of Non-Convexity

This appendix contains a proof of non-convexity of the solution space for the general bilevel wood supply problem. It may be helpful to recall that, for a convex set of feasible solutions, any linear combination (i.e. convex combination) of two solutions (e.g. $0.5x_1 + 0.5x_2$) will yield a third feasible solution. The basis for our proof of non-convexity is to show, using a simple counter-example, that this property does not hold for all general bilevel problem instances.

Specifically, we describe three solutions to our hypothetical counter-example problem. These three solutions lie along a line segment in solution space. The endpoints of this line segment are bilevel-feasible, however the mid-point is infeasible. By definition, this solution space cannot be convex. Because the counter-example problem is an instance of the general bilevel problem, we can conclude that the general bilevel problem can be non-convex. Any optimisation algorithm for the general bilevel problem would therefore have to assume non-convexity of the solution space, or risk terminating prematurely at a local optimal solution.

Also, it may be helpful to recall the definition of the general bilevel problem solution space. For a wood supply solution to be bilevel-feasible, it must of course be both upper- and lower-level feasible. Furthermore the wood supply must be entirely, and willingly, consumed by the profit-maximising agent in the lower-level model. The second solution of our solution triplet (i.e. the midpoint of our line segment in bilevel solution space) is bilevel-infeasible because it does not respect the second condition for feasibility, *viz.* the agent will not willingly consume the entire wood supply, as it is more profitable for him to leave one unit of hardwood unconsumed.

We now describe the counter-example problem instance as follows. The context for our counter-example is a simple setup where the principal offers a supply of both softwood and hardwood to the agent. Similarly to the dataset we use in our case study (see §2.2.3.1), the agent is actually composed of two independent sub-agents (i.e. softwood and hardwood lines) which must be independently profitable. Each sub-agent maximises his own profit (i.e. is not willing to reduce his profit for the benefit of the other). There are three types of transformation processes: boards, paper, and cogeneration. The *boards* process is equally profitable for both lines (+50 \$/m³ for both softwood and hardwood), and has a transformation capacity of two input units for each line. The *paper* process is profitable for both lines, but at different rates (+50 \$/m³ for the softwood line, +10 \$/m³ for the hardwood line); it has a transformation capacity of three input units for each line. The *cogen* process is marginally profitable for the softwood line (+1 \$/m³), and marginally unprofitable for the hardwood line (−1 \$/m³); it has a transformation capacity of one input unit for each line.

Transforming inputs using the paper process requires utilisation of a common resource for both lines (for example, this could correspond to pulp digester capacity). The common resource has a limited capacity of 6 units, which quickly becomes saturated. There is a difference in line-wise efficiency for the utilisation of the common resource. The softwood line uses two units of the common resource for each unit of input transformed, whereas the hardwood line uses one unit of the common resource for each unit of hardwood consumed (for example, this could correspond to softwood chips requiring twice as much time to digest as hardwood chips). Problems with non-convexity of the solution space may arise when this common resource becomes saturated.

We illustrate the counter-example in Figure 2.7. We use the symbols S and H to illustrate shared

resource capacity utilisation by softwood and hardwood lines, respectively. We show three optimal agent resource allocations, corresponding to three different wood supply offers from the principal. These three wood supply offers correspond to three distinct solutions, which form a line segment in bilevel solution space. The midpoint of this line segment is bilevel-infeasible.

The starting point of the line segment in solution space is a wood supply offer of 4 units of softwood and 4 units of hardwood (see Figure 2.7, Solution 1). The optimal allocation of the softwood supply is 2 units to boards and 2 units to paper. The optimal allocation of the hardwood supply is 2 units to boards and 2 units to paper. Shared paper resource capacity is saturated, with 4 resource capacity units utilised by the softwood line and 2 units utilised by the hardwood line. The agent consumes the entire offer, for a profit of \$320, therefore this point is bilevel-feasible.

The endpoint of the line segment in solution space is a wood supply offer of 6 units of softwood and 2 units of hardwood (see Figure 2.7, Solution 3). The optimal allocation of the softwood supply is 2 units to boards, 3 units to paper and 1 unit to cogen. The optimal allocation of the hardwood supply is 2 units to boards. Shared resource capacity is saturated, with all 6 resource capacity units utilised by the softwood line. Once again, the agent consumes the entire offer, for a profit of \$351, therefore this point is also bilevel-feasible. This also corresponds to the global optimal solution for this problem.

The midpoint of the line segment in solution space is a wood supply offer of 5 units of softwood and 3 units of hardwood. The optimal allocation of the softwood supply is 2 units to boards and 3 units to paper. The optimal allocation of the hardwood supply is 2 units to boards. Shared resource capacity is saturated, with all 6 resource capacity units utilised by the softwood line. The agent does *not* voluntarily consume the entire offer—his maximum profit of \$350 for this wood supply offer is achieved by leaving one unit of hardwood supply unconsumed (the agent avoids allocating the remaining unit of hardwood to the marginally unprofitable hardwood cogen process). Thus, the midpoint solution is bilevel-infeasible.

Given that all three solutions are located along a line in solution space, infeasibility of the intermediate point proves non-convexity of this solution space¹⁸.

¹⁸By definition, given a line segment whose endpoints lie inside a convex space, it is not possible for any point along this line segment to lie outside the convex space.

Wood Supply Offer		Line Capacity Utilisation		Shared Resource Capacity Utilization		
		Softwood	Hardwood			
Solution 1	4 softwood units 4 hardwood units	Boards			S S S S H H	
		Paper				
		Cogen				
Solution 2	5 softwood units 3 hardwood units	Boards			S S S S S S	
		Paper				
		Cogen				
Solution 3	6 softwood units 2 hardwood units	Boards			S S S S S S	
		Paper				
		Cogen				

Figure 2.7 – Simple counter-example illustrating non-convexity of bilevel problem solution space

2.C Lower-Level Model Implementation Notes

We present a more detailed description of our simulation framework’s underlying data model. The LogiLab platform is used to model both the agent-anticipation mechanism in the bilevel model and to simulate actual agent fibre consumption in the second stage of the iterative rolling-horizon replanning simulation process. The LogiLab data model can be described as a network of abstract processors¹⁹ connected by product flows. Each processor node represents a business unit in the value creation network (e.g. sawmill, pulpmill, end-product-client, etc.).

At the upstream (source) end of this network is the interface between upper and lower level models. Outputs from the first planning period in the upper level²⁰ model (i.e. outputs $o \in O$) represent raw material that can be transformed by the lower-level network. At the downstream (sink) end of the lower level network, external clients are willing to pay exogenously determined unit prices to satisfy a bounded demand for each end product. Profits are induced by pulling a subset of potential fibre supply through the network to satisfy a subset of end client demand.

When raw upper-level model outputs are first pulled into the lower network, they must go through one of several front-line processor nodes that convert the raw wood supply into species-wise assortments of

¹⁹An abstract processor consumes inputs and resources, and produces outputs. Abstract processors can be topologically connected to form a network.

²⁰In the context of our bilevel model, the terms *upper level* and *lower level* refer to principal and agent decision variables, respectively.

logs. These front-line processors simulate the interface between the forest and the mills (i.e. the process of harvesting and delivering logs to mills, including transportation cost, which can vary depending on the forest zone from which the raw volume inventory was procured).

Raw upper-level volume is classified by species, and this species-wise distinction may (optionally) be maintained as the outputs are pulled into the lower-level network, depending on configuration of front-line processors. For example, in the case of our test dataset, the front-line processors are configured to convert raw upper-level volume into assortments of either hardwood or softwood logs of various sizes²¹.

Due to the abstract nature of the lower-level processor implementation, it is possible to simulate any combination of divergent and convergent product flows. Strictly divergent networks correspond to special case 1, and can be solved to global optimality using Algorithm 1. Special case 2 occurs when the network includes convergent product flows, but no joint capacity constraints are saturated. Although somewhat more difficult to detect, special case 2 is not problematic and can also be solved to global optimality using Algorithm 1. For special cases 1 and 2, each product line $o \in O$ can be treated as an independent subproblem.

The product-wise subproblems can be represented using the lower-level model by disabling all non-targeted outputs²². We can then easily solve each subproblem to obtain the optimal subproblem solutions without having to explicitly locate intermediate inflection points of profit function $p(v_o)$ (see Figure 2.2). This corresponds to the maximum volume that the agent can be expected to consume for a given output o , which we use as an upper bound on harvest level for output o in the final step of Algorithm 1. In other words, optimal solutions of the output-wise subproblems can be used to define valid (and sufficient) cuts for the solution space of the bilevel optimisation problem (for special case instances).

We aggregate all fibre flows into the agent model into a number of input lines, corresponding to the species groups the principal uses to express AAC (e.g. our test case has *hardwood* and *softwood* lines), and require fibre flows from each of these lines to be independently profitable using *line-wise profitability constraints*. The purpose of the line-wise profitability constraints is to model a common situation in many real-world value creation networks.

Subsets of the agent network may be independently owned and managed, and tend to specialise in processing certain species (e.g. hardwood or softwood). Demand for a given species group may be limited by local processing capacity, exogenous end-client demand, or exogenous market prices (i.e. production will cease before capacity is saturated if cumulative unit procurement and processing costs exceed unit revenues, for a given product). The line-wise profitability constraints ensure that all mills that process a certain species group (i.e. each line) ceases production before it drops below maximum profit. The basic LogiLab model formulation assumes centralised network planning (i.e. maximises profit for the entire network)—the model would, were it not for the line-wise profitability constraints, induce one or more lines to continue production beyond the profitability threshold if this was beneficial

²¹Our upper-level dataset uses a more fine-grained classification of tree species, which is aggregated into *hardwood* and *softwood* log types by the front-line processors.

²²Non-targeted outputs can be disabled by manipulating input parameters of the upper level model, setting conversion efficiency and conversion cost of front-line processors to null values. Instead of converting wood supply units to log assortments, the modified front-line processors now consume all non-targeted outputs at zero cost, which blocks non-targeted outputs from further flow through the lower-level model network.

to the network as a whole.

The line-wise profitability constraints are implemented by updating the species-wise upper bounds set by the principal in the first stage, in the event that AAC exceeds the maximum volume that a given line can profitably consume. To determine these maximum line-wise profitable volumes, we simply solve the output-wise submodels (with non-targeted outputs disabled) described in bilevel solution methodology (see §2.2.2).

2.C.1 Computational Experiment Dataset

This section provides more information on the synthetic dataset we used as input to the lower level model for the computational experiments. The case study dataset does not represent an actual agent network, however these parameters were synthesised from realistic data that was compiled for previous research projects realised by the FORAC Research Consortium. See §1.2 for a schematic representation of the value creation network used in the computational experiments.

The lower level data model allows for an arbitrary number of external fibre suppliers with each supplier potentially having distinct input parameters. Our test dataset features a single fibre procurement source, thus all input units of fibre in the lower-level model have the same cost. In reality, procurement cost may vary due to a number of factors (e.g. distance from mill, accessibility, terrain, choice of harvesting system and silviculture prescription, species mix, etc.). However, modelling variable procurement cost was not necessary given the objectives of our computational experiments, and would only have served to obfuscate the results. We model the forest (i.e. external supply of fibre for the network) as a business unit, which encapsulates processes that convert raw fibre to assortments of logs. The forest business units produce three types of softwood logs (small, medium, large) and one type of hardwood log.

The three types of softwood logs can flow to any of the three softwood sawmills in the network. The softwood sawmills produce four types of softwood lumber (2×3 , 2×4 , 2×6 , 2×8) and softwood chips. Softwood lumber can be sold to a single softwood lumber external customer.

The hardwood logs are processed by the single hardwood sawmill, which produces a single type of hardwood lumber and hardwood chips. Hardwood lumber can be sold to a single hardwood lumber external customer.

Both hardwood and softwood chips may flow from the sawmills to the paper mill, where they can be converted to paper. The paper can be sold to one of two paper external customers. There is no external customer for chips.

The lower level model includes *storage* process at each business unit. Raw volume can be stored at roadside in the forest. Sawmills have storage processes for logs, lumber and chips. The pulpmill has storage processes for chips and paper. Finally, each external client has a storage process for the products they accept. Periodic storage costs are modelled as a proportion of cumulative product cost, which depends on the amount of processing that a unit of product has undergone. Thus, storage costs increase as products move through the network, with the least expensive storage cost being at source nodes and most expensive storage costs being at sink nodes.

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Chapter 3

Exploring Bilevel Wood Supply Policy Space

In Chapter 1, we introduced a two-stage rolling-horizon replanning simulation framework. Using this framework and a realistic test dataset, we simulated *status quo* interaction between the principal (i.e. government steward of public forest) and the agent (i.e. industrial fibre consumer). We concluded Chapter 1 by observing that the classic wood supply optimisation model formulation does not predict wood supply failure under certain circumstances¹, and conjectured that the risk of wood supply failure could be mitigated by extending the classic wood supply model to explicitly anticipate industrial fibre consumption behaviour.

In Chapter 2, we presented a bilevel wood supply model formulation that anticipates fibre consumption, and developed a solution algorithm to solve a special case of the problem to global optimality. We compared performance of classic and bilevel wood supply models using the simulation framework and test dataset introduced in Chapter 1. We concluded that the bilevel model can mitigate the risk of wood supply failure, although there is some residual instability in long-term wood supply.

Together, the iterative simulation framework and the bilevel model constitute both a methodological and technological advance in distributed wood supply planning. In this chapter, we use our bilevel model and iterative simulation framework to explore the bilevel wood supply policy space. Using a scenario-based analysis, we test the impact, on stability and magnitude of the wood supply, of varying five different aspects of forest management policy. We implement these tests by varying model input parameters controlling AAC attribution level, planning horizon length, tightness of even-flow constraints, variability of fibre demand, and centralisation of agent network planning. Simulation results provide some new insight into wood supply planning policy space, and reveal several directions for further research.

3.1 Methodology

We organised simulation results into five scenarios series. Each scenario simulates 30 sequential two-stage rolling-horizon replanning cycles, using the bilevel model and iterative framework presented in previous chapters.

The first scenario series tests the impact of varying AAC attribution level on bilevel wood supply

¹Specifically, the classic wood supply model fails to predict wood supply failure when industrial fibre consumption is a species-skewed subset of AAC.

stability and magnitude. Partial allocation of AAC (i.e. withholding a portion of the AAC determined by the wood supply optimisation model) is a common policy instrument used by government planners. The reference bilevel scenario simulates 100% attribution of AAC at each planning cycle, however in theory the principal has total control over fibre allocation, and is free to deviate from wood supply model output when attributing timber licenses to the agent. It is not clear how the bilevel model will react to various AAC attribution levels after repeated rolling-horizon replanning cycles. We test a number of attribution levels between 110% and 60% of softwood AAC. The purpose of the over-attribution scenarios is to validate the hypothesis that systematically exceeding AAC has a negative effect on wood supply. Conversely, the purpose of the under-attribution scenarios is to validate the hypothesis that allocating only a subset of AAC can improve wood supply stability. Note that we only test the effect of varying *softwood* AAC attribution level. Saturated hardwood fibre capacity constraints in our test dataset already limit bilevel softwood AAC, so reducing hardwood AAC attribution level below 100% would not constitute a rational policy alternative for our test instance, and would make results more difficult to interpret. Consequently, hardwood AAC attribution level is held constant at 100% of bilevel AAC for all scenarios in this series.

The second scenario series tests the impact of varying principal and agent planning horizon lengths on stability and magnitude of bilevel wood supply. The reference scenario features a 30-period planning horizon for the principal, and a 1-period planning horizon for the agent. Results provide insight into the relevance, and potential benefits, of maintaining long principal planning horizons and of providing incentives for the agent to lengthen his planning horizon. Although the length of the planning horizon for wood supply modelling varies among jurisdictions, the choice of horizon length tends to be very stable within jurisdictions. Planning horizon length is typically not regarded as a policy instrument, however it is one of the basic input parameters that is controlled by the principal. This scenario series provides insight into bilevel model response to varying principal and agent planning horizon lengths, after repeated rolling-horizon replanning cycles.

The third scenario series tests the effect of varying the tightness of the species-wise even-flow constraints in the wood supply model. The reference scenario allows 5% variability between the highest and lowest periodic species-wise harvest levels throughout the 30-period planning horizon (see §1.3.1 for a detailed mathematical formulation of the long-term planning model). We test constraint gap settings between 0% and 50%. The typical 5% gap value is not clearly supported by scientific evidence. This scenario series provides some insight into the effects of a sustained-yield wood supply policy using our bilevel analysis framework.

The fourth scenario series simulates stochastic variation of agent fibre consumption at each planning cycle. The reference scenario simulates deterministic fibre consumption (i.e. perfect anticipation of the volume of fibre that the agent will consume in the first period of each planning cycle). In practice, it is highly unlikely that the principal can achieve perfect anticipation, therefore this scenario series tests the sensitivity of wood supply magnitude and stability to stochastic variability in fibre demand. We implement this scenario by randomly determining the proportion of AAC consumed at each planning cycle, which we sample from a uniform distribution on the interval $[1.1, 0.9]$ (i.e. $\pm 10\%$ of AAC). By allowing the agent to exceed AAC in certain periods, this scenario series simulates a potential relaxation of current AAC attribution policy, which tends to interpret AAC as a strict upper bound on fibre consumption volume.

The fifth and last scenario series simulates relaxation of the line-wise agent profitability constraints. In the reference scenario, these constraints model the distributed nature of agent fibre procurement planning behaviour (i.e. the fact that subsets of the business units that make up the value creation network are managed as independent profit centres), and play an important role in inducing the species-skewness in the fibre consumption behaviour. Relaxing these constraints in the test scenario allows us to simulate more collaborative fibre procurement planning behaviour in the agent. In the test scenario, the softwood line can choose to subsidise the capacity-bound hardwood line to push excess hardwood supply through a negative-net-value storage process, which opens up the possibility of harvesting valuable softwood that would otherwise be locked up in hardwood-rich mixed-wood stands. By potentially mitigating the problematic species-skewness of the fibre consumption profile (which limits AAC in the reference scenario), the collaborative agent scenario opens up the possibility for the principal to offer higher AAC. Although the test scenario requires the agent to modify his fibre consumption planning behaviour (i.e. implement some form of cross-line subsidy), the potential for sustainable wood supply increases is in the best interest of both principal and agent, and represents an interesting opportunity to develop policy instruments to incite collaborative agent planning and benefit sharing.

We use the same case study dataset throughout this thesis, including the present chapter. When compiling this dataset, we deliberately configured the agent’s value creation network with limited processing capacity for hardwood fibre. This induces the species-skewed negative consumption bias described in Chapter 1, showing that softwood fibre consumption can be indirectly constrained by hardwood processing capacity despite sufficient agent capacity to profitably process the entire softwood AAC. Although tension between hardwood and softwood fibre consumption is common in practice, there are many other potential problematic principal-agent configurations that we have not been explored here. Our objective in this thesis is to clearly document one potentially problematic (albeit common) case, and to propose a mathematically optimal and computationally tractable extension to the classic wood supply model to address this case.

Although the simulation results presented here cannot be directly generalised to other cases, our rolling-horizon simulation framework and bilevel modelling methodology were deliberately designed to be potentially applicable to any *distributed* wood supply planning problem (i.e. where the roles of forest land owner and industrial fibre consumer are played by independent agents). Collectively, the scenarios presented in this chapter show that our iterative framework and bilevel model can be used to provide valuable insight into wood supply policy space. For example, some areas are characterised by asymmetrical industrial demand between spruce and fir², which would induce principal-agent problems similar to the one described in our case study. Other areas feature an over-abundance of low-quality fibre that, for various operational and political reasons, must be harvested to gain access to the high-quality subset of the wood supply.

All simulation results presented in this chapter use the test dataset described in Chapter 1. Unless specified otherwise, all scenarios use the bilevel model to determine wood supply at each planning cycle and simulate perfect anticipation of agent fibre consumption volume.

The reference scenario for all series is the base bilevel scenario³, which we first introduced as *Scenario 3*

²Spruce tends to be more highly valued than fir, due to longer kiln-drying times required for fir.

³For the reference scenario, the principal uses the bilevel model to determine AAC, and the agent is allowed to replan

in Chapter 2 (see §2.2.3.3). We include the reference scenario in all scenario series, and specify where it can be found in each sequence (it changes position and name, depending on the context).

3.2 Results

Table 3.1 lists the five scenario series, cross-referenced to document sub-sections.

Scenario Series Description	Section Reference
Varying softwood AAC attribution	§3.2.1
Varying planning horizon length	§3.2.5
Varying tightness of even-flow constraints	§3.2.3
Stochastic fibre demand variation	§3.2.4
Relaxation of line-wise profitability constraint	§3.2.5

Table 3.1 – Description of scenario series with cross-references

3.2.1 Varying Softwood AAC Attribution

This scenario series tests the effect of varying the proportion of bilevel softwood AAC that the principal attributes to the agent in each planning cycle. This series has a direct potential application to real-world cases—the principal controls the wood supply, and is not obligated to allocate 100% of AAC to timber licencees⁴.

We use a shorthand notation to refer to particular scenarios (e.g. *hw100sw110* refers to the scenario with 100% of hardwood AAC allocated and 110% of softwood AAC allocated). Scenario *hw100sw100* corresponds to our reference scenario (i.e. 100% attribution of both hardwood and softwood AAC). We simulate six alternate allocation levels, viz. 110%, 105%, 95%, 90%, 80%, and 60% of softwood AAC. Figure 3.1 shows simulation results for the seven scenarios in this series.

Over-allocating softwood AAC (i.e. scenarios *hw100sw110* and *hw100sw105*) has a negative effect on long-term wood supply stability, and a positive effect on wood supply magnitude. However, the effect is moderate—we can conclude that the bilevel wood supply model is moderately sensitive to AAC over-allocation level for our test case.

Under-allocating softwood AAC (i.e. scenarios *hw100sw95*, *hw100sw90*, *hw100sw80* and *hw100sw60*) has a positive effect on wood supply stability, and a negative effect on wood supply magnitude. At the 60% allocation level, wood supply stability is completely restored, albeit at a relatively high cost in terms of wood supply magnitude. Withholding a large fraction of softwood AAC indirectly creates a form of safety stock, which offsets the temporal wood supply decline observed in the reference scenario⁵.

harvesting on a one-period horizon, choosing the profit-maximising subset of available wood supply. Due to the optimal formulation of the bilevel model and perfect anticipation of volume consumption, the agent always chooses to harvest the entire wood supply. However, the agent may select to harvest this volume from a different combination of forest types than what was prescribed in the first period of the principal’s optimal solution.

⁴Provincial governments policy on public forest in Canada typically holds back a small fraction of AAC (around 5%) with justification that this hold-back policy satisfies the need for some form of safety buffer in the wood supply (e.g. to protect against risk of wood supply disruption from forest fires, insect outbreaks, blowdown, and other uncertain events).

⁵As discussed in Chapter 2, the slow decline in AAC observed in the reference scenario is attributable to the agent planning his own harvest schedule in the second stage of each planning cycle, which results in a slightly different mix of stand types than those featured in the first period of the optimal wood supply solution. The (deterministic) optimal wood supply solution, being located at the boundary of feasible solution space, is highly prone to infeasibility, and sensitive to deviations from the optimal solution.

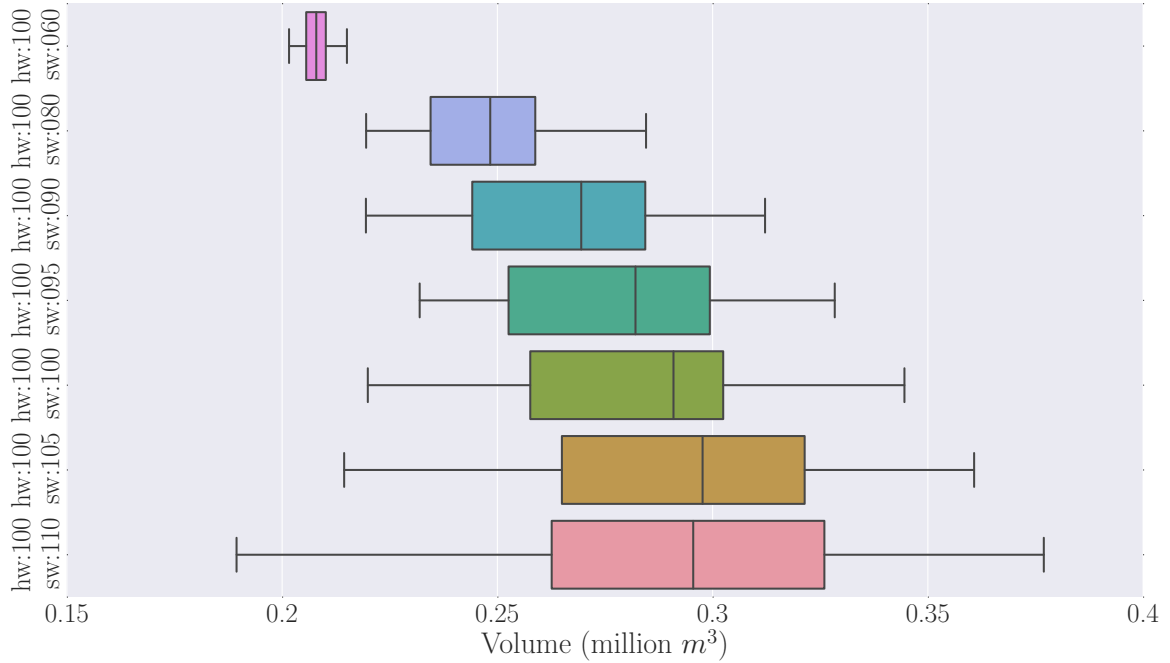


Figure 3.1 – Bilevel wood supply for a range of softwood AAC attribution levels

3.2.2 Varying Planning Horizon Length

This scenario series tests the impact of varying planning horizon length for both principal and agent. We test all combinations of horizon lengths for values 30, 20, 10, 5, and 1 periods. Agent horizon is never allowed to be longer than principal horizon, which leaves 15 valid horizon length combinations. We use a shorthand notation to refer to particular scenarios (e.g. *p30a01* refers to the scenario with a 30-period principal horizon and a 1-period agent horizon). Scenario *p30a01* is our reference scenario for this series, and corresponds to the *status quo* horizon lengths used in Quebec today. Simulation results, which we present in Figure 3.2, reveal interesting patterns on two levels.

On the first level, longer planning horizons improve stability of long-term wood supply. This effect is most pronounced when comparing scenarios with matched principal and agent horizon lengths (i.e. scenarios *p30a30*, *p20a20*, *p10a10*, *p05a05*, *p01a01*). This supports the common belief that long modelling horizons help stabilise solutions from long-term wood supply models (although we use a bilevel wood supply model here, which is not common practice).

On the second level, the relative effect of principal horizon length on wood supply stability becomes gradually less pronounced as the agent horizon shortens, with the least effect observed for single-period agent horizons. For example, shortening the principal horizon length in the reference scenario from 30 to 10 periods (i.e. scenarios *p30a01*, *p20a01*, *p10a01*), has relatively little impact on wood supply. This would indicate that the destabilising effect of a short agent planning horizon (due to residual systematic drift that is not anticipated by the bilevel model) overwhelms the stabilising effect of lengthening the principal horizon.

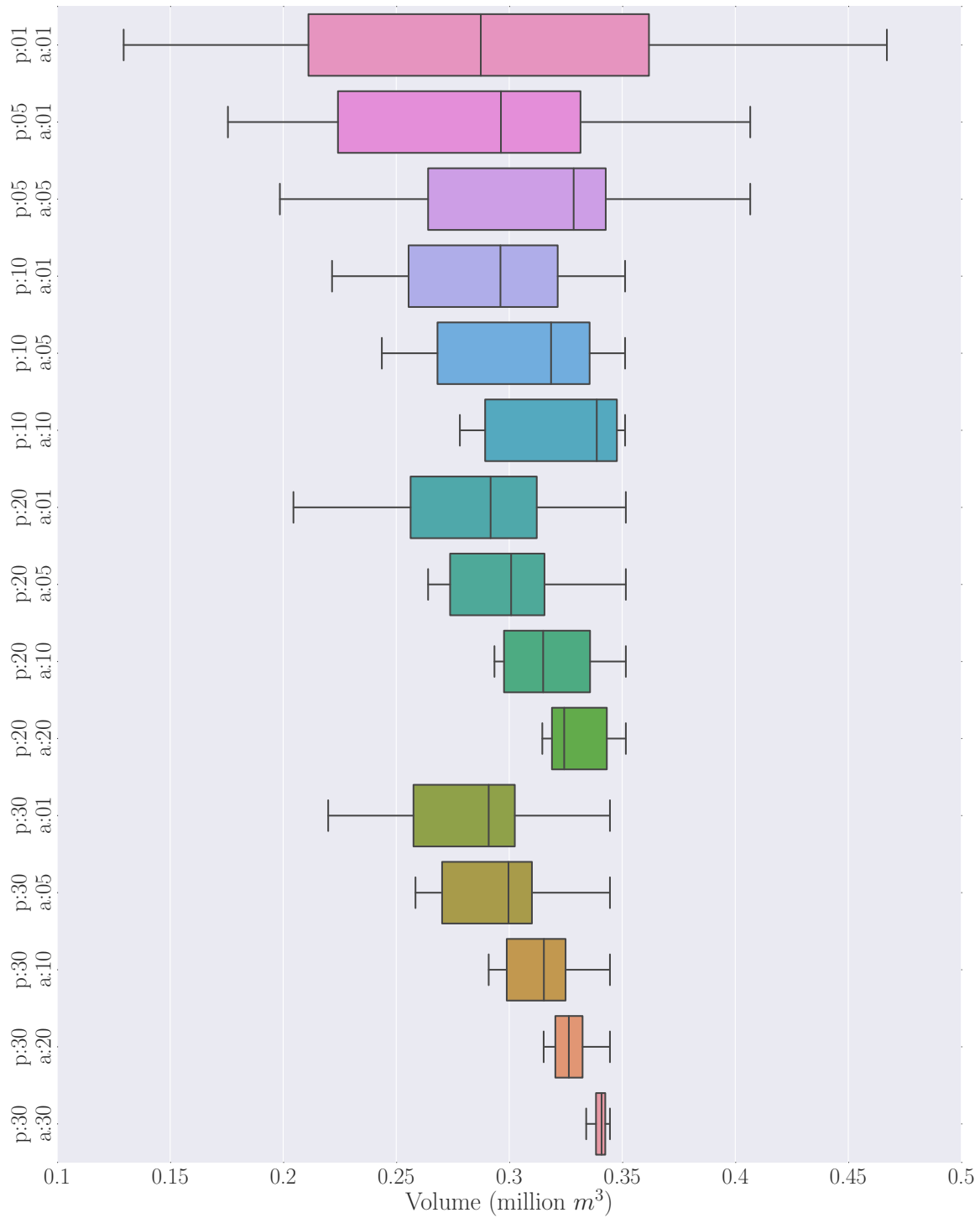


Figure 3.2 – Bilevel wood supply for a range of combinations of principal and agent planning horizon lengths

If one focuses on the $p30$ scenarios (i.e. $p30a01$, $p30a05$, $p30a10$, $p30a20$, $p30a30$), results provide some insight into the opportunity cost (in terms of both the magnitude and stability of long-term fibre supply) of maintaining the *status quo* distributed wood supply planning process, where a relatively

short-sighted agent is responsible for final implementation of harvesting operations. There is a parallel here with the *price of anarchy* concept in game theory (Shoham and Leyton-Brown, 2009; Nisan et al., 2007). The contrast between reference scenario *p30a01* and best-case scenario *p30a30* can be interpreted as an upper-bound estimate of the price of maintaining the *distributed* nature of the wood supply planning process (i.e. allowing the agent to plan his own harvesting, relative to a scenario where the principal controls both long-term wood supply planning and short-term fibre procurement planning processes).

We specify that this is an *upper bound* estimate, as scenario *p30a30* makes rather optimistic assumptions regarding the ability of the principal to produce long- and short-term feasible fibre procurement plans using a simple two-stage planning process as simulated here. Recall that we present simulation results from highly aggregated models. In practice, as-implemented fibre procurement plans are the end product of a complex hierarchical planning process, which iteratively refines plans until they are operationally feasible. Much of the detailed information required to efficiently produce feasible fibre procurement plans is absent from the high-level simulations we present here. To preserve both long- and short-term feasibility, the two-stage planning process (as simulated here) would have to be extended to include an *iterative* dimension—the long-term plan would have to be iteratively constrained, based on feasibility feedback from the (disaggregated) short-term planning level, until the first period of the long-term plan converges with the short-term plan. The process of iteratively constraining the long-term wood supply plan would have either a null (best-case) or negative effect on final AAC, hence our qualification of the contrast between scenarios *p30a30* and *p30a01* as an *upper bound* estimate of the price of distributed wood supply planning.

In practice, the agent (i.e. the industrial fibre consumer) has both the motivation and the technical expertise required to plan and implement economically efficient short-term fibre procurement plans. It is not clear that the principal has either the proper motivation or the technical capability (recall the information asymmetry that induces the principal-agent problem in the first place) to take over short-term fibre procurement planning (i.e. become an industrial fibre supplier, rather than simply a steward of the public forest resource) without negatively affecting industrial profitability (i.e. stability of fibre cost and value-creation potential). Given the tight profit margins typical of the forest products industry in many jurisdictions, it seems unlikely that the principal could completely exclude the agent from the fibre procurement planning process without destabilising the economic sustainability of the forest industry. In fact, the inverse situation may prove to be ideal—economically efficient wood supply planning may be attainable only if realised by the agent (Haley, 1985; Luckert, 1991; Kant and Berry, 2005; Haley and Nelson, 2007).

Simulation results suggest that the principal could improve long-term wood supply stability by inciting the agent to adopt a longer planning horizon. The case where principal and agent horizon lengths are equal is equivalent to the principal subsuming the agent (i.e. principal has total control of fibre procurement activities, including final selection of stands to be harvested). In reality, the principal has limited power to incite the agent to extend his planning horizon beyond one period, so forcing an extension of the agent’s planning horizon in the bilevel anticipation may in fact introduce a problematic bias into the model.

Development of such incentive policies, using game-theoretic analysis of the decoupling point between principal and agent and the *price of anarchy* concept, represents an interesting direction for further

research. Taken from a slightly different perspective, our simulation results hint at potential benefits of a third-party fibre procurement planning integrator, as described in Azouzi et al. (2012).

Note that computational effort required to solve wood supply models can increase geometrically with the number of planning periods for certain model formulations. This is mostly due to the combinatorial explosion of potential future system states for large numbers of planning periods⁶. Our test wood supply model solves very quickly (i.e. approximately 10 seconds), in part due to the computationally efficient SilviLab solver engine, however production models using commercial wood supply modelling software may require significantly more computational effort to solve similarly-sized instances. These long solution times may limit the number of scenarios that can realistically be run by government wood supply analysts, which may in turn limit the quality and scope of wood supply analyses. Our simulation results hint at a potentially interesting tradeoff between computational effort and solution quality. For our test case, principal planning horizon can be shortened by a factor of 3 (i.e. from 30 to 10 periods) with minimal effect on wood supply stability⁷. Exploratory wood supply scenarios, or sensitivity analyses requiring large numbers of simulations, could potentially be run using shortened model horizons. Assuming limited budget for wood supply analysis resources, there is potential to either reduce the cost of analysis with minimal impact on quality, or to improve the quality of the analysis at similar cost.

Finally, simulations results from this scenario series suggest that adopting a unified horizon for long- and short-term fibre procurement planning may preclude some of the solution instability we observe in the DWSPP. For example, this could involve simultaneously shortening the long-term horizon and lengthening the short-term horizon, such that they meet in the middle at some acceptable compromise. For the sake of discussion, a 25-year (i.e. 5-period) unified horizon length might be a good starting point for further analysis—this would correspond to scenario *p05a05*.

3.2.3 Varying Tightness of Even-flow Constraints

This scenario series tests the effect of varying the tightness of even-flow constraints in the upper-level model (see §2.2.1.1 for a mathematical formulation of the upper-level model). The reference scenario *hw005sw005* simulates a 5% even-flow constraint gap for both hardwood and softwood. We simulate five alternate tightness values (0%, 1%, 10%, 20%, and 50%). We use a shorthand notation to refer to particular scenarios (e.g. *hw001sw001* refers to the scenario with a 1% even-flow constraint gap for both hardwood and softwood). Figure 3.3 shows simulation results for the six scenarios in this series.

Wood supply stability degrades slightly when the even-flow constraints are tightened in scenarios *hw001sw001* and *hw000sw000*. Although this observation is somewhat counter-intuitive, these results are consistent with the tendency for deterministic models to be sensitive to small deviations from the optimal solution. Tightening the even-flow constraints apparently makes the wood optimal wood supply solutions less robust to the slight deviations we simulate in the second phase of each simulated planning cycle. Conversely, wood supply stability improves slightly when the even-flow constraints are

⁶Geometric explosion of solution time and memory requirements, as a function of number of planning periods, is particularly problematic when the wood supply is formulated using a *Model II* approach. For example, this is the case for the Remsoft Woodstock modelling platform, which is used extensively by government wood supply planning authorities in Canada.

⁷We suspect that the amount of planning horizon compression that the principal can safely implement, without inducing unacceptable increases in risk of wood supply failure, is linked to a number of factors that will vary between problem instances (e.g. timing of the critical period and, tightness of even-flow-constraints, etc.).

tightened in scenarios *hw010sw010* and *hw020sw020*, although we also observe a slight reduction in wood supply magnitude.

Overall, results show that the bilevel model is relatively insensitive to the tightness of the even-flow constraint, as shown by the relatively small change in wood supply stability and magnitude in scenario *hw050sw050*. Relaxing even-flow constraint tightness will tend to widen the feasible bilevel solution space, which may open up possibility for interesting trade-offs in more complex upper-level model formulations (e.g. multi-objective formulations designed to explore tradeoffs between ecosystem, social and economic values).

The use of even-flow constraints in wood supply models has been the subject of debate, with some experts advocating their relaxation or outright exclusion (Gunn, 2009; Luckert and Williamson, 2005). Our simulation results show that tight even-flow constraints may not be an effective policy instrument if the objective is to stabilise long-term wood supply in a distributed planning environment, where the principal has limited control over the execution of the harvest schedule.

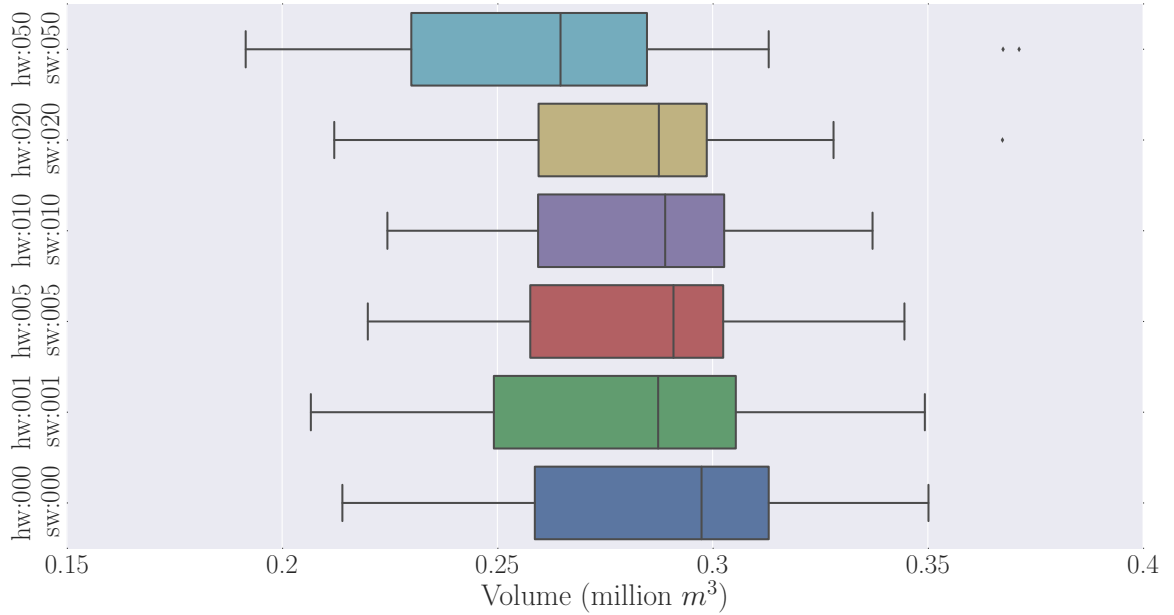


Figure 3.3 – Bilevel wood supply for a range of even-flow constraint tightness levels

3.2.4 Stochastic Agent Fibre Demand

The reference scenario optimistically simulates perfect anticipation of agent fibre consumption volume. In practice, we expect some error in the anticipation of fibre demand. This scenario series tests the sensitivity of wood supply to controlled stochasticity in agent fibre demand.

To implement this scenario series, we simulate randomly-generated AAC consumption levels in the second stage of each planning cycle. More specifically, we randomly generate the proportion of AAC consumed in each cycle by sampling from a uniform distribution on the interval $[0.9, 1.1]$ (i.e. $\text{AAC} \pm 10\%$). This is similar to the range and distribution of historic variation in AAC consumption in Canada between 1990 and 2012 (see Figure 2.1). Note that the data presented in Figure 2.1 was

compiled for the *status quo* situation, where the wood supply models systematically over-estimate actual fibre demand. The simulation results presented here assume that agent fibre demand is distributed symmetrically about AAC—we have shown that our bilevel model virtually eliminates the species-skewed negative consumption bias, therefore it is reasonable to assume any residual error in agent fibre demand estimation will be centred on AAC (i.e. free of bias).

Figure 3.4 presents results from 20 random replicates of the experiment, comparing replanned AAC for the reference (i.e. deterministic) and test (i.e. stochastic) demand scenarios. For the stochastic scenario, we plot the 2σ error band (i.e. 95% of expected variability is contained within the error band).

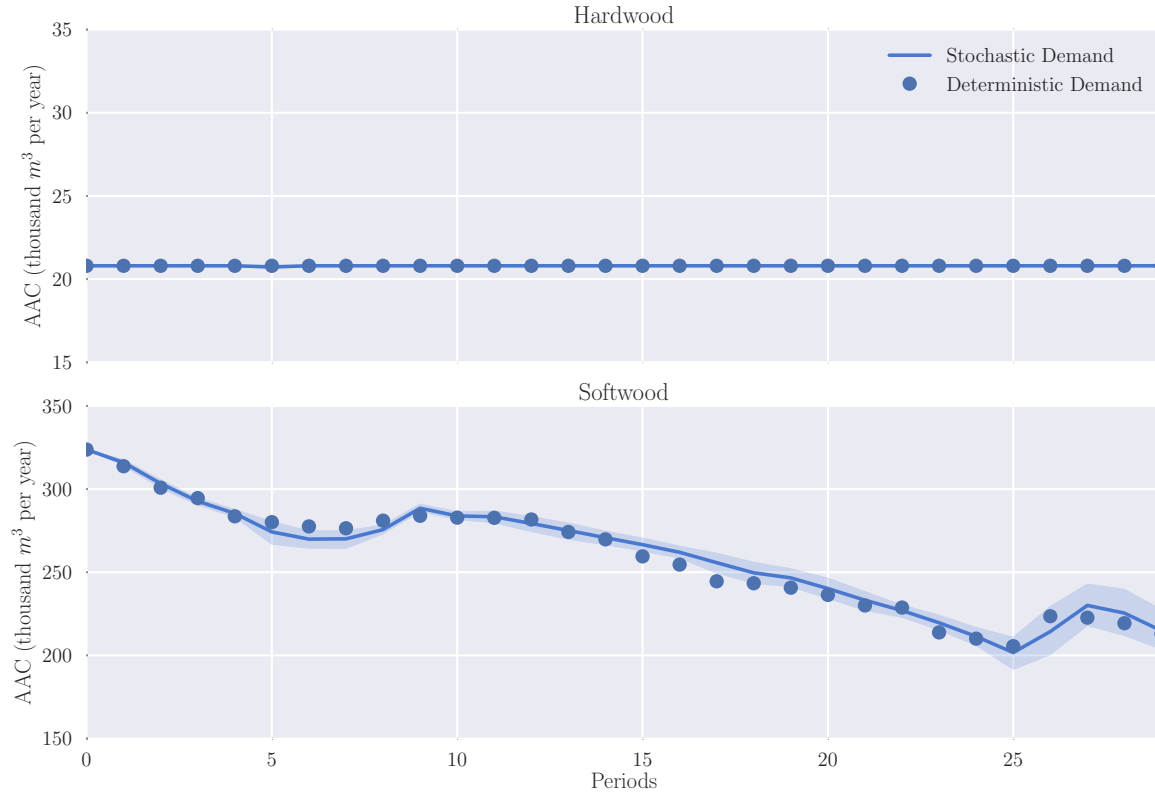


Figure 3.4 – Bilevel wood supply under deterministic and stochastic demand

These simulation results show that error in anticipation of demand within the bilevel model does not induce a systematic drift effect, when stochastic agent fibre demand is uniformly distributed on an interval centred on re-planned AAC. This result has important policy implications, in terms of potential feasibility of implementing a bilevel wood supply modelling solution in practice—although we can expect any agent-anticipation mechanism will never be perfect (i.e. there will always be residual information asymmetry between the principal and the agent), this error in the agent-anticipation function is not necessarily problematic *if it is not biased* (i.e. mean fibre consumption is randomly distributed on an interval centred on AAC). Thus, the principal could potentially allow the agent to exceed AAC in periods of peak demand and lag behind AAC in periods of depressed demand, without necessarily compromising the stability and magnitude of the wood supply.

By providing a certain amount of short-term flexibility, such a policy could potentially allow the agent capture more short-term value creation opportunities, thereby improving mean profit without

compromising the sustainable forest management objectives. Implementing this type of flexibility within the short-term fibre procurement planning process may be key to capturing ephemeral value-creation opportunities (Gautam et al., 2014).

3.2.5 Relaxation of Line-wise Profitability Constraint

Our base bilevel scenario imposes a *line-wise profitability* constraint⁸ on the agent sub-model.

For our test dataset, hardwood fibre consumption is constrained by low hardwood fibre processing capacity⁹. Flow conservation constraints in our model limit harvest volume to consumed volume (i.e. all harvested fibre must be consumed). In the reference scenario, the combination of several elements limits sustainable bilevel AAC, *viz.* low hardwood processing capacity, high softwood fibre demand, line-wise profitability constraints and flow conservation constraints.

Our test dataset features an alternative agent process that could potentially consume hardwood fibre (beyond the capacity of the single hardwood sawmill). This alternative hardwood consumption process can be described as a *storage* process, which has a negative net unit value (i.e. the agent has to pay a constant unit cost to push fibre through this process). The line-wise profitability constraints prevent any fibre from flowing through this process in the reference scenario, as any utilisation of this process would have a negative effect on hardwood line profit.

If the line-wise profitability constraints are relaxed in the agent model (i.e. if we simulate centralised agent fibre procurement planning, which is the default configuration of the LogiLab model), the cost of pushing a small volume of hardwood fibre through the storage process may be offset by profits from the large volume of additional softwood fibre that is made available¹⁰. This improves alignment between forest inventory volume and industrial fibre consumption, which indirectly allows the principal to increase bilevel AAC.

Figure 3.5 shows simulation results for a scenario where we relax the line-wise profitability constraints in the agent model. The collaborative scenario simulates perfectly integrated agent fibre procurement behaviour, and makes it possible for the hardwood line to operate at a reduced profit if this is profitable for the network as a whole. The perfect integration assumption is optimistic—the collaborative scenario represents an upper bound, *ceteris paribus*, on the value of collaborative planning. Taken from the perspective of the principal, the difference in value of the two scenarios presented here represents an upper bound on the cost the principal is willing to pay (in the form of incentives, subsidies, or other policy instruments) to incite the agent to adopt collaborative fibre procurement behaviour. Azouzi et al. (2012) discuss potential benefits of adding a third-party integrator agent to forest products value creation networks. Our methodology for simulating the collaborative agent scenario could potentially be used to determine an upper bound on the fee the principal would be willing to pay a hypothetical third-party integrator agent to oversee fibre procurement operations.

⁸See §2.C

⁹Hardwood logs must pass through a single hardwood sawmill, whose annual processing capacity is limited to approximately one third of initial hardwood AAC (using the classic wood supply model).

¹⁰The initial softwood to hardwood ratio in the forest inventory is approximately 10:1 for our test dataset, so we can expect that approximately 10 additional units of positively-valued softwood will be made available to the system for every unit of negatively-valued hardwood that is pushed through the system. This provides a considerable amount of leverage in the model, even if the marginal loss on every unit of stored hardwood exceeds the marginal gain on every addition unit of softwood that can be processed.

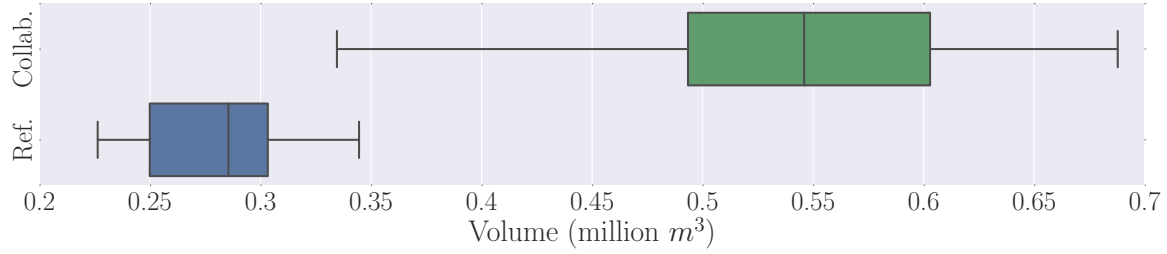


Figure 3.5 – Bilevel wood supply under distributed and centralised planning

Collaborative fibre procurement planning almost doubles the average fibre consumption (92% increase) relative to the reference bilevel scenario. Implementing intra-agent collaborative fibre procurement behaviour is beyond direct control of the principal, but he has a clear incentive¹¹ to lobby the agent to align fibre consumption capacity with maximum potential wood supply.

For our test case, the increase in hardwood fibre consumption under collaborative fibre procurement planning has a positive-feedback effect on long-term fibre supply. The principal may now harvest the problematic mixed-wood stands in the first few planning cycles. These mixed-wood stands, which are left standing in the reference scenario, can now be replaced with pure softwood plantations. In the medium term, this has the effect of partially re-aligning the inventory of standing timber with industrial fibre demand. In practice, current agent behaviour more closely resembles the greedy agent behaviour simulated in the reference scenario.

The collaborative scenario simulates willingness of the hardwood line to operate at a reduced profit when this benefits the rest of the network. Some form of collaborative benefit-sharing agreement would have to be in place for this setup to be economically optimal for the hardwood sub-agent. Although the potential for collaborative benefit-sharing has been discussed in the literature, for forestry and other supply-chain contexts (Audy et al., 2012; Lehoux et al., 2011; Forget et al., 2008; Dudek and Stadtler, 2005; Frisk et al., 2010; Guan and Liu, 2011; Beaudoin et al., 2010), benefit-sharing in the context of collaborative fibre procurement planning has received relatively little attention. Furthermore, collaborative benefit-sharing fibre procurement agreements in a distributed planning environment seem to be rare in practice, and evidence of such is mostly anecdotal. Elaboration of benefit-sharing agreements to help balance asymmetrical species-wise fibre demand, and the game-theoretic conditions under which these agreements might be stable, represents a promising direction for further research.

Simulation of collaborative fibre procurement planning allows the principal to increase bilevel AAC, and brings fibre consumption levels closer to the maximum biophysical level that can be sustained by anticipated forest growth. Note that wood supply for the collaborative scenario is relatively volatile, in contrast with the slow and steady decline of wood supply in the reference scenario. Both scenarios make similarly optimistic assumptions regarding the ability of the principal to anticipate agent behaviour (i.e. perfect anticipation of fibre consumption *volume*, although the agent may select an unanticipated mix of forest types in his harvest plan executed in the second stage of the simulation at each planning cycle). The aforementioned volatility in the collaborative scenario may be linked to increased fragility of the wood supply model as the optimal upper-level solution approaches the biophysical production

¹¹As indicated by the objective function of the wood supply model, which maximises long-term sustainable fibre consumption.

limit of the forest. We conjecture that robustness of these solutions could be improved (i.e. wood supply volatility reduced) by extending the formulation of the bilevel model to include some form of explicit safety-stock management mechanism. The development of an efficient safety-stock management mechanism, and the determination of minimal buffer levels required to stabilise wood supply under fibre demand uncertainty, represent promising topics for further development.

3.3 Discussion

Our innovative combination of a two-stage rolling-horizon simulation framework and a bilevel wood supply model provides some heretofore unavailable insight into the distributed wood supply planning problem. The results presented here have some interesting policy implications, and suggest directions for future research.

The first scenario series (see §3.2.1) tests the effect of varying the proportion of softwood AAC that is attributed to the agent at each planning cycle. It is not surprising that results show long-term wood supply is more volatile at higher attribution levels, and more stable at lower attribution levels. However, it is interesting to note that a policy of systematic over-attribution of AAC (e.g. scenario *hw100sw110*) has a relatively moderate and predictable negative effect on wood supply (i.e. the gradual decline in wood supply that can be observed in reference scenario *hw100sw100* is accentuated). The last scenario in this series completely restores wood supply stability, but requires a 40% reduction of softwood AAC attribution¹². For political reasons, both the principal and the agent may like to see short-term AAC levels maintained as high as possible. We suspect that withholding 40% of AAC may meet with considerable opposition, and may not be perceived as an attractive policy option. It may be possible to develop more robust model formulations that preserve long-term wood supply stability characteristics of scenario *hw100sw060* at higher AAC levels, for example by integrating explicit safety stock management mechanisms into the upper level model.

The second scenario series (see §3.2.2) examines the relationship between principal and agent planning horizon lengths and long-term wood supply. Results show that longer planning horizons tend to improve both the magnitude and the stability of wood supply. Conversely, wood supply magnitude and stability tends to degrade as a function of the difference in principal and agent horizon lengths—for a given principal horizon length, the lowest mean wood supply and highest wood supply variability are observed for single-period agent planning horizon. For the reference scenario (i.e. principal plans on 30-period horizon and agent plans on 1-period horizon), some of the potential benefit of the principal forecasting wood supply 30 periods ahead is lost when the agent replans harvesting using a short (i.e. 1-period) horizon. For example scenario *p10a10* dominates the reference scenario in terms of both stability and magnitude of the wood supply—perfect execution of a 10-period wood supply plan is better than imperfect execution of a 30-period wood supply plan. Although the bilevel formulation generally improves wood supply stability, relative to the classic model, the bilevel optimal solution are nonetheless highly sensitive to deviations during execution. Extending the agent planning horizon tends to improve coherence between the stages in our simulated planning process, which has a positive

¹²Note that softwood AAC for scenario *hw100sw060* is greater than softwood AAC for the reference scenario (*hw100sw100*), and that the 40% reduction in attribution level for scenario *hw100sw60* is relative to softwood AAC for that scenario. Average softwood fibre consumption for scenario *hw100sw060* actually drops by 26% relative to reference scenario *hw100sw100*. This 26% drop can be seen in Figure 2.4, by comparing the mean planned volume of scenario 3 and the mean executed volume of scenario 5.

effect on both stability and magnitude of the wood supply. Although we caution against premature generalisation from our single test case, these preliminary results warrant further investigation into the development of policy instruments to incite the agent to lengthen his planning horizon. Also, these results suggest that wood supply modelling horizons could potentially be shortened considerably in some cases.

The third scenario series (see §3.2.3) varies the tightness of even-flow constraints. Surprisingly, tighter even-flow gaps tend to have a negative effect on wood supply stability, although this effect is relatively moderate for our test case. In practice, even-flow gap settings may tend to be tighter than necessary (typically around 5%, as in our reference scenario), possibly destabilising long-term wood supply and over-constraining short-term value-creation opportunities. Relaxing even-flow constraints may constitute a promising policy instrument for adding flexibility and stability to highly-constrained wood supply planning environments, although we caution against generalising results from our case study and recommend further testing on other datasets.

The fourth scenario series (see §3.2.4) tests the effect of adding stochastic variability in fibre demand in the second stage of the simulation at each planning cycle. We conclude that allowing fibre consumption to vary in a bounded range on either side of bilevel AAC does not necessarily compromise wood supply sustainability, as long as demand anticipation error is unbiased. This is a promising area for further wood supply policy research, as allowing agent fibre consumption to occasionally exceed AAC (e.g. when markets conditions are favourable) could potentially improve planning process agility (i.e. enabling the agent to capture ephemeral short-term value creation opportunities, thereby potentially improving mean profitability).

The fifth scenario series (see §3.2.5) shows that wood supply in the reference scenario is bound by the line-wise profitability constraint. For our test case, bilevel wood supply could potentially be doubled if the agent adopts more collaborative fibre-procurement behaviour. Assuming that wood supply scarcity is a problem in certain regions, collaborative agent fibre procurement represents a very promising topic for further study.

3.4 Conclusion

The simulation results presented in this chapter explore bilevel wood supply policy space. We show the response of wood supply to variation in five dimensions of policy space, providing valuable insight into the performance of the bilevel model in a rolling-horizon replanning context.

In some cases, the results confirm our intuitions (e.g. wood supply stability can be improved by withholding a portion of AAC). In other cases, the results are more surprising (e.g. tightening even-flow constraints in the wood supply model has a slight destabilising effect on wood supply).

Although the bilevel model helps to stabilise the long-term wood supply¹³, most bilevel scenarios exhibit some form of residual systematic drift (i.e. gradual decline of wood supply at each replanning cycle). Residual drift is moderate, compared to the highly unstable *status quo* scenario (see Figure 2.5, scenarios 1 and 3), but a systematic decline in long-term wood supply is nonetheless incoherent with the even-flow constraints in the wood supply model. Although AAC decline for the reference scenario

¹³See Chapter 2 for an in-depth comparison of long-term performance of classic and bilevel models.

is moderate (approximately 1.3% decline per planning cycle), the cumulative result after 30 planning cycles is a 40% decline in initial wood supply. This decline is not predicted by the bilevel model in any of the 30 cycles of rolling-horizon replanning, but is revealed by our rolling-horizon replanning simulation approach. We were able to simulate stable wood supply in two cases (scenario *hw100sw060* and scenario *p30a30*), which is promising. Overall the results presented here show that our combination of a bilevel wood supply model and a rolling-horizon replanning simulation approach constitutes a powerful new approach for exploring wood supply policy options.

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Conclusion

This chapter summarises contributions of the thesis, describes directions for further research, and presents some final remarks.

Contributions

This thesis has a twofold objective. The first objective is to demonstrate that ineffective linkages between long- and short-term forest management planning are currently compromising the credibility of the distributed wood supply planning process—we addressed this objective in Chapter 1. The second objective is to propose improved linkage design, thereby improving the credibility of the distributed wood supply planning process (DWSPP)—we addressed this objective in Chapter 2. Chapter 3 explores the possibilities of the novel wood supply simulation framework developed in Chapters 1 and 2, providing insight into bilevel wood supply policy space and revealing several potential directions for further research.

Chapter 1

Chapter 1 presents the distributed wood supply planning problem (DWSPP), which we describe from a game-theoretic perspective as an instance of the principal-agent problem. The *status quo* wood supply planning process fails to consider the principal-agent relationship that exists between government and industry, and ignores historical fibre consumption data showing that some portions of AAC are never consumed by industry. Consequently, the classic wood supply planning model systematically over-estimates periodic fibre consumption volume for current and future time periods.

We present an iterative two-phase modelling framework, which we use to simulate the net result of this negative fibre-consumption bias after repeated rolling-horizon replanning cycles. We show that the classic wood supply model fails to predict potential wood supply failures under certain circumstances, notably when fibre demand is a species-skewed subset of AAC. Thus, we conclude that the classic wood supply optimisation model formulation does not constitute a rational basis for the implementation of sustainable forest management.

These wood supply failures, which have not been discussed in the literature, were revealed by our iterative simulation methodology. We therefore recommend that a similar methodology (i.e. iterative two-stage rolling-horizon replanning simulation) be implemented, in practice, as part of the distributed wood supply planning process (regardless of wood supply model formulation).

Finally, we conjecture that extending the classic wood supply model to explicitly anticipate industrial

fibre consumption behaviour could reduce risk of long-term wood supply failure. More specifically, we propose that the principal-agent relationship between government and industry could be modelled mathematically as a bilevel optimisation problem.

Chapter 2

Chapter 2 presents a bilevel extension of the classic wood supply model. We show that the bilevel solution space is non-convex, which makes the general case difficult to solve to global optimality. We identify a convex special case, and develop a methodology to solve this special case to global optimality.

Using the two-stage simulation framework and dataset presented in Chapter 1, we describe a computational experiment comparing long-term wood supply stability for classic and bilevel models. We hypothesise that the bilevel model will yield a more stable wood supply. Results from this case study confirm that the bilevel model mitigates the risk of wood supply failure, although we observe some residual instability in long-term wood supply. We were able to eliminate this residual instability in our simulations by holding back a large proportion of AAC (an indirect form of buffer stock management), albeit at a high cost in terms of wood supply magnitude.

The computational experiment also allows us to compare the computational effort required to solve the classic and bilevel models, using an input dataset of realistic size and complexity. We observe that the bilevel model approximately doubles¹⁴ the computational effort to solve the wood supply model, which we do not consider to be problematic, given that wood supply plans tend to be updated infrequently.

Chapter 3

Chapter 3 explores the bilevel wood supply policy space. Using the same setup as the computational experiment presented in Chapter 2, we simulate several alternative scenarios, which we group into five scenario series. Each series contains a number of scenarios, testing a range of values for a selected dimension.

The first scenario series simulates the effect of varying the proportion of softwood AAC attributed to the agent. We simulate a range of attribution levels¹⁵ varying between 110% and 60%. We confirm that wood supply stability is inversely related to AAC attribution level (i.e. wood supply is more stable when a smaller proportion of bilevel AAC is attributed to the agent). Holding back a portion of bilevel AAC creates a form of buffer stock in the standing timber inventory, which helps mitigate the residual instability associated with the distributed nature of the wood supply planning process. Thus, we were able to simulate stable long-term wood supply using a combination of bilevel modelling and holding back a large (i.e. 40%) proportion of AAC at each cycle. Conversely, over-attribution tends to destabilise long-term wood supply.

The second scenario series varies principal and agent horizon lengths, and confirms that wood supply stability and magnitude are positively correlated with planning horizon length and negatively correlated with the difference between principal and agent horizon lengths. These results provide evidence in support of long planning horizons (e.g. 30 periods) in wood supply models, although it may be possible to use significantly shorter horizons (e.g. 5 or 10 periods) in some cases with only minor negative effect

¹⁴For our test dataset.

¹⁵This range includes three scenarios presented in Chapter 2: 100% (scenario 3), 80% (scenario 4), 60% (scenario 5).

on wood supply stability and magnitude. It may be possible to eliminate for the negative effects of shortening planning horizons by adding carefully chosen end-state constraints to the upper-level model, however this has not been tested. We also conclude that both stability and magnitude of bilevel wood supply are highly sensitive to even slight deviations from first-period optimisation model solutions.

The third scenario series varies the tightness of even-flow constraints. Results show that the bilevel wood supply model is surprisingly insensitive to even-flow constraint tightness. Although wood supply stability deteriorates slightly when constraints are relaxed, this effect is surprisingly moderate. Under certain circumstances, relaxing even-flow constraints may provide much-needed flexibility in the planning space, without compromising long-term wood supply sustainability.

The fourth scenario series, which simulates stochastic variation of fibre demand at each planning cycle, shows that allowing industrial fibre consumption volume to vary in a bounded range on either side of AAC does not necessary compromise long-term wood supply sustainability. In the context of bilevel wood supply planning, this represents an interesting potential policy alternative to the common interpretation of AAC as a strict upper bound on harvest levels.

The fifth scenario series simulates that bilevel wood supply could potentially be increased¹⁶ if the agent adopts more collaborative fibre procurement planning behaviour. By subsidising consumption of a relatively small volume of negatively-valued hardwood fibre, the agent can potentially gain access to a large volume of (otherwise unavailable) positively-valued softwood fibre. These results highlight an interesting opportunity for sustainably increasing bilevel wood supply beyond current levels.

Overall, Chapter 3 shows that a combination of our rolling-horizon simulation framework and our bilevel wood supply model constitutes a powerful new tool for exploring the impact of various policy options on wood supply stability and magnitude.

Further Research

This thesis represents a pioneering work on the distributed wood supply planning problem (DWSPP). We describe a novel two-stage rolling-horizon replanning simulation framework, present heretofore undocumented failure modes of the classic wood supply model, and propose an improved wood supply model formulation. This section summarises several open questions and opportunities for further research that were identified throughout this thesis.

Alternative Instances

Chapter 1 presents a case study, based on a synthetic dataset of realistic size and complexity. Simulation results presented in this chapter support our conclusion that incoherence between projected and actual fibre consumption may jeopardise long-term wood supply sustainability. Although we can reasonably conjecture that similar results would be obtained for other datasets with similar characteristics (i.e. species-skewed negative industrial fibre consumption bias), results from our case study cannot be directly generalised to other cases without further testing. Simulation results presented in Chapter 3 show

¹⁶Wood supply is almost doubled for our test case, although results may vary for different problem instances. Simulating more centralised (i.e. collaborative) agent fibre procurement behaviour constitutes a relaxation of the line-wise profitability constraint, and relaxing constraints in an optimisation model tends to have a either a positive (best-case) or null (worst-case) effect on objective function value. Thus we can generally expect increases in intra-network agent collaboration to have positive effect on bilevel wood supply.

that bilevel wood supply predictions are sensitive to a number of key input parameters. Consequently, we recommend replicating the case study from Chapter 1 using alternate datasets, thereby providing richer insight into wood supply failure modes under the *status quo* distributed wood supply planning process.

Model Complexity

The wood supply model used in our computational experiments is a simplified version of a real government wood supply model. The original model featured a range of harvest treatment options (including some partial cut treatments), however we retained only the clearcut option for our set dataset. We also omitted a number of area-based constraints that appeared in the original model (e.g. minimum old-growth area in each period, maximum area affected by plantations in each period, etc.), retaining only the even-flow constraints. Although these simplifications were useful and justified in the context of our pioneering efforts to isolate the long-term effects of incoherent distributed hierarchical planning, we recommend further testing with an integral (i.e. unsimplified) wood supply model dataset¹⁷.

Fibre Procurement Cost Variability

Our test dataset models procurement cost using static unit cost parameters. In reality, wood procurement cost is highly variable. Common sources of wood supply procurement cost variability include: harvest treatment prescription (e.g. clearcut versus partial cut), harvesting machine system (e.g. full-tree system versus cut-to-length system), stand attributes (e.g. mean stem size, basal area, block topography, block shape, soil sensitivity to rutting, species mix and number of log assortments required, etc.), spatial distribution of blocks within harvest sectors and harvest sectors within the forest (e.g. does harvesting machinery need to be moved with a truck, or is the next block near enough that it can self-propel?), landscape constraints (e.g. green-up delays, or other regulations constraining location and timing of harvest activity), seasonal effects (e.g. weight limits can limit load sizes, which increases unit transportation cost, etc.), block distance from existing roads, block distance from mill gate, and stumpage cost variations (e.g. government may reduce or increase stumpage cost on some blocks to incite or deter harvesting).

These factors are not explicitly modelled in the computational experiments, yet in reality these are all considered by the agent during his fibre procurement planning. Integrating some or all of these factors into a bilevel wood supply model may significantly affect anticipation of industrial fibre consumption behaviour, which in turn may affect AAC determined by the principal (if using the bilevel wood supply model). We recommend extending the agent-anticipation model to include sources of fibre procurement cost variability, and testing the impact of these extensions on long-term bilevel wood supply stability.

Nested Supergames

We present the distributed wood supply planning problem from a game-theoretic perspective, and model principal-agent interaction as a non-repeated (i.e. one-shot) game, *viz.* principal determines AAC, agent consumes profit-maximising subset of AAC. The bilevel model could be extended to include an iterative negotiation dimension, which corresponds to a *repeated game* or *supergame* in game theory. Modelling the distributed wood supply planning process as a supergame may converge on socially

¹⁷Using our two-stage rolling-horizon replanning framework.

optimum equilibrium solutions, which could be superior to the selfish optimal behaviour we model here (as a one-shot game). This supergame concept could also be applied at a larger scale, to model anticipation of upcoming planning cycles (and, potentially, memory of past planning cycles). Finally, both supergame scales could potentially be nested (i.e. iterative negotiation within each planning cycle, anticipation of negotiations in future planning cycles), although we anticipate that implementing such a nested-supergame wood supply model poses serious technical challenges.

Multi-Period Agent Anticipation

We model agent behaviour using a single-period (i.e. 5-year) planning horizon. In reality, industrial fibre procurement planning typically spans two or more periods. Thus, extending the bilevel wood supply model to include multi-period anticipation of agent behaviour should produce more accurate results. This represents an interesting direction for further research.

Computational Tractability of General Case

We can solve the special (i.e. convex) case of the bilevel problem in approximately twice the computation effort required to solve the classic wood supply model. We present an enumeration-based method to solve the general (i.e. non-convex) case of the bilevel problem, which, although technically valid (i.e. will converge on global optimal solution if all possible feasible solutions are inspected), is not computationally tractable for realistically-sized problems. The general case might be rendered tractable if the enumeration method presented were enhanced (e.g. with an intelligent branching heuristic, or with an alternative cutting plane algorithm).

Variable (Stochastic) Exogenous Prices

Our test dataset assumes fixed exogenous prices for end-products in the agent network (e.g. lumber, paper, chips, etc.), whereas in reality end-product prices are highly volatile and difficult to predict based on historical data. Testing the impact of stochasticity in exogenous end-product prices on bilevel wood supply represents an interesting opportunity for further investigation.

Collaborative Agent Behaviour

Our bilevel model includes line-wise profitability constraints, which force hardwood and softwood lines to be independently profitable. In Chapter 3, we relax these line-wise profitability constraints, and show that bilevel wood supply could potentially be doubled if the agent adopts a more collaborative fibre procurement behaviour. Although the principal does not have the power to compel the agent to collaborate, he may be able to encourage this type of behaviour using policy incentives. Identification and testing of such collaborative policy incentives, and development of game-theoretically stable benefit-sharing strategies, represents a promising direction for further study.

Alternate Bilevel Wood Supply Model Formulations

We present a bilevel extension of the classic wood supply model. The only difference between classic and bilevel wood supply model implementations is the addition of the agent-anticipation mechanism. Although this was useful in the context of our computational experiment, further modification of the

classic wood supply model could potentially generate globally improved wood supply solutions. For example, the long-term objective function maximises volume and the short-term objective function maximises profit. Modifying the long-term objective function to maximise value may yield more coherent solutions overall. We recommend exploration of alternate bilevel wood supply model formulations.

Two-Stage Rolling-Horizon Replanning Simulation Framework

The two-stage rolling-horizon replanning simulation framework we developed has played a key role in this project: the framework made it possible to measure the difference between projected and actual trajectories of forest system state after multiple (distributed) planning cycles, which in turn led to the conclusion that the current planning process was failing to ensure wood supply sustainability. We therefore recommend that a similar approach (i.e. simulation of future rolling-horizon replanning cycles) be implemented in practice as part of the standard wood supply planning process, as this may provide early warning of impending (and potentially catastrophic) wood supply failures that are not currently predicted by the *status quo* planning process.

Final Remarks

We present a number of scenarios comparing the performance of classic and extended wood supply optimisation models after repeated two-stage rolling-horizon replanning simulation cycles. We conclude that our bilevel model formulation, if implemented in practice, could potentially reduce the risk of wood supply failures. By extension, this would improve the credibility of sustainable forest management planning. We hope the work presented in this thesis will incite a critical review of current wood supply planning practice.